Survey Research 2019-08-07

2019 K-Innovation ODA Program with Azerbaijan

Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan

STEPI SCIENCE AND TECHNOLOGY POLICY INSTITUTE

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2019 K-Innovation ODA Program with Azerbaijan

Title	Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan
Institute	Science and Technology Policy Institute (STEPI)
Partner Country	Republic of Azerbaijan
In Cooperation with	Azerbaijan National Academy of Sciences - High Technologies Park (ANAS-HTP)
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Executive Summary

Azerbaijan, officially known as the Republic of Azerbaijan and called the Land of Fire and the Pearl of the Caucasus, is located in South Caucasus and on the shores of the Caspian Sea. It shares land borders with Russia, Iran, Turkey, Armenia, and Georgia. The country became an independent state in 1991 after gaining its sovereignty from Russia. Azerbaijan is classified as an upper middle-income country with vast oil and gas resources. The large amount of revenues accumulated in the state oil fund made the country's economy very sensitive to the price of oil.

To solve the economic problems mainly caused by the government's heavy dependence on the oil industry and very weak manufacturing output, the Azerbaijan government has been working hard to improve the structure of the manufacturing industry and developing other areas of non-oil sector such as agriculture, transport, and communication technologies. Furthermore, in the national development plan "Azerbaijan 2020: view to the future," the Azerbaijan government highlighted the necessity to improve the institutional condition, strengthen scientific and technological potential, and train high-caliber personnel.

The Azerbaijan National Academy of Sciences (ANAS) is the leading organization for developing, proposing, and implementing science-related programs, plans, and strategies within the governance hierarchy. ANAS has established ANAS-HTP in accordance with the presidential decree in November 2016. By establishing ANAS-HTP in 2016[1], the vision was to ensure strong science and industry cooperation and establish better national innovation ecosystem and knowledge economy through the commercialization of science. These will

[1] ANAS (2016), Official documents, Order of the President of the Azerbaijan Republic on the establishment of the Park of High Technologies of the Azerbaijan National Academy of Sciences of (ANAS), http://science.gov.az/news/open/4508

all be achieved with the proper functioning of incubation and acceleration programs and Technology Transfer Office along with an equipped Laboratory and the plant area.

In this context, ANAS requested the Korean government to provide policy consultation to ANAS-HTP for its institutional capacity development in the field of technology transfer and commercialization in 2018. Thus, STEPI as an STI policy think tank in Korea assumed the role of implementing agency for this project to provide consultation and training to ANAS-HTP and other innovation actors of Azerbaijan for the period 2019 and 2020.

The project has the main objective of building the institutional capacity of ANAS-HTP and in terms of technology transfer and commercialization. In addition, this project aims to strengthen STI policy capacity through training for ANAS-HTP and other main innovation stakeholders including researchers, academicians, and industrialists.

For this project, more than 20 Korean experts and practitioners have been involved as resource persons to deliver Korean knowledge and experiences effectively in the areas of TT/TC policy, entrepreneurship, management of STI parks and TTOs and technology financing, etc.

The main activities of this project consist of fact-finding mission, brainstorming workshops among stakeholders, training, and dissemination workshop. To do so, the STEPI team visited Baku twice and invited 6 key personnel of ANAS-HTP and other experts through a selection process to Korea for training. After the training, these trainees were assigned to share the lessons learned during the training and make policy suggestions at the dissemination workshops held in October in Baku. Two training participants have shown their extraordinary learning and application capabilities by developing a program for tech entrepreneurship and created a global ICT startup.

CONTENTS

List of Tables

List of Figures

CHAPTER 1

Project Overview

Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan

Chapter 1. Project Overview

1. Introduction

Azerbaijan, officially called the Republic of Azerbaijan is located in the South Caucasus region squeezed in between Russia and Iran. Declaring independence from the Soviet Union in 1991, Azerbaijan became a sovereign state with vast oil and gas resources on the one hand and serious challenges in transforming their political and economic heritage into new governance order on the other.

Azerbaijan is classified as an upper middle-income country and characterized by its vast amounts of hydro-carbonate and other natural resources. The economy of Azerbaijan is mostly dependent on revenue from these resources. It is strategically located in South Caucasus and on the shores of the Caspian Sea.

The economy now strongly depends on one resource — oil — and the oil sector is a major contributor to Azerbaijan's GDP. Thus, after the world economic crisis in 2008 and continuing financial crisis in 2011, the economy of Azerbaijan faced serious challenges, namely the double devaluation of the Azerbaijan manat and a sharp reduction in oil prices. Reliance on the stability of the global oil market has given rise to inevitable difficulties in developing Azerbaijan's economy, since the turnaround of oil money has drastically decreased during the last five years.

The oil and gas industries of Azerbaijan are core industries that account for 45~50% of GDP and over 90% of export. In addition, Azerbaijan is strategically located on the coast of Caspian Sea, and the country's strategic significance is high in the international energy market as the place for the production and stopover of crude oil and natural gas.

Since the financial crisis in 2008 and 2011, Azerbaijan has been trying to break away from its excessive dependence on the oil industry to enable major social･economic development, and the government is promoting policies based on shifting to a knowledge-based economy, overcoming the financial crisis due to the low oil price, diversifying the industrial structure, developing fields other than oil, etc.

Especially, the Azerbaijan government is continuing efforts in strengthening scientific and technological innovation to reinforce national competitiveness, enable development through school-work links in the technical transfer and commercialization sectors, and implement policies to achieve industrial diversification.

Since oil and gas resources are not permanent sources of income for sustainable economic progress, the government of Azerbaijan has focused on the diversification of national economic sectors. Nowadays, the country is using its current oil and gas resource wealth on developing its non-oil sectors, namely the newly emerging transportation, agriculture, tourism, and ICT sectors.

The government aims to improve the situation of the national economy in 11 sectors within the framework of the Strategic Roadmap as well as other state development programs. During the last 10 years, Azerbaijan reached the status of global leader in terms of economic growth ratio. Thanks to a successful oil strategy, the infrastructure has been modernized, non-oil sectors developed, social welfare improved, and strategic currency reserves have been accumulated.

Among several state development programs and strategies, “strategic roadmap on the national economy and main sectors of the economy” is currently the major development program of Azerbaijan, with focus on the development of STI, education, and innovation in different fields of the national economy.

In addition, there are several programs and strategies designed to ensure the competitiveness of the national economy and non-oil products, expansion of social welfare, sustainable economic development of the country, and STI capacity development.

Major state programs related to STI issues include: "A decree on a Development Concept. Azerbaijan – 2020: Outlook for the future"; "National Strategy for Science Development in the Republic of Azerbaijan in 2009-2015"; "National Strategy for Information Society Development in Azerbaijan in 2014-2020"; "State Program for the Development of Industry for 2015-2020"; and "Strategic Roadmap on the Production of Consumer Goods of Small and Medium-sized Enterprises in the Republic of Azerbaijan," etc. The programs and strategies mainly focus on STI capacity development as a driving force for economic growth to improve the non-oil sectors of the economy. Its measures are STI policy development and management/operation of STI parks, technology transfer and commercialization, etc.

To this end, the Azerbaijan National Academy of Sciences (ANAS) established the High Technologies Park (HTP)[2] in 2016 for conducting scientific research, developing the high-technology industry, and improving the STI system. Reporting directly to the president's administration, ANAS is the leading organization in Azerbaijan to develop, propose, and implement science-related programs, plans, and strategies within the governance hierarchy.

ANAS-HTP proposed the training program on STI policy capacity development in 2017 as the STEPI-IICC K-Innovation ODA program for 2018. This program was approved by the Korean government.

The K-Innovation ODA Program performs in-depth analysis on the actual demand of the partner nation to use this analysis result for sharing the methodology of implementing the policy in detail and to focus on the development of practical capabilities.

This project is a successful case of developing the result of the local workshop on the 2017~2018 preliminary demand survey into a two-year policy advisory project.

As described earlier, Azerbaijan has been pursuing economic development recently based on science and technology innovation for the goal of breaking free from the existing resource (oil)-dependent economy. Especially, it is focused on supporting the development

[2] As an organization exclusively for science and technology in Azerbaijan, ANAS plans national science & technology development policies to execute the relevant activities, and various research studies are being conducted to contribute to the development of national science & technology.

of the program for activating technical transfer and commercialization.

In the preliminary demand survey, ANAS-HTP took interest in introducing the development experience and implementation method of Korea regarding technical transfer and commercialization program development and operation system improvement for constructing the innovation system and to reinforce the industry-university-institute collaboration system in Azerbaijan, proposing collaboration with STEPI. Accordingly, both governments approved the promotion of this project.

Especially, the purpose is to support capability reinforcement including technical transfer & commercialization strategy, process, program, operation, and management by the major Technology Transfer Center (TTC) in Azerbaijan.

Korea’s various policy experiences for promoting technical transfer and commercialization and the methodology related to technical commercialization and technology startup are passed down to improve understanding and strengthen the implementation capability of the technical transfer personnel and industry-university-institute collaboration experts in Azerbaijan.

Ahead of the 2018 training program, a workshop on training needs analysis titled “*2017 Training Workshop on ‘Science Technology Innovation Policy and Industry’ for Azerbaijan*” was held by STEPI in cooperation with ANAS on August 7~10, 2017 in Azerbaijan to clarify the needs of ANAS-HTP. The workshop was arranged to support STI capacity development for ANAS-HTP’s staff, researchers, government officials who are involved in STI policy planning, innovative technology development, technology parks, and R&D management, and relevant experts from the private industrial sector. After this workshop, ANAS and STEPI discussed topics on a) sharing of Korea’s economic development experience, b) analysis of the STI policy and system, and c) STI cooperation issue.

As a result, ANAS came up with an agenda covering the improvement of the operation and management system of ANAS-HTP and development of programs facilitating technology transfer and commercialization.

Based on the pre-demand survey and capacity analysis of the trainees, STEPI has organized a capacity development training workshop in 2018 on the following topics: a) Strategic Management of Innovation Cluster; b) Technology Park Operation and Management: Korean TP; c) SME Support Programs of Gyeonggi Technopark; and d) Tech Entrepreneurship and Commercialization.

As follow-up of the training in 2017 and 2018, ANAS requested the Korean government to further support for the promotion of technology transfer and commercialization in Azerbaijan through policy consultation. This consultation is composed of a kick-off workshop in Baku, an invitational training, and a dissemination workshop.

As the first stage of the project, the Korean researcher team specified the demand for the 2019-2020 project at the kick-off meeting with the relevant stakeholders and conducted fact finding on the status of technology transfer ecosystem in Azerbaijan from February 16 to 23, 2019.

In order to promote the organizational and institutional development of ANAS-HTP, ANAS-HTP and STEPI analyzed the need for capacity development on HRD in the field of Technology Transfer and Commercialization. In particular, the Azerbaijan research team requested the training for sharing Korean experiences and practical skills on TT and TC.

From Azerbaijan, seven practitioners were invited to Korea for an intensive Capacity Development Workshop. The Azerbaijan experts' team consisted of the appropriate trainees in the fields of TT and TC in Azerbaijan including ANAS-HTP. To select the right targeted group and to enhance effective training, STEPI planned and considered the application process, and ANAS recommended the applicants.

The 2019 training program was designed to improve the relevant knowledge and work skills in developing the TT and the TC system at the institutional level. The training consisted of 6-day intensive curriculum with lectures, Q&A, group discussions, TT action planning, study visits and cultural visits, etc. from May 27 to June 1, 2019.

In the fourth quarter of this year, STEPI and ANAS-HTP will create opportunities for disseminating the outcomes of the 2019 policy consultation in Azerbaijan.

STEPI will draw up an implementation action plan through a dissemination workshop aimed at developing TT/TC programs and strengthening linkage with Industry-University – Institute Collaboration.

The dissemination of the outcomes of 2019 was held on October 16-17 by STEPI and ANAS-HTP.

The dissemination workshop consisted of lectures and Q&A, case studies of the best practices and discussions of Korean experts (Technology Entrepreneurship and Ideation-based Startup, Role of Technopark for Technology Commercialization and Chungnam TP Case Study, Korean Technopark (K-TP) Model as an Integrated Business Platform: GBTP Case), and Azerbaijan experts' reporting and presentation (cases of application and improvement of business capacity after the invitational training for STI capacity development in Korea)

ANAS-HTP is currently in the stage of establishing strategy development & planning, and the effectiveness of our policy advice is expected to be very high in organizing the implementation strategy.

Pending issues are the requirement of the legal system to support incentives for innovation-driven enterprises, ANAS-HTP infrastructure to support projects in the cutting-edge technology field, development of open and innovative ecosystem in partnership with large companies to implement the innovation projects, globalization of TT/TC, reform of the training system for supporting the technical project for youths, development of technical transfer office･manager･broker･entrepreneur･researcher, expansion of provision of consulting service to innovative SMEs, and provision of science & technology training exchange program to enhance awareness and participation of the parties related to STI.

2. Objectives

As part of the K-Innovation ODA Program, 「Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan for 2019-2020」 is jointly conducted by STEPI and ANAS in partnership with the Azerbaijan Innovation Ecosystem Actors. STEPI-ANAS jointly conducts a diagnosis of technology transfer ecosystem and shares the Korean experience relevant to Azerbaijan in order to support the promotion of technology transfer and commercialization in Azerbaijan. The program duration is from January 1, 2019 to December 31, 2020, or approximately 2 years.

The purposes of this consultancy (2019-2020) are as follows:

- Improving the operation and management system of technology transfer and commercialization in Azerbaijan (2019)
- Supporting technology transfer and commercialization program development in Azerbaijan (2020)
- Setting the technology transfer and commercialization capacity building

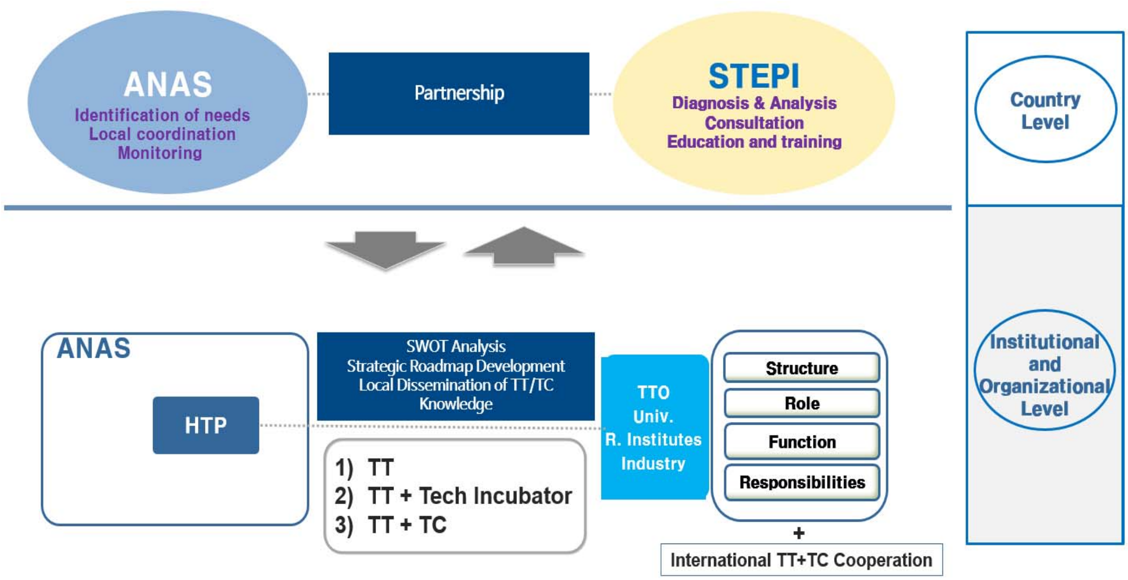
More specifically, this project is aimed at effectively making an implementation action plan through the kickoff workshop, STI capacity development workshop (invitational training program), and dissemination workshop aimed at developing TT/TC programs and strengthening linkage with Industry-University–Institute Collaboration/Discussion with policymakers, practitioners, researchers of TT- and TC-related ministries, institutions, research centers.

3. Project Framework

In general, the aim of this project is to strengthen the capacity of the ANAS-HTP managers with respect to the development of programs to link research results to innovation and support its resident companies and startups as a center for technology transfer and commercialization. For the effective implementation and dissemination of the knowledge obtained through training and workshops during the project period, STEPI formed a partnership with the Presidium of Azerbaijan National Academy of Sciences (ANAS) to faciliate subprograms and activities in both countries.

As part of efforts to promote the culture of technology commericialization and entrepreneurship in Azerbaijan, STEPI and ANAS tried to create a platform for Azerbaijan’s diverse stakeholders such as universities, research institutes, firms, and TTOs for sharing TT- and TC-related knowledge and Korean experiences. Furthermore, STEPI assisted ANAS-HTP in drafting their management strategies to play a pivotal role as a national technology incubator in the nation.

[Figure 1-1] Conceptual Framework of the Project



The framework of the research is as follows:

1) Joint survey research on the status of technical transfer and commercialization in Azerbaijan

Joint research is performed by ANAS and STEPI researchers on the status of technical transfer and commercialization in Azerbaijan and on the innovation ecosystem

Prioritize deriving the field for collaboration to improve the technical transfer and commercialization system and to reinforce the capability of the technical transfer organization

2) Consultation on the development of the operation strategy by ANAS-HTP

To establish the operating strategy of ANAS-HTP as a major science and technology complex in Azerbaijan, the management and operating experience of a major technopark in Korea is shared to derive the program appropriate for the local environment.

Investigation and diagnosis of the status of the local science and technology complex are performed by domestic experts and STEPI researchers in the technical transfer and commercialization field.

Invitational training is held in Korea for the main decision makers and technical transfer personnel of ANAS-HTP and local experts in science and technology innovation and technology startup to improve understanding and establish consensus on the main functions and role of the science and technology complex and on the business support program.

A local workshop is held for sharing and mutually learning with the Azerbaijan government and parties related to the industry-university-institute the knowledge and experiences related to technical transfer and commercialization as obtained through the invitational training in Korea.

3) Establish a Korea-Azerbaijan institution and an expert network on technical transfer and commercialization

Establish a collaboration network between the organizations related to technical transfer and commercialization in Azerbaijan and Korea Technopark (Chungnam TP, Gyeongbuk TP, etc.), universities specializing in technical transfer, commercialization, and startup (Hanbat National University, Hoseo University, etc.), and research institutes (KIST, ETRI, etc.).

Improve understanding of the local condition and seek opportunities for future collaboration through the participation of domestic specialist institutions and experts in domestic invitational training and local workshop.

The contents of the project in the first and second years are as follows:

1) 1st Year (Year 2019): Based on the research and analysis on the status of the technical transfer and commercialization ecosystem in Azerbaijan, the goal is to develop the business support program for the Korea Technopark and provide the methodology for operating experience, technology commercialization, and startup.

- (Kick-off meeting and field survey, Feb. 16~23, 2019): Confirmation on the scope of the project, field survey to identify the status of the innovation ecosystem in Azerbaijan

- (Invitational workshop in Korea to reinforce capabilities, May 25 ~ Jun. 2, 2019): Derivation of the methodology for technical transfer/commercialization and technology startup, transfer experiences by the Korea Technopark and TLO (Technology Licensing Organization), and exchange study and knowledge through participation by local experts

- (Local workshop to disseminate the results and field survey, Oct. 12~19, 2019): Share and disseminate knowledge for the related parties in Azerbaijan, derive improvement methods on the level of application to the actual work, and concretization of the implementation plan for the following year (Year 2020).

2) 2nd Year (Year 2020): The goal is to establish and realize the roadmap and implementation strategy for technical transfer and commercialization in Azerbaijan.

- (1st Business Trip) Confirm the framework and methodology for analyzing the roadmap on technical transfer/commercialization in Azerbaijan, perform additional field survey.
- (Empowerment Workshop) Joint preparation of the draft roadmap on technical transfer/commercialization in Azerbaijan, learning with the participation of experts from the recipient country
- (2nd Business Trip) Share the draft roadmap on technical transfer/commercialization in Azerbaijan and disseminate the outcome.

4. Project Team

4.1 Research Team

The research team is composed of 6 researchers. Ms. Eun Joo Kim from STEPI is the project manager, and Ms. So Hyun Kwon from STEPI is the project coordinator. The main research topic of Prof. Jong-in Choi is technology commercialization, entrepreneurship education and startup, and incubation management. Prof. Sang Hyuk Suh is an expert in the field of technology marketing based on the ecosystem of technology transfer and commercialization. Likewise, Mr. Seong-Kun Kang and Dr. Seong-Keun Park are the experts in transferring the knowledge and sharing the expertise and experience for management technopark and for technology businesses with advanced business incubation.

[Table 1-1] STEPI Research Team

Name	Organization	Position & Area of Expertise	Contact Information
Eun Joo Kim	STEPI	Head of External Cooperation and Public Relations/ PM (Project Manager)	ejkim@stepi.re.kr
So Hyun Kwon	STEPI	Researcher at the Division of Global Innovation Strategy/ PC (Project Coordinator)	shkwon@stepi.re.kr
Sang Hyuk Suh	Hoseo University	Professor/ Technology Transfer Promotion Policy and System in Korea, Technology Transfer and Commercialization of Public R&D Organization in Korea, Technology Marketing	suh8777@hoseo.edu

Name	Organization	Position & Area of Expertise	Contact Information
Jong-in Choi	Hanbat National University	Professor/ Technology Entrepreneurship, Startup and Incubating	jongchoi@hanbat.ac.kr
Seong-Kun Kang	Chungnam Technopark	General Manager/ Technopark Status, Role of Technopark for Technology Commercialization and Chungnam TP Case Study	ksg@ctp.or.kr
Seong-Keun Park	Gyeong-Buk Technopark	Team Manager/ Korean Technopark (K-TP) Model as an Integrated Business Platform: GBTP Case	park.songkun@gmail.com

4.2. Local Research Team

ANAS-HTP and TT/TC stakeholders are responsible for this project in Azerbaijan. They are in charge of promoting the organizational and institutional development of ANAS-HTP and capacity development on HRD in the field of TT/TC and supporting the operation & management of the business incubation center, technology transfer office, STI parks, etc.

[Table 1-2] Azerbaijan Team

Name	Organization	Position & Area of Expertise	Contact Information
Samir Mammadov	ANAS-HTP	Director	samirmammadov70@gmail.com
Bunyamin Seyidov	ANAS	Chief, Department of Relations with Eastern Countries, ANAS Presidium/ Contact Point/ Global STI Partnership Programs Development	bunyamin.seyidov@science.az

Name	Organization	Position & Area of Expertise	Contact Information
Mammad Baghir-zada	ANAS-HTP	Deputy Director on Strategic Development/ Strategic Planning on HTP	m.baghirzada@ameaytp.az
Zaur Akhmedzadeh	ANAS-HTP	Head, Technology Transfer and Business Incubator Department/ Law in the field of TT	z.akhmedzadeh@ameaytp.az
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Samir Veliev	International Bank of Azerbaijan (OJSC)	Economist at the Economic Research Department under the Board of Directors/ Economic Analysis on Innovation and Industry	Samirveliev@yahoo.com
Aysel Abdullayeva	Azerbaijan State University of Economics (UNEC)	Lecturer in Innovation Risk Management at UNEC Business School/ Research on Innovation, Entrepreneurship, and Marketing	abdullaeva.aysel@yahoo.com
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Sevinj Iskandarova	Khazar University	Lecturer/ Korean Culture	Sevinc-iskenderova@hotmail.com

5. Main Activities

5.1 Activity 1: Kick-off and field study (February)

- Specify the demand for the 2019-2020 project
- Stakeholders meeting and fact finding on the status of technology transfer ecosystem in Azerbaijan

5.2 Activity 2: Reports on Azerbaijan's TT ecosystem and Korean experiences (March-April)

- Based on the first field study, the local consultant and Korean experts will prepare reports on the status of Azerbaijan's TT and Korean experiences

5.3 Activity 3: Capacity Building Workshop in Korea (Invitational Training Program) (October)

- Workshop held in Seoul, Korea with the specific topics confirmed at the first field study
- The period and beneficiary of the training will be discussed at the Kick-off Meeting
- The participants will develop Action Plans for the Promotion of TT in Azerbaijan and present them at the dissemination workshop

5.4 Activity 4: Dissemination Workshop in Azerbaijan (October)

- Presentations on Action Plans for the Promotion of TT and Tech Incubation Program by Azerbaijan's counterpart team
- Presentations on Korean experiences and policy lessons for Azerbaijan

5.5 Activity 7: Final Reporting

- Publish the final report of the project

6. Project Schedule

The entire project schedule is as follows:

[Table 1-3] Project Schedule

Activities	Jan. 2019	Feb. 2019	Mar. 2019	Apr. 2019	May 2019	Jun. 2019	Jul. 2019	Aug. 2019	Sep. 2019	Oct. 2019	Nov. 2019	Dec. 2019
Review of the Proposal and Discussion of Project Plans (Video Conference)												
Kick-off Meeting and Field Study in Azerbaijan												
Preparation for Capacity Building Workshop (Selection of Trainees)												
Capacity Building Workshop in Korea												
External Review on Interim Report												

Activities	Jan. 2019	Feb. 2019	Mar. 2019	Apr. 2019	May 2019	Jun. 2019	Jul. 2019	Aug. 2019	Sep. 2019	Oct. 2019	Nov. 2019	Dec. 2019
Interim Reporting Workshop												
Draft Final Report and Preparation for the Dissemination Workshop												
Dissemination Workshop and Site Visit in Azerbaijan												
Final Reporting Workshop												
Submission of Final Report												
Plan for 2020 Project												

On February 16 (Sat.) ~ 23 (Sat.) (including departure and entry dates), STEPI researchers promoted the 2019 Project Kick-off Workshop and field survey* to materialize the project scope activities and discuss the work plan (draft) with ANAS.

* Industry-University-Institute collaboration organizations related to the innovation system in Azerbaijan (ANAS-HTP, Sumgait Technologies Park, Incubation & Acceleration Center INNOLAND, ADA University and Baku Engineering University, etc.) were inspected to identify the status of technical transfer and commercialization in Azerbaijan.

The aim of the K-Innovation ODA Program is to become a convergent program (Education & Training + Consulting: including investigation, analysis, and research) as well as kick-off local workshop, domestic invitational training, and local workshop for disseminating the outcome, to be included in the project scope activity and work plan (draft) in Azerbaijan.

ANAS-HTP is still in the initial development stage of organization and system, so the main interests lie in establishing the strategy for science and technology for the organization, strengthening the capabilities for managing and operating the science and technology innovation complex, and improving the technical transfer and commercialization program and relevant operation and management systems, etc.

In the local kick-off workshop, detailed demands are derived, and HTP will share the Korean experience in the development of practical capabilities for technical transfer and commercialization to develop into an organization exclusively for technical transfer.

Based on this, STEPI promoted the invitational training on reinforcing capability in technical transfer and commercialization in Korea for seven Azerbaijan innovation system personnel from May 25 (Sat.) to June 2 (Sun.) (including departure and entry dates).

The workshop for expanding awareness through empowerment training and for disseminating the outcome through mutual collaborative research will be held locally in Azerbaijan during October.

7. Project Outputs

STEPI provided the following deliverables:

- **Deliverable 1:** Project Proposal (January 2019)
- **Deliverable 2:** Presentation materials and report of the Kick-off Meeting (February 2019)
- **Deliverable 4:** Workshop book and Certificate of STI Capacity Building Workshop (May-June 2019)
- **Deliverable 5:** Workshop book and Certificate of the Dissemination Workshop (October 2019)
- **Deliverable 6:** Final report (October-November 2019)

CHAPTER 2

Status Analysis of STI Policy and Technology Transfer Ecosystem in Azerbaijan

Policy Consultation on the Promotion of
Technology Transfer and Commercialization in
Azerbaijan

Chapter 2. Status Analysis of STI Policy and Technology Transfer Ecosystem in Azerbaijan

1. General Status of Azerbaijan

1.1 Macroeconomic Indicators

Azerbaijan's GDP in the early 1990s was just around 60 billion USD (GDP, PPP (current international $), which even decreased sharply to 25.5 billion USD in 1995 due to conflicts and devastation of economic value chains. It took GDP a decade to reach 60 billion USD levels. Thanks to the oil industry boom, Azerbaijan started growing very rapidly in the late 2000s and recorded GDP of 161 billion USD in 2015. Note that Azerbaijan's annual GDP growth in 2006 stood at 34.5%, which was one of the highest growth indicators globally[3]. Because of the rapidly decreasing oil prices and challenging macroeconomic situation, the Azerbaijani currency, manat (exchange rate to US dollar – 1 manat=0.588 cent), has devalued by almost double since 2015[4], and the GDP growth rate also hit rock bottom with -3.1% per annum, according to the World Bank data[5].

In Azerbaijan, curbs on oil production were offset by stronger non-oil sector activity and fiscal stimulus measures. Azerbaijan is currently improving its macroeconomic situation, and it is forecast to grow by 3.8% and 3.2% in 2019 and 2020, respectively[6]. GDP per capita

[3] World Bank Data (2018), https://data.worldbank.org/country/azerbaijan

[4] Center for Economic and Social Development (2017), "Devaluation of Azerbaijani National Currency: Causes and Consequences," http://cesd.az/new/wp-content/uploads/2015/03/Azerbaijan_National_Currency_Devaluation2.pdf

[5] World Bank Data (2018), https://data.worldbank.org/country/azerbaijan

[6] World Bank (2018). Global Economic Prospects, June 2018: The Turning of the Tide? Washington, DC: World Bank. doi: 10.1596/978-1-4648-1257-6

was 15,847 USD in 2017[7].

Currently, the total tax rate in Azerbaijan is 39.8% of commercial profits, and a new tax minister has recently announced further cuts in the tax burden for entrepreneurs and businesses to boost economic activities[8].

Inflation rate fluctuates from year to year but is usually above 10% annually. The recent data shows a 15.9% inflation rate[9].

Unemployment rate is around 5%. Out of the 5 million total labor force, only less than 10% of unemployed people have advanced education level. The industry employs 14% of total employment, with self-employment accounting for 69% of the total employment[10].

Around 20% of the GDP comes from total natural resources: 17% from oil and 3% from natural gas[11]. From its expenditures, the government spends more than 20% on military and 7-8% on education. Science budget is in an unfortunate situation in this regard because it gets only 0.2% of GDP, which is mainly spent on salaries and infrastructure[12].

Of the total GDP, trade volume is around 90%, and FDI inflow and outflow were found to be 7% and 6%, respectively. 68% of merchandise imports are composed of manufactured imports. Note, however, that only 3.3% of merchandise exports come from manufactured exports due to the dominant dependence on oil and gas exports[13]. The biggest attention drawer is high-technology exports. The already weak contribution of manufactured exports to the total is very low, but high-technology exports account for only 2% of that of weak manufactured exports.

[7] World Bank Data (2018), https://data.worldbank.org/country/azerbaijan

[8] Ibid.

[9] Ibid.

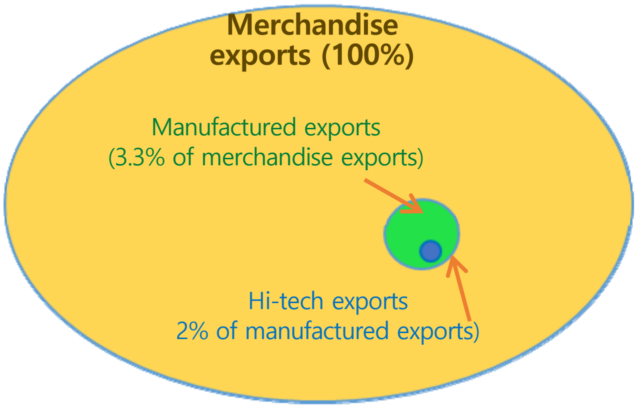
[10] World Bank Data (2018), https://data.worldbank.org/country/azerbaijan

[11] Ibid.

[12] Ibid.

[13] Ibid.

[Figure 2-1] Share of High-Technology Exports in Merchandise Export Share



Azerbaijan is ranked 57th in the ease of doing business index[14]. The World Economic Forum report on Global Competitiveness index[15] shows that Azerbaijan is ranked 35th globally. In the Global Innovation Index, however, Azerbaijan only came in 82nd, barely improving its situation in recent years[16].

Compared to natural resource extraction, the economy relied considerably on the small share of other value-added economic factors. Challenges in the non-oil sectors, including monopolistic competition and closely linked businesses, as well as weaker institutional settings had hindered the nurturing of a stimulating environment for entrepreneurship. Despite showing rapid increase in "Doing Business" reports, the Azerbaijani economy still lacks strong SMEs, let alone true entrepreneurs.

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14 World Bank, Doing Business 2018, http://www.doingbusiness.org/~/media/WBG/DoingBusiness/Documents/Annual-Reports/English/DB2018-Full-Report.pdf

15 World Economic Forum (2017), Global Competitiveness Report, http://reports.weforum.org/global-competitiveness-index-2017-2018/

16 Global Innovation Index (2018), https://www.globalinnovationindex.org/gii-2018-report

1.2 Demography and Human Resources

With 9.8 million population in 2017, Azerbaijan can be considered a country with a young population because the population aged above 65 is only about 6% of the total population. The population aged 15-64 accounts for 70.7% of the total. Thanks to the high level of school attendance and 10-year compulsory education, the literacy rate of the adult total (% of people aged 15 and above) is 99.7%. The total labor force is 5 million.

Tertiary school enrollment is 27% of the total. The student-teacher ratio in tertiary education was 1:10 in 2016. 15% of the total population above 25 has at least a bachelor’s degree or equivalent educational attainment.

The government spends 12-13% of the total expenditures on tertiary education and 55-56% on secondary education, spending 17% of the GDP per capita per student[17].

An internal survey of the Azerbaijan National Academy of Sciences revealed that, in contrast to the younger population, more than 50% of staff members are aged above 60 years. This indicates the rapidly aging academe due to generational breaks and delays in the country[18].

1.3 Governance

Azerbaijan is a presidential democracy with the strong power consolidated in the President’s office. This power is further strengthened after the Referendum held in 2016. The President is the head of state and government with the power to announce the dissolution of the Parliament and Cabinet anytime[19].

The Parliament of Azerbaijan, “Milli Majlis,” has a single chamber with 125 seats, and the ruling party is the overwhelming majority with proxy parties. Among the 15 committees of

[17] World Bank Data (2018), https://data.worldbank.org/country/azerbaijan

[18] Survey for internal use only (2016)

[19] President’s official webpage, “Presidential Power,” https://en.president.az/president/power

the Parliament, 2 committees, namely “Economic policy, industry, and entrepreneurship” and “Science and Education,” correspond to STI and entrepreneurship issues[20].

The Cabinet of Ministers is led by the Prime Minister who has relatively less power. It is composed of 18 ministries, 8 state committees, 5 state services, and 2 state agencies[21].

The Ministry of Education manages all education layers. Note, however, that science and research issues are managed under the Azerbaijan National Academy of Sciences (ANAS), the largest scientific organization. The role and structure of ANAS will be further explained later.

The Ministry of Economy is responsible for the overall economic policy and economic development, overseeing industrial parks as well as the newly created SME Development Agency.

The Ministry of Transportation, Communication, and High Technologies is responsible for high technology issues, including hi-tech parks in the country.

The Science Development Fund under the President of the Republic of Azerbaijan provides grants to science and research projects. Its annual budget in 2018 is 4 million manat (2.3 million USD). This number had been halved since the economic recession that started in 2016[22].

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[20] Milli Majlis official webpage (Parliament), http://www.meclis.gov.az/?/az/content/181

[21] Nazirlər Kabinetinin Tərkibi (Structure of the Cabinet of Ministers), http://www.cabmin.gov.az/page/post/343/

[22] Science Development Foundation, “Financing of the Foundation,” http://www.sdf.gov.az/en/generic/menu/Detail/102/menu//

1.4 National Development Plan and Strategy

The heavily oil-dependent Azerbaijan encounters several challenges in economic diversification and in building an innovative, sustainable economy. The government aims to improve the situation of the national economy in eleven sectors within the framework of the Strategic Roadmap as well as other state development programs.

Azerbaijan is a resource-rich country at the crossroads of Europe and Asia. Its huge accumulation of energy resources gives additional economic power to develop the country after the dissolution of the Soviet Union. It is also why the country has a shot in pulling off one of the most dramatic economic turnarounds ever attempted in 2008. That same year, Azerbaijan was cited as one of the top 10 reformers by the World Bank's Doing Business Report[23].

Azerbaijan was among the first oil-producing countries in the world, generating half of the world's oil supply at the beginning of the XX century. The State Oil Company of the Azerbaijan Republic (SOCAR) is involved in exploring oil and gas fields, producing, processing, and transporting oil, gas, and gas condensate, marketing petroleum and petrochemical products in domestic and international markets, and supplying natural gas to the industry and the public in Azerbaijan. This state organization has 3 production divisions, 1 oil refinery, and 1 gas processing plant as well as a deep-water platform fabrication yard and 1 institution with 23 subdivisions operating as corporate entities under SOCAR. In addition to this, SOCAR conducts various petroleum activities abroad particularly in in Georgia, Turkey, Romania, Switzerland, Germany, Ukraine, and Nigeria[24].

According to the State Statistical Committee of the Azerbaijan Republic, production of energy products in 2016 involved crude oil and natural gas at 41 050,4 thousand tons and 18 717,6 million cubic meters, respectively. In addition to this, 85% of the total production of crude oil was exported in 2016, and that of natural gas is 43%. Furthermore, only 10,6

[23] The World Bank, “Most Improved in Doing Business 2009,” http://www.doingbusiness.org/reforms/top-reformers-2009/

[24] Official website of the State Oil Company of the Azerbaijan Republic (SOCAR), www.socar.az

thousand tons of the produced raw oil and 436,9 million cubic meters of natural gas were consumed for own usage[25].

Thanks to the oil sector's development, Azerbaijan experienced real GDP growth during the first decade after winning its independence in 1991. In 1999, the Azerbaijani government created the State Oil Fund of Azerbaijan (SOFAZ) as a fund wherein Azerbaijan accumulates and manages all oil and gas revenues.

The economy now strongly depends on one resource -- oil -- and the oil sector is a major contributor to Azerbaijan's GDP. Thus, after the world economic crisis in 2008 and continuing financial crisis in 2011, the economy of Azerbaijan had been facing serious challenges such as the double devaluation of the Azerbaijan manat and sharp reduction of oil prices. The country's reliance on the global oil market stability has given rise to inevitable difficulties in the development of Azerbaijan's economy, since the turnaround of oil money has decreased sharply during the last five years.

Since oil and gas resources are not permanent sources of income for sustainable economic progress, the Azerbaijan government focused on the diversification of national economic sectors. Nowadays, the country is using its current oil and gas resource wealth on developing its non-oil sectors such as newly emerging transportation, agriculture, tourism, and ICT sectors.

During the last 10 years, Azerbaijan has been the world's number one in economic growth ratio. Thanks to a successful oil strategy, infrastructure has been modernized, non-oil sectors developed, social welfare improved, and strategic currency reserves have been accumulated.

Despite the unfavorable economic situation after internal financial challenges, the business and economic climate in Azerbaijan is slowly improving. Actual changes appeared with the 2016 approval of the "Strategic roadmap on the national economy and main sectors of the economy"[26] by the President of Azerbaijan for the sustainable and

[25] State Statistical Committee of the Azerbaijan Republic, https://www.stat.gov.az/source/balance_fuel/?lang=en

[26] President.az (December 2016), "Decree of the President of the Republic of Azerbaijan on the approval of the Strategic roadmap on the national economy and its main sectors," http://www.president.az/articles/21953

competitive development of the non-oil sector in Azerbaijan. The roadmap indicates the pivotal direction of economic policy progress of the country.

The tactical Roadmap on the National Economy and Key Sectors of the Economy of Azerbaijan is a complex of programs designed to ensure the competitiveness of the national economy and non-oil products, expansion of social welfare, and sustainable economic development of the country.

The main aim of the Strategic Roadmap for the short-term period till 2020 is to achieve economic diversification by increasing the benefits of the heavy industry and mechanical engineering, agriculture, and tourism as well as ICT technologies and innovations.

This strategic document identifies the way of expansion of the 11 economic sectors of the country:

- Development of the oil-gas industry (including chemical products);
- Production and processing of agricultural products;
- Production of consumer goods at the small and medium-sized entrepreneurship level;
- Development of heavy industry and machinery;
- Development of specialized tourism industry;
- Development of logistics and trade;
- Development of affordable housing;
- Development of vocational education and training;
- Development of financial services;
- Development of telecommunications and information technologies;
- Development of utility services (electricity and thermal energy, water, and gas). [27]

[27] Azertag, "Azerbaijan endorses strategic roadmap for the development of national economy and its main sectors," https://azertag.az/en/xeber/Azerbaijan_endorses_strategic_road_maps_for_development_of_national_economy_and_main_economic_sectors-1016958

Implementation of the abovementioned roadmap is planned in 3-time slot periods: 2020 — short term, till 2025 — medium term, and long-term period after 2025. Likewise, the execution of this strategy in steps involves three methods to achieve the set of goals. They include optimization of existing assets, creation of a competitive sector, and provision of financial support and international collaboration.

Moreover, the Center for Analysis of Economic Reforms and Communication was tasked with monitoring, evaluation, and communication between the stakeholders involved during implementation of strategically aspiring reforms.

Strategic Roadmaps covers global trends for each focused area, 360-degree diagnostics of the economy and SWOT analysis of the current situation, actions to be undertaken, required investments, and key indicators as well. In particular, public investments will act as a catalyzer, with the private sector serving as a locomotive for economic development.

In accordance with the abovementioned issues, the Strategic Roadmap focuses on five major strategic targets from the national economic and sustainable perspectives. First, Azerbaijan will ensure fiscal sustainability and formulate monetary policy based on the floating exchange regime. This will support macroeconomic stability and ensure balance between real and financial sector as well as fiscal and monetary system.

Second, it is an improvement of activities of legal and state organizations to ensure the dynamic development of the economy. Implementation of privatization will strengthen private sector participation by fostering more efficient practices from private businesses, which will eventually lead to better service and products of public services, lower prices, and less corruption.

The third goal focuses on human capital development, which will eventually end up with the increase of employment rate and provision of balanced regional development.

Fourth are the improvement of business environment through tactical incentives in law regulations, protection of intellectual property rights, and grafting of investment capabilities and ICT technologies. This will promote the transparent and sustainable development of the ecosystem with poverty reduction and high share of the private sector in the GDP.

Lastly, focusing on ICT technologies and innovations will envisage almost all sectors of the national economy and the formation of a new knowledge base model of development of the country with competitive industrial and scientific sectors. "At the same time, the share of Azerbaijani industrial production will be increased in the regional market, and achieving the long-term objectives (by 2025) will considerably boost the competitiveness of the country at the regional level. The measures designed for the period until 2025 will turn the country into a center of machine engineering. The post-2025 objectives envision making the country an integral part of the global value chains."[28]

To sum up, the Strategic Roadmap was intended to improve a non-oil structure of the economy and ease the dependence of macroeconomic stability on the world oil market.

1.5 STI and Education System

With 9.8 million population and considerably high level of literacy rate, Azerbaijan has many opportunities to strengthen its STI and education performance. The historical path of development and formulation of policies show certain challenges that can be solved through effective reforms to the modern world.

Due to the long-existing Soviet heritage, science and research are mostly separate from education and teaching in Azerbaijan. Universities usually focus on teaching bachelors, master's, and few doctorate programs. ANAS and other research entities almost completely concentrate on conducting basic and fundamental research along with doctoral studies. All levels of education -- preschooling, schooling, TVET, higher education, and extra-schooling – are under the Ministry of Education. According to the 2017 data:

1750 preschooling entities (0.05% private) provide education for 118685 children.

[28] Azernews, "Strategic Roadmap to catalyze economic changes." https://www.azernews.az/business/108646.html

4452 general schools (4427 public, 25 private) provide education to 1461748 pupils and employs 156852 teaching staff.

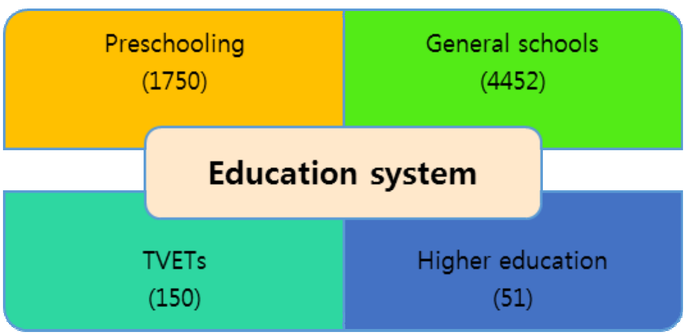
Azerbaijan has 150 TVET colleges and schools that offer hundreds of specialties. Within the State Strategy on the Development of Education in the Republic of Azerbaijan, there was a special strategic roadmap for the TVET sector supported by the European partners.

51 higher education entities (38 public, 13 private) offer bachelor's degree (to 36126 students) and master's degree (to 5098 students) to 163779 students. 11.7% of students study at private higher education organizations. The largest student population is in Baku State University. Azerbaijan has 169 students per population of 10000. Among these students, the majority prefers studying education and teaching programs, followed by economics and management and technical and technological programs. Only 2436 students study agricultural programs.

These 51 universities hire 14500 professors and teaching members, 5690 of whom have PhDs or higher.

In general, 1,872,888 pupils, students, and doctorates study in Azerbaijan[29].

[Figure 2-2] Education Institutions



Before the adoption of the Law on Education in 2009, there was no officially recognized research university in Azerbaijan. Nonetheless, there are certain intentions to transform

[29] State Statistics Committee of the Republic of Azerbaijan (2018), Education, Science, and Culture, https://www.stat.gov.az/source/education/?lang=en

some universities into a research concept. For example, Azerbaijan State University of Economics, [30](UNEC), and ANAS[31] currently work on establishing research universities.

3370 international students are getting their education from Azerbaijani universities as of 2017. Currently, 2197 Azerbaijani students study in various foreign countries with Azerbaijani government support and scholarships. The Azerbaijani government sponsors 8 students to study in the Republic of Korea as of 2017[32].

1.6 R&D and STI System

The lack of a full-fledged science policy is the reason for the continuous marginalization of research and development organizations, e.g., ANAS, branch institutes, and high education centers.

Out of 110, 77 scientific organizations and 33 higher education entities offer PhD degrees to 2182 students. These 110 belong to different ministries, committees, and organizations including ANAS, which owns 44 of them. According to the 2016 data, the largest share of PhD students is hosted by the higher education entities under the Ministry of Education and ANAS with 1165 and 532, respectively[33].

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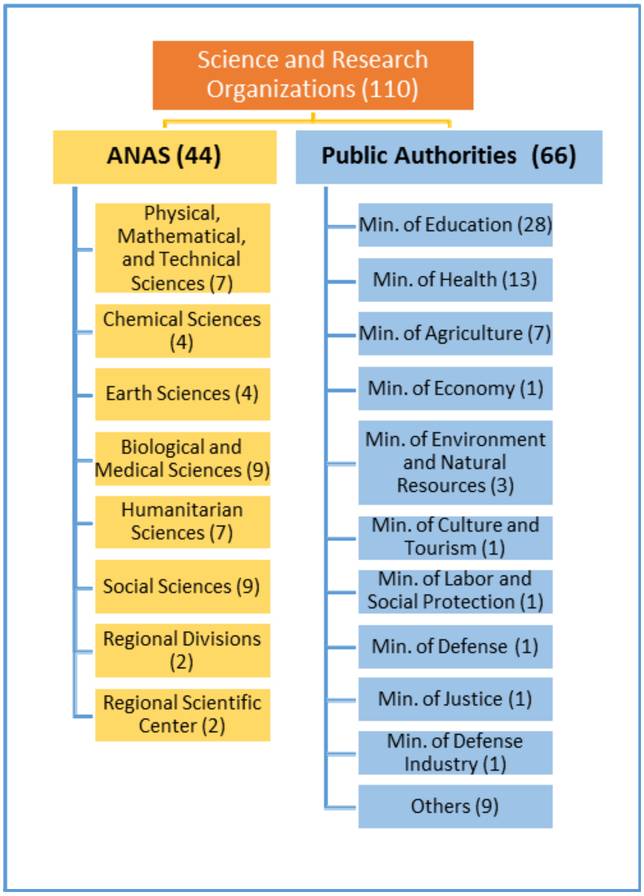
30 Etibar Aliyev (2015), "An Example of the Successful Cooperation of the University of Economics," http://unec.edu.az/en/ugurlu-emekdasligin-iqtisad-universiteti-numunesi/

31 "Academician Akif Alizadeh speaks at a joint scientific session of the Collegium of AR Ministry of Education" (2016), http://www.nauka.gov.az/news/open/4754

32 State Statistics Committee of the Republic of Azerbaijan (2018), Education, Science, and Culture, https://www.stat.gov.az/source/education/?lang=en

33 Ibid.

[Figure 2-3] Science and Research Organizations



Compared to 9.3 million manat in 2000, the government spent 110.2 million manat in 2016. R&D expenditures stood at 127.2 million manat, again accounting for 0.2% of GDP. The government sector spent 106.3 million manat (83% of the total) on R&D compared to 11.6 million manat (9.1%) by the higher education entities and 6.8 million manat (5.3%) by the private sector. Most of the R&D expenditures are directed to fundamental research (67 million manat), development (32 million manat), and applied research (21 million manat)[34].

The number of entities conducting R&D was 137 in 2000, increasing to as many as 148 in 2009. Afterward, this number decreased to 135 in 2016. Out of 135 R&D entities, 88

[34] Ibid.

belong to the public sector. 9 of them are owned by the private sector, and 38 of them are higher education entities[35].

Note, however, that the number of people working at these R&D entities increased from 15809 to 22527 in the respective period. Out of 22527 people, 15548 are researchers, 2019 are technicians, 2574 are support teams, and 2386 are other workers. 43% are male, and 57% are female researchers. Moreover, around 10000 scientific-teaching people work for higher education entities that implement R&D projects, but they are not employed by science-research units[36].

The following shares of 15548 researchers are observed among science fields: 31.6% natural sciences, 12.6% technical, 13.8% medical, 4.6% agricultural, 15.1% public, and 22.1% humanities. In terms of their age distribution, statistics show that only 13.8% are younger than 30. Ages 30 ~ 60 account for 60% of the total number of researchers. 8.7% are older than 70 years[37].

Patent applications by residents have rapidly declined from 254 in 2000 to 144 in 2016. Only 19 non-resident patent applications had been registered in 2016. On the other hand, non-residents' industrial design applications have dramatically increased since 2010. Compared to 44 such applications in 2010, Azerbaijan registered 618 industrial design applications in 2016[38].

Patent commercialization data is not fully available. Internal information reveals low indicators in recent years, however.

The participation of Azerbaijan in international, especially European, STI communities and calls are at one of the lowest levels. For example, figures for both project applications and winning projects are considerably low compared with the other Eastern Partnership

[35] State Statistics Committee of the Republic of Azerbaijan (2018), Education, Science, and Culture, https://www.stat.gov.az/source/education/?lang=en

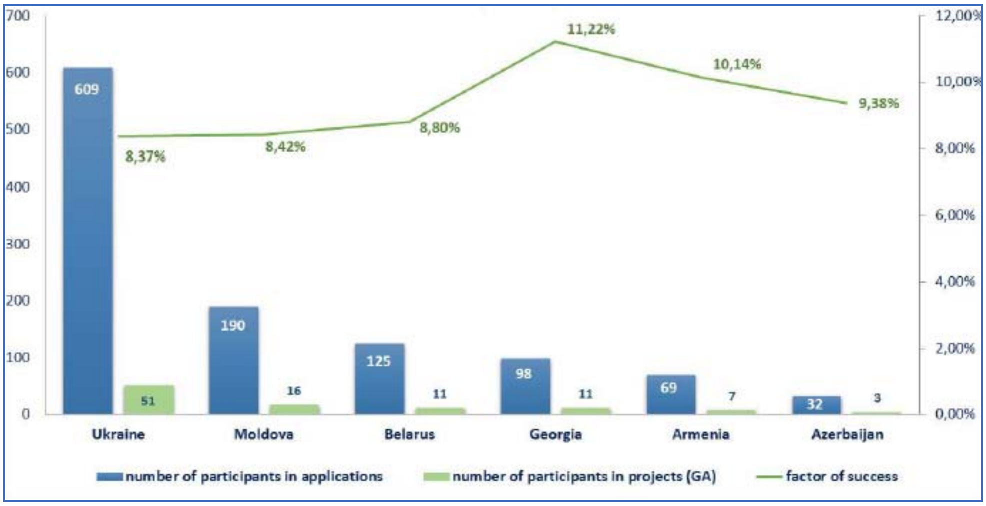
[36] State Statistics Committee of the Republic of Azerbaijan (2018), Education, Science, and Culture, https://www.stat.gov.az/source/education/?lang=en

[37] Ibid.

[38] Ibid (also [38] World Bank Data (2018), https://data.worldbank.org/country/azerbaijan)

framework program HORIZON 2020[39].

[Figure 2-4] Eastern Partnership Participation in the HORIZON 2020 Framework Program



As the largest scientific-research organization operating in almost all fields of science, ANAS had conducted fundamental and applied research, project and technological works, experimental research, and prototyping as well as construction projects and scientific-research services worth 55.5 million manat in all in 2016, 50.1 million of which came from its own funding (public)[40].

The number of internationally recognized papers and level of citations from Azerbaijan is low compared to peer countries. The number of articles in scientific and technical journals by Azerbaijan has sharply decreased compared to 2012. While 684 papers got an opportunity to be published in such journals in 2012, this number was lowest in 2014 with only 399 before increasing to 480 in 2016[41].

Overall, the Global Innovation Index 2018 ranks Azerbaijan's R&D indicators 79th globally.

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39 Kenan Aslanli (2017), "Ways of improving the innovation policy in Azerbaijan," Entrepreneurship Development Foundation (EDF), http://edf.az/ts_general/eng/layihe/IIED/policy-papers/kenan-aslanli_innovation.pdf

40 State Statistics Committee of the Republic of Azerbaijan (2018), Education, Science, and Culture, https://www.stat.gov.az/source/education/?lang=en

41 Ibid.

According to the Index, Azerbaijan scored considerably low in particularly important areas of STI: 110th in intellectual property receipts (% of total trade) and industrial designs by origin, 108th in knowledge creation, 97th in scientific-technical articles, 110th in knowledge impact, 107th in citable documents, and 118th in creative goods export (% of total trade)[42].

All these show that there is a long way ahead to enhance the STI potential and apply effective tools to deliver noticeable output.

[42] Global Innovation Index 2018, https://www.globalinnovationindex.org/gii-2018-report

2. STI Policy Status in Azerbaijan

2.1 STI Policy Governance and Strategy in Azerbaijan

According to Doing Business 2019, Azerbaijan, officially called the Republic of Azerbaijan, had been ranking "among the 10 economies with the most notable improvement in the world"[43]. These successful political and economic transforming reforms focus on the enhancement and diversification of non-oil sectors, including STI.

The heavily oil-dependent economy of Azerbaijan encounters several challenges in economic diversification and building of innovative, sustainable ecosystem. The government aims to improve the situation of the National Innovation System (NIS) through indirect and direct reforms and development programs. There is no single responsible central authority to develop and conduct the STI policy. Note, however, that the newly established Department of Innovative Development and E-government Issues of the President's Administration is a strategically leading authority for the development of STI and innovation ecosystem. Moreover, the Innovation Agency (est. in 2019) under the Ministry of Transportation, Communication, and High Technologies seems to be leading the way in the system.

During the Soviet era, Azerbaijan had achieved successful STI results in specialized areas like mathematics, chemistry, geology, and some other priority fields. The lack of single vision and policy strategy as a challenge could be one of the reasons Azerbaijan was outperformed by relatively similar countries, e.g., Belarus, Ukraine, Georgia, etc.

Based on the facts on the existing strategies, and given the lack of science policy vision, the country needs economic diversification, and it has to draw attention to innovation in the non-oil sectors. It will lead Azerbaijan to become a knowledge economy, in contrast to

[43] Azertag (October 2018), "Doing Business 2019: Azerbaijan implemented a record number of reforms globally," https://azertag.az/en/xeber/Doing_Business_2019_Azerbaijan_implemented_a_record_ number_of_reforms_globally-1210273

today's oil economy.

According to Global Innovation Index 2018, however, Azerbaijan "ranks 82nd"[44] out of 126 countries. Moreover, "Azerbaijan ranks very high in indicators such as "Cooperation between universities and industry" at 33rd place and "Cluster development" at 34th place"[45]; thus highlighting improving results compared to the previous years.

"The National Innovation System (NIS) ensures the efficiency of achievement, development, dissemination, and use of the knowledge in the country. Subdomains that have to be regulated in coordination by innovation policy include education, scientific research, trade, finance, and industry."[46] Therefore, a country aiming to be a leading global hub for creating value-added innovations has to reform its STI policy unavoidably to improve technologies, and then transfer it to priority industries. In this context, strengthening innovation and knowledge base with the help of research institutes (like ANAS), higher education entities, training centers, and private firms is extremely crucial.

In 2009, UNESCO consultants visited key institutions and met with the country's main stakeholders to review Azerbaijan's strategies and policies for science, technology, and innovation (STI). This resulted in the preparation of a report, "Formulation of STI Strategy and STI Institutional Capacity Building in Azerbaijan." As a roadmap for 2011-2015, its main components were the research role of ANAS, funding mechanisms and structures for STI, prioritization and collaboration with the industry and public sector, popularization of science, and international cooperation[47]. Nonetheless, the report did not deliver the expected outcomes due to weak implementation.

According to local experts, "To bring the GDP share of innovation (R&D) expenditures up to 1.5%, effective university-industry relationships and public-private partnership

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44 AzVision.az (July 2018), "Azerbaijan improves in Global Innovation Index 2018 rating," https://en.azvision.az/news/89797/azerbaijan-improves-in-global-innovation-index-2018-rating.html

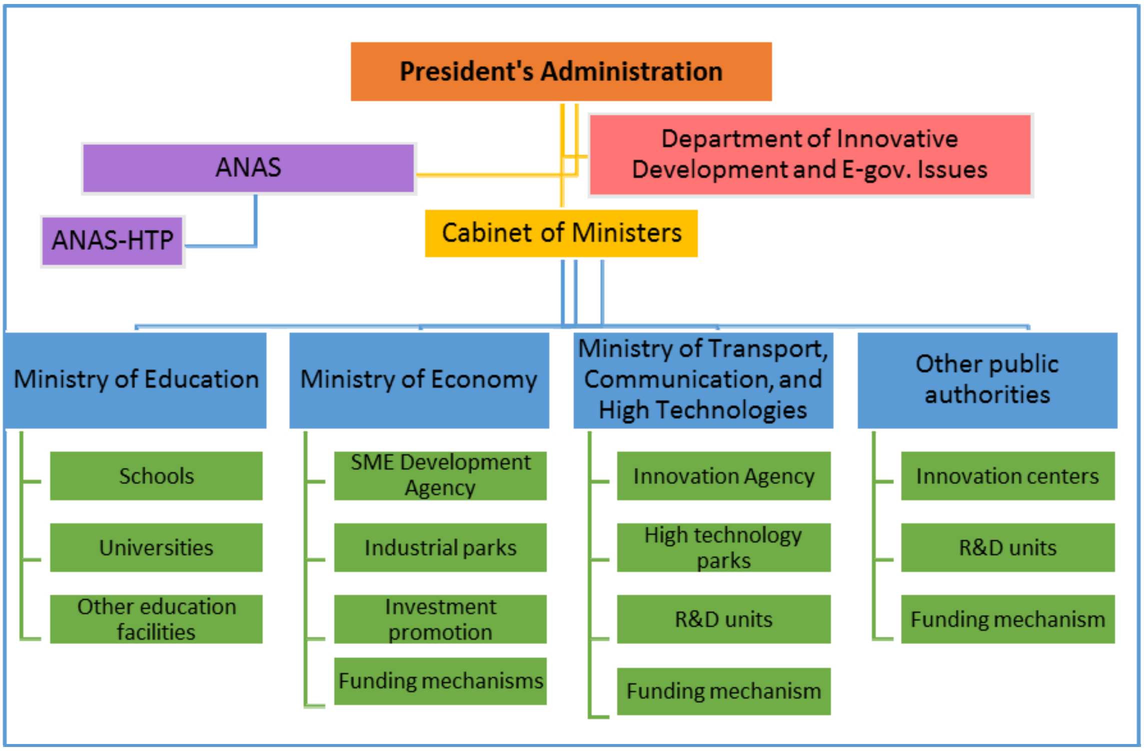
45 Ibid.

46 Kenan Aslanli (2017), "Ways of improving the innovation policy in Azerbaijan," Entrepreneurship Development Foundation (EDF), p.6, http://edf.az/ts_general/eng/layihe/IIED/policy-papers/kenan-aslanli_innovation.pdf

47 UNESCO (2015), Science Report, Country Studies – Azerbaijan, http://www.unesco.org/new/en/natural-sciences/science-technology/sti-policy/country-studies/azerbaijan/

promotions in innovation actions may be key goals of the government's innovation policy."[48]

[Figure 2-5] STI Governance Authorities



Source: Authors' contribution/updated in April 2019

As described above, different authorities implement different roles and responsibilities within their own framed environments. Most of the state development programs and strategies, e.g., development program for entrepreneurship, national strategy for the development of sciences, etc. cover STI aspects rather separately and vaguely.

- President's Administration

Moreover, since 2018, the Department of Innovative Development and E-government

[48] Kenan Aslanli (2017), "Ways of improving the innovation policy in Azerbaijan," Entrepreneurship Development Foundation (EDF), p.13, http://edf.az/ts_general/eng/layihe/IIED/policy-papers/kenan-aslanli_innovation.pdf

Issues of the President's Administration[49] has been focused on the issuance of a strategic development plan for the establishment of an innovation ecosystem in the country, including technological transfer and technology commercialization. Surely, the relevant ministries and public authorities are assigned to particular clauses of state programs and strategies, but coordination among them is still mostly weaker. Despite having strategies and plans in most areas, those were usually neither well-justified nor well-associated with STI directly.

Since 21.10.2009, as per the amendments to Decree #526 of the President of Azerbaijan "On the Establishment of the Science Development Foundation (SDF) under the President of the Azerbaijan Republic,"[50] it was, in fact, disbanded (by becoming a self-financing entity) within additional measures for improvements of the public administration[51]. SDF, like Korea's NRF, previously supported scientists and specialists through research grants. It funded fundamental, applied, invention (innovative), and mega projects and building of scientific infrastructure. Nonetheless, SDF did not usually became competent in correctly "picking the winners" due to particular interests, and sustainability and accountability also left a lot to be desired, although the online platform shows certain data, which became the reason for the disembodiment with the future reorganization of its activity in a more transparent, efficient way.

- Azerbaijan National Academy of Sciences (ANAS)

Reporting directly to the President's Administration, the Azerbaijan National Academy of Sciences (ANAS) is the leading organization for developing, proposing, and implementing science-related programs, plans, and strategies within the governance hierarchy. The ANAS Presidium meets biweekly to discuss and monitor ANAS activities.

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[49] President.az, "Structure of the President's Administration," https://en.president.az/administration/structure

[50] SDF (2009), Official document (in Azeri), Decree of the President of Azerbaijan Republic on the Establishment of the Science Development Foundation (SDF) under the President of Azerbaijan Republic, http://sdf.gov.az/development/uploads/resmi_senedler/arp_serencam_526.pdf

[51] SDF (2019), Official document (in Azeri), Decree of the President of Azerbaijan Republic on additional measures directed toward improvements of public administration in the Republic of Azerbaijan, http://sdf.gov.az/development/uploads/resmi_senedler/arp_ferman_14.01.2019.pdf

ANAS established ANAS-HTP in November 2016 and started popularizing science but lacked in many other components of development of innovations and its components to enhance research capacity, to transfer its research outcomes to industries, and to increase revenues from technology commercialization. By establishing ANAS-HTP in 2016[52], the vision was to ensure strong science and industry cooperation and establish better national innovation ecosystem and knowledge economy through the commercialization of science. All of these will be achieved with the proper functioning of incubation and acceleration programs and Technology Transfer Office along with an equipped Laboratory and the plant area.

ANAS-HTP tenants (~15 residents: ANAS Experimental Industrial Plant, Buta Armor, AzMonbat, Alqoritmika, DN Technologies, Millers Oil, Azeltech, ETP, Alximik, etc.), are provided with seminars and training as well as opportunities to participate in events, laboratory and prototype production support, patent application support, etc.

Moreover, ANAS launched Master's programs[53] for the first time ever to fill the gap between generations and attract youth to science and research. Very recently, ANAS also declared its plan to establish a research university[54].

- Azerbaijan National Academy of Sciences-High Technologies Park (ANAS-HTP)

The Azerbaijan National Academy of Sciences High Technologies Park (ANAS-HTP) was created on 8 November 2016 by Decree №. 2425 of the President of Azerbaijan.

The main purpose of HTP is - the provision of state support for the sustainable development and competitiveness of the economy, expansion of branches of innovation and high technologies on the basis of scientific researches and technological achievements, and creation of modern complexes for research and development of new technologies.

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[52] ANAS (2016), Official documents, Order of the President of the Azerbaijan Republic on the establishment of the Park of High Technologies of Azerbaijan National Academy of Sciences (ANAS), http://science.gov.az/news/open/4508

[53] ANAS (2015), Master's studies in ANAS kick off, http://www.science.gov.az/news/open/2440

[54] "Academician Akif Alizadeh speaks at a joint scientific session of the Collegium of AR Ministry of Education." (2016), http://www.nauka.gov.az/news/open/4754

In May 2017, ANAS-HTP finished all the required registration process in the government and started its activities.

The Azerbaijan National Academy of Sciences High Technologies Park operates in two places and has two offices: Industrial zone (main office) located on a 21-hectare area, not far away from the center of Baku; and City residence located at the academic city of Baku at the city center.

Industrial Zone (Main Office) of ANAS-HTP	City Residence of ANAS-HTP
	

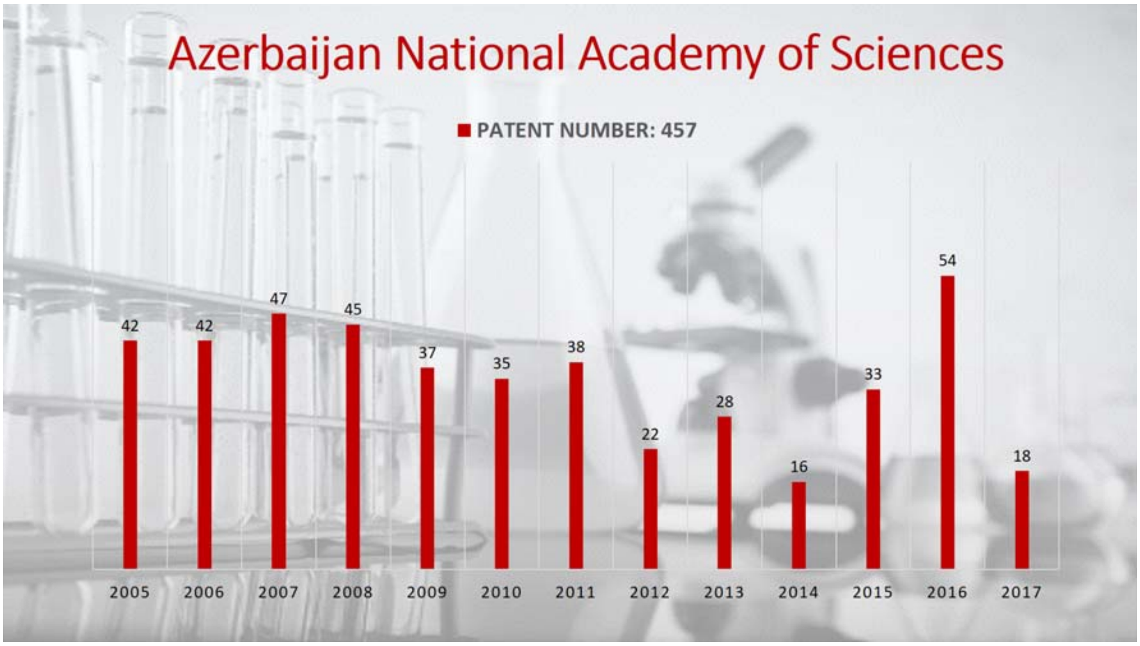
ANAS-HTP is a favorable place for investment. It is exempted from profit tax, property tax, land tax, VAT (import), and custom clearance. It also provides services (utility, security, communications (ICT), logistics/railway terminal, land zone for construction, and storage.

ANAS-HTP has 11 resident companies. The activities of residents are universal: civil (petrochemicals, cleanser-chemicals, healthcare, education projects, battery utilization, drone) and military (Quadchapters, simulations, bulletproof vests, etc.). The products of residents are innovative and science-based and have been sold in the local/global market.

[Figure 2-6] 11 Resident Companies of ANAS-HTP and Their Products

ANAS has forged a partnership with the World Intellectual Property Organization (WIPO). The patent bank of ANAS has been established, and ANAS transferred this patent bank to HTP.

[Figure 2-7] Number of Patents of ANAS



As an advanced and accredited testing center, ANAS SSATC (Azerbaijan National Academy of Sciences Strategic Scientific Analytic Testing Center) offers the following tests:

- Gas and LPG testing;
- Crude oil testing:
- Analysis of toxic elements in toys, shoes, textile, polymeric materials, furniture, and paint layers.

The tasks and objectives of ANAS-HTP include Technology Transfer, which is being implemented in our technology park and at the Academy of Sciences; it attempts to commercialize these technologies. ANAS has about 500 patents, and transfer of these patents is also one of our tasks. In the Business Incubation of ANAS-HTP, we have supported various startups. Some of the startups got promotion, and one of our startups became a resident of our technology park, but some startups could not develop.

One of the issues and problems that should gradually be resolved in the field of

Technology Transfer and Commercialization, both in Azerbaijan and in the ANAS-HTP, is the issue of Financial Support for Technology Transfer -- that is, the search for funds for the development of projects. The main problem here is the search for sponsors. Sponsors do not always believe in any project or in its future. Sponsors naturally do not want to risk investing their money. In order to convince the sponsor to give money for the project, one needs to be able to present the project correctly, write a good business plan, and study the potential market for future products well. At the workshop in May 2019 in Seoul, Korean experts covered these issues in their lectures.

One of the important topics in the field of technology transfer and technology parks activity is legislation (laws, decrees, rules, etc.). Azerbaijan has the appropriate “Regulations on Technoparks” dated 15 May 2014, “Resolution of the Cabinet of Ministers on the Approval of Regulations for the Registration of Residents of the Technology Park” dated 8 July 2015, and “Order of the Cabinet of Ministers on the High Technology Park of the Academy of Sciences” dated 16 November 2016 and “Order of the President of Azerbaijan on the establishment of the High Technology Park of the Academy of Sciences” dated 8 November 2016. At the present time, the legislation of Azerbaijan in this field needs a "Law on the Activity of Technology Parks." This law could reflect issues related to technology transfer, technology commercialization, and support for startups. The topic of the appropriate legislation was also tackled and considered at the workshop in Seoul, and the Korean experts answered our questions.

The ANAS High Technologies Park has the so-called “right of legislative initiative.” Various state bodies apply to our Hi-Tech Park in case there are drafts of legislative acts, and they ask us to come forward with our initiatives. Not long ago, an important regulatory enactment has been adopted in Azerbaijan -- the “Resolution of the Cabinet of Ministers on the approval of Rules for placing government orders for products manufactured in the ANAS High Technologies Park.” ANAS-HTP also participated in the preparation of this document, which is a very important document. When ANAS-HTP propounds a legislative initiative, we refer to the experience of technologically developed countries, including the Korean experience as well.

All of the abovementioned suggest that ANAS-HTP is developing and trying to apply the skills and knowledge it gains by participating in various workshops and conferences, such as the workshops organized by STEPI.

- Ministries

Since the Soviet period, the Ministry of Education has been managing the education layers, but it has weaker links with science and industry. No clear vision is seen as to how to link with the stakeholders of the innovation ecosystem. The Ministry of Economy establishes industrial parks in various sectors, e.g., chemistry, urban management, etc. in the relevant areas of Azerbaijan. It is mostly focused on the industrial aspects of the parks with almost no dedicated research base.

Recently established by the Decree of the President of Azerbaijan Republic dated December 28, 2017[55], however, the Small and Medium Business Development Agency under the Ministry of Economy supports the development of small and medium-sized businesses in the country, provides a range of services to SMEs, and coordinates and regulates the services of state bodies in this field.

The Charter and structure of the Agency were approved by the Decree of the President dated June 26, 2018[56]. According to the Charter of the Agency, the purpose is to strengthen the role of SMEs in the economy of the country through the expansion of participation of SMEs in the regulation of SME policy, application of flexible management system, and effective coordination mechanisms that are widely used and implemented in international practice, increase of competitiveness of local SMEs, expansion of access to financial resources and improvement of institutional support mechanisms, coordination of the activities of public and private entities in this field, and, last but not least, creation of favorable conditions for entrepreneurship development in the region, and local and

55 President.az (2017), Official document, "Decree of the President of the Republic of Azerbaijan on the further management improvement of small and medium-sized entrepreneurship," https://president.az/articles/26657

56 President.az (2018), Official document, "Decree of the President of the Republic of Azerbaijan on Ensuring the Activity of the Small and Medium Business Development Agency of the Republic of Azerbaijan," https://president.az/articles/29169

foreign investment attraction to this economic area.

The Agency includes the Office of the Agency, SME associations, development centers, and State-Entrepreneurship Partnership Development Center as well as development fund.

The Ministry of Transport, Communication, & High Technologies mostly concentrated on ICT-related projects and innovations in its High Technologies Park. It has drawn flak for failing to establish the projected 50-hectare HTP zone on an island in sub-Baku. Currently, there is a changed plan to build a relatively modest park area in Baku in the near future. The Ministry also created the State Fund for the Development of Information Technologies (2012), which was intended to provide startup funding for innovative and applied S&T projects in the ICT fields through equity participation or low-interest loans.

- Innovation Agency

Moreover, as a continuation of the recent reforms, the Innovation Agency under the Ministry of Transport, Communication & High Technologies was established on November 6, 2018 by the Decree of the President of Azerbaijan Republic[57]. It encompasses the State Fund for the Development of Information Technologies and the Hi-Tech Park LLC. Currently, the two organizations are being consolidated under the Innovation Agency.

Nevertheless, the establishment process of the Innovation Agency includes preparation; the scope of its activities will include assisting local business entities in acquiring up-to-date technologies and technological solutions as well as organizing technology transfer, supporting innovation-oriented scientific research, encouraging innovative projects (including startups) by funding them through grants, loans, and equity investments (including venture capital funding), and, last but not least, promoting the innovative initiative. It is still not clear what the vision and strategy of the Innovation Agency on supporting technology commercialization are.

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[57] Ministry of Transport, Communication & High Technologies official website (2018), "President signs decree on the establishment of Innovation Agency under the Ministry of Transport, Communication & High Technologies," http://mincom.gov.az/en/view/news/432/prezident-neqliyyat-rabite-ve-yuksek-texnologiyalar-nazirliyinin-tabeliyinde-innovasiyalar-agentliyinin-yaradilmasi-haqqinda-ferman-imzaladi/

- Other organizations

Other public authorities also conduct scientific research, experimental developments, and, as a new trend, establishing small, tailored innovation projects is planned ahead. The Knowledge Foundation under the President of the Republic of Azerbaijan was established in 2014 “to enhance scientific, technical, socio-economic, and humanitarian knowledge” but was disbanded recently.

Likewise, there are some centers that are focusing on the implementation of social innovations in Azerbaijani society. The State Agency for Public Service and Social Innovations (popularly known as ASAN Service) under the President of the Republic of Azerbaijan is an example of successful state innovation tool to bring public services transparently and effectively into a single place across regions in 20 centers. Moreover, Innovations Center LLC, ABAD public legal entity, E-government Development Center public legal entity, and Innoland Incubation and Acceleration Center are subordinate bodies of ASAN Agency.

One of the subordinate bodies of ASAN Agency is “Innovations Center” LLC, which supervises Innoland Incubation and Acceleration Center on innovations and improvements of information systems and resources. A goal of the recently established "INNOLAND" Incubation and Acceleration Center is “supporting the establishment of startup ecosystem and encouraging innovation and development of the private sector in Azerbaijan and outside of its borders.” Innoland includes an incubation, acceleration, and coworking space and the Tech Academy. This Center is the first in the country to consolidate services for startups and existing companies. Moreover, Innoland has up-to-date infrastructure for qualified tech support.

Another recent reform was the establishment of “Azerbaijan Industrial Corporation” Open Joint-Stock Company (OJSC) in November 2017[58]. The core purpose is increasing the effectiveness of the management system of government property, establishing beneficial cooperative relations among state-owned entities, reforming the reporting and control

[58] AIC OJSC (2018), official website, About the Corporation, http://ask.gov.az/en/mission

system, and achieving the industrial production growth of value chains of the non-oil sector[59], which are key indicators of increasing government intervention in the development of the industry.

Affiliated companies of the Corporation are “Azeraluminium” LLC, “Azerpambiq ASK” LLC, “Azertutun ASK” LLC, "ASK Xidmat" LLC, "ASK Idman Kompleksi" LLC, "ASK Shusha" LLC, "ASK Shaki Sharab" LLC, "Azeripak" LLC, “ASK Izolit” LLC, “ASK Tekstil” LLC, “ASK Security” LLC, "ASK Ayaggabi Fabriki" LLC, and “ASK Heyvandarlig” LLC.

The Corporation has two special divisions within its organizational structure: Research, Innovation, and Technology Division, and; Startup Support Division. Though these divisions have not yet started active works, they aim to support TT/TC and encourage innovations for the industry.

Furthermore, almost all state organizations have established innovation centers or tech laboratories. For instance, the Ministry of Transport, Communication & High Technologies established the Information and Communication Technology Application and Training Center, or simply ICT Lab. “By Decision #389 of the Cabinet of Ministers of the Republic of Azerbaijan, the "Electronic Government Training Center" LLC was established under the Ministry of Communication & High Technologies on December 17, 2015. Pursuant to Decision #292 of the Cabinet of Ministers dated July 12, 2018, the "Electronic Government Training Center" LLC was renamed "Information and Communication Technology Application and Training Center" LLC.[60] Key activities of the Lab are training on ICT, business and management, services related to e-government, and application of ICT projects. Moreover, special IoT Lab for developing innovations established by Microsoft (http://iktlab.az/mslab/), Networking Lab by Cisco, IT Tech Lab, and Multimedia Studio are part of the ICT Lab with broader opportunities for the development of IT systems as well as programming and coding in Azerbaijan.

Likewise, Social Innovation Lab is an active and leading social Startup Accelerator &

[59] Ibid.

[60] ICT Lab (2018), Official webpage, About, http://www.iktlab.az/about

Incubator center in Azerbaijan established in August 2016[61]. It brings a solution to different business and social problems in cleantech, edtech, food & beverage, hospitality, coworking, etc. through innovative approaches to building sustainable business models and globally impactful solutions.

Furthermore, there are many other specialized Academies, such as STEP IT Academy (https://itstep.az), Code Academy (http://code.edu.az/), etc., which have the same IT programming direction in their portfolio for nurturing future programmers, developers, and IT engineers.

2.2 Legal Infrastructure

According to the Global Innovation Index 2018, Azerbaijan's score on "Institutions" is ranked 71st. Although, business environment is ranked 38th globally, political and regulatory environments scored relatively lower at 88th and 94th, respectively.[62]

Along with the rank of business environment and Azerbaijan ranking 35th in the Global Competitiveness Report 2018 by the World Economic Forum[63], and it is sometimes criticized for being less representative of the reality.

Azerbaijan is also judged according to monopolistic market structure, rule of law, access to finance for risky businesses, i.e., entrepreneurship, etc. For example, the Global Innovation Index 2018 ranks Azerbaijan 101st in "Ease of getting credit" and 100th in "Intensity of local competition."[64]

There is no single legal act on science, research, technology, and innovation issues. The Law on Entrepreneurial Activity is in force, regulating entrepreneurial issues in general.

[61] Crunchbase, About Social Innovation Lab, https://www.crunchbase.com/organization/social-innovation-lab#section-overview

[62] Global Innovation Index 2018, https://www.globalinnovationindex.org/gii-2018-report

[63] World Economic Forum (2017), Global Competitiveness Report, http://reports.weforum.org/global-competitiveness-index-2017-2018/

[64] Global Innovation Index 2018, https://www.globalinnovationindex.org/gii-2018-report

The Law on Innovations has been drafted, and it is still waiting for adoption in the future.

The Competition Code is also in the phase of final drafting and is expected to pass the Parliament in 2020. This critical legal act is still missing – the Competition Code has been rejected several times, and it has some political economy explanations, i.e., protection of certain interest groups. Recently, media outlets shared information that Parliamentary discussions on the draft Competition Code are again expected to be possible, along with the establishment of the Authority for Competition[65]. This point was also confirmed by Economy Minister Shahin Mustafayev at the forum "Taxes, Transparency, and Development"[66] held in February 2019 in Baku. Nevertheless, the adoption of the Competition Code is highlighted in the "Strategic Roadmap for the Production of Consumer Goods at the Level of Small and Medium Entrepreneurship" as another government bureaucracy is expected to oversee the market.

Moreover, since January 1, 2019, new amendments to the tax legislation came into force. These changes consider the interest of the business sector, and they had been developed in accordance with the strategic roadmaps. "The changes to the Azerbaijani Tax Code … will cover five important areas"[67]: ensuring the transparency of taxation; expanding the taxable base; improving tax administration; supporting the development of entrepreneurship; and increasing the economic efficiency of benefits and exemptions. Entrepreneurial support has high priority for the country's economic development as well as "reduction of the scales of the "shadow economy" and ensuring of transparency of the economy"[68]. Moreover, according to the adjustments, simplified tax throughout Azerbaijan will decrease to 2%, and entrepreneurs along with SMEs that will operate in clusters will be exempted from income tax, land tax, and profit tax for a period of seven years; startups will also receive benefits in the form of exemption from profit tax and income tax for a

[65] Azernews, (July 17, 2018), " Agency for Competition to be created in Azerbaijan," https://www.azernews.az/news.php?news_id=134939&cat=business

[66] Azernews.az (February 2019), "Competition Code may appear in Azerbaijan soon," https://www.azernews.az/business/145866.html

[67] Azernews.az (November 2018), "Minister: Changes in tax legislation to cover five areas in 2019," https://en.azvision.az/news/96582/minister-changes-in-tax-legislation-to-cover-five-areas-in-2019.html

[68] Ibid.

period of three years. The abovementioned recent tax changes will contribute to the development of the non-oil sector.

The government also updates its FDI and overall investment laws and policies to increase foreign investments not only quantitatively but also qualitatively, i.e., to attract investments in higher value-added industries.

Adopted in 2016, the Law of on Science[69] is the main legislation act to frame the principles and policy vision of the government in STI. With a strong legal base, Azerbaijan aims to offer equal conditions to stakeholders in scientific innovation activities and stimulate their functionality through favorable funding and investment terms and by reforming STI priorities according to the market demands, establishing innovation areas, etc.

On June 14, 2019, the President of Azerbaijan approved the Rules on Using the Funds of the Innovation Agency under the Ministry of Transportation, Communication & High Technologies. The funds of the Innovation Agency are directed toward innovation activities and innovation initiatives, supporting local entrepreneurs in absorbing and transferring cutting-edge technologies and technology solutions as well as supporting domestic and foreign scientific research projects and innovative projects (and startups). The fund will also be used to give grants, preferential loans, and equity investments (venture capital as well). The Rules also specify 4 forms of funding:

1. **Investment funding** – This includes equity investments, venture funding, buying shares, etc. Note, however, that the Innovation Agency cannot become the major shareholder in invested enterprises.

2. **Preferential loans** – Through the approved list of credit institutions, it provides enterprises with preferential loans of up to AZN10 million (approximately USD5.86 million) for a 3~10-year period. Though the Rules underline that the Ministry of Transportation, Communication & High Technologies defines the criteria for such loans, it is not clear yet. One important aspect of such loans also includes the

[69] Law of the Republic of Azerbaijan on Science (in Azerbaijani), http://science.gov.az/uploads/PDF/Elm_haqqinda_Azerbaycan_Respublikasinin_Qanunu.pdf

relationships among the entrepreneur, credit institution, and government agency, with the credit institution giving loans to entrepreneurs at its own risk.

3. **Grants** – The Innovation Agency provides grants to entities based on the results of calls for applications. Contest participants are selected by an independent jury according to the defined criteria. It has been previously experienced by the State Fund for the Development of Information Technologies but failed to prove its effectiveness.

4. **Other forms of funding** – Such funding aims to support entrepreneurs in developing, adopting and transferring modern technologies (i.e., TT and TC), financing the creation of innovation-related centers and entities, funding university research infrastructure development, innovation and IPR-related costs and supports, and innovation events and capacity building. More importantly, it also includes attracting funds from foreign innovators and ideas, researchers, and experts to Azerbaijan.

The Ministry of Transportation, Communication & High Technologies will regularly supervise and monitor this funding scheme.

Since there is no specific legislative act (law, code, etc.) on technology parks, the currently existing ones, e.g., Sumgait Technologies Park, Hi-Tech Park Azerbaijan, and ANAS-HTP, had been established as per a direct decree of the President based on an exemplary charter on technology parks.[70] All the fiscal and administrative incentives are granted to technology parks and regulated pursuant to those Presidential decrees. Along with fruitful office conditions, these stimulus package includes 7 years of exemption from almost all tax duties for the tenants.

Moreover, there are some strategic documents that put forward certain relevant conditions and steps to be implemented by the authorities:

[70] Exemplary Statute on the Technology Park (in Azerbaijani), https://www.president.az/articles/11748

- Strategic Roadmap from the National Economic Perspective of the Republic of Azerbaijan
- National Strategy for the Development of Information Society During the Period 2014-2020
- Development Concept “Azerbaijan 2020: Look into the Future.”

2.3 State Development Programs

Azerbaijan is a country with solid scientific potential for the development of different fields of STI, TT, and TC. This direction has been actively discussed during the last years in the country and is currently at the top of priorities in the agenda of the Government. Thus, science and ICT technologies are well-developed and smoothly adopted in the modern reality of Azerbaijan.

For instance, Azerbaijan is one of the limited members of the World Space Club, and it is successfully developing the E-Governance field with deep penetration into different usage areas. According to the UNDP Country Office and Azerbaijan International Development Agency (AIDA), Azerbaijan agreed to introduce and develop the information-communication technologies sector in Afghanistan[71].

Though the “Strategic Roadmap on the National Economy and its Main Sectors” is currently the major development program of Azerbaijan, there are past and present several state development programs and strategies with focus on the development of STI, TT/TC, and innovation in the different fields of the national economy.

For instance, in December 2012, the President signed a decree on the “Development Concept. Azerbaijan – 2020: Outlook for the Future.” The concept covers the main strategic goals of development policy in all areas of the country, and it will be realized

[71] AIDA, “Government of Afghanistan sees Azerbaijan’s experience in e-Governance as a great development model,” http://aida.mfa.gov.az/en/news/65/

through various state programs[72]. The document ensures and stresses guidelines for balanced, innovative, and sustainable development of the economy, improves the social welfare of the population, provides an efficient management system of law, human rights, and freedom, and ensures the involvement of civil society in public life as well.

[Figure 2-8] Major State Programs Related to STI Issues

Strategic Roadmaps on the national economy and main sectors of the economy	Development Concept. **Azerbaijan – 2020**: Outlook for the future	The State Program on the **Education of Azerbaijani Youth in Foreign Countries** in the years 2007-2015
The State Program on **Informatization of the Education** System of Azerbaijan Republic for 2008-2012	State Strategy for the **Development of Education** in the Republic of Azerbaijan	National Strategy for **Information Society Development** in Azerbaijan in 2014-2020
State Program for the **Development of Industry** for 2015-2020	National Strategy for the **Science Development** in the Republic of Azerbaijan in 2009-2015	Further Improvement of Management of the **Small and Medium-sized Entrepreneurship** in Azerbaijan

Moreover, in order to improve the quality and quantity of human resources for the specified qualifications and programs as well as to ensure the education of Azerbaijani youth on the required specializations to transform them into an important component of the country's development, the "State Program on the Education of Azerbaijani Youth in Foreign Countries During the Period 2007-2015" was adopted and approved on April 16, 2007. The main objectives of the State Program are: (1) to increase the total number of Azerbaijani youth to 5,000 to acquire foreign education during the period 2007-2015; (2) to improve the entire process of selecting and sending students abroad on priority specializations; (3) to identify specializations, qualifications, and requirements in accordance with the economic development and labor market; (4) to strengthen the

[72] Trend News Agency, "Azerbaijani President approves "Azerbaijan 2020: Look into the Future" Development Concept," https://en.trend.az/azerbaijan/politics/2104451.html

scientific and scientific-pedagogical potential of the country to increase the export potential of Azerbaijani intellectual property; and (5) to develop the human capital of the country through the further improvement of labor productivity to increase competitiveness and sustainable development.

The "State Program on the Informatization of the Education System of Azerbaijan Republic for the Period 2008-2012" was developed for the purpose of creating a unified national education environment with the effective application and usage of up-to-date ICT technologies at all levels of education and to provide fair access to qualified education for all population. The Program was approved by the Presidential Decree in June 2008[73]. Moreover, this State Program was developed based on the "National Strategy for Information and Communication Technologies for the Development of Azerbaijan Republic (2003-2012)" approved by the President of Azerbaijan in February 2003.

The following are the results of implementation of the Program: (1) ICT infrastructure has been developed in the education system (improvement of ICT facilities of schools and institutions); (2) expansion of introduction of e-learning technologies (new training forms, as distant education was applied); (3) electronization and informatization of management of an education system (introduction of electronic teaching and methodological resources as well as usage of ICT technologies in the curriculum and teaching process); and (4) strengthening of staff's capacity with regard to Azerbaijan's educational system (improvement of knowledge and skills of pedagogical and administrative-managerial staff of schools and institutes, involvement of local experts' potential).

The key purpose of the "State Strategy for the Development of Education in the Republic of Azerbaijan," which was approved in October 2013[74], is to create a qualified, based-on-external-experience educational system through the introduction of up-to-date ICT and innovative technologies.

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[73] Ministry of Education of Azerbaijan Republic, "State Program on the Informatization of the Education System of Azerbaijan Republic for the Period 2008-2012," http://edu.gov.az/az/page/83/599

[74] President's Administration, "State Strategy for the Development of Education in the Republic of Azerbaijan," http://www.e-qanun.az/framework/27274

Moreover, it is important to meet the world's leading education standards and transform contemporary higher education system into education – research – innovation centers. The other strategic directions are: (1) development of effective education content and curriculum through the addition of innovative training methods and technologies; (2) formation of a novel education management system based on public – business partnership; (3) development of innovative education infrastructure with a transparent, sustainable funding mechanism.

In 2016, with the involvement of the Ministry of Transportation, Communication & High Technologies in the implementation process, the "National Strategy for Information Society Development in Azerbaijan for the Period 2014-2020" was approved by the Presidential decree. The National Strategy is implemented in two stages and will be accompanied by public programs[75] to encourage the hi-tech industry to provide competitive, export-oriented IT products as well as the latest innovative solutions and applications. Moreover, this National Strategy will develop the governance competencies of the economy by strengthening the innovation and IT potential of Azerbaijan, having considered all existing experiences and recommendations of the International Telecommunication Union (ITU), European Union (EU), and UNESCO[76].

The objectives also include: (1) enhancing the scientific and technical potential of hi-tech technologies -- nano, bio, nuclear, and cosmic technologies -- according to the dissemination of novel knowledge and innovative skills into society; (2) strengthening state – business – citizen partnership in the establishment of knowledge society as well as expanding international cooperation from the point of view of new markets; and (3) developing technoparks, business incubators, and innovation structures for the development of innovation-based knowledge and skills.

To support industrial development and application of industrial innovations, the "State Program for the Development of Industry for the Period 2015-2020" was approved in

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75 Azernews, "Information Society Strategy to be Implemented in Two Stages," https://www.azernews.az/business/65843.html

76 Azernews, "Information Society Strategy to be Implemented in Two Stages," https://www.azernews.az/business/65843.html

December 2014. Increasing the competitiveness of the industry through innovations and technology transfer, introduction of special economic zones, innovative clusters, and new industrial enterprises in priority fields as well as improving industrial infrastructure will positively influence the industrial capacity of the country and stimulate the attraction of local and foreign investments into the non-oil industry, serving as pivotal reasons for the authorization of the State Program. Moreover, the objectives of the State Program are:

- To establish industrial and technology parks along with industrial sites;
- To strengthen cooperation between state and private sectors for the purpose of development of non-oil industrial export;
- To strengthen the government's support for industrial innovation activity;
- To encourage the application of innovative technologies at industrial enterprises[77].

All of the abovementioned in the framework of the State Program on industrial development will stimulate the innovative industrial initiatives of regions, and the development and promotion of innovative, diversified industrial clusters will eventually transform a non-oil industry into the core driver of economic growth and become the main source of national export growth.

Likewise, industrial innovations would not be successful without appropriate scientific support. Thus, integrating science into this process is supported by the "National Strategy for Science Development in the Republic of Azerbaijan During the Period 2009-2015" with the following key goals:

- Expanding the role of science in the economic development of the country;
- Ensuring the participation of science and its fields in the transfer of innovations and technology;
- Expanding research on fundamental and applied sciences;

[77] "State Program for the Development of Industry for the Period 2015-2020," http://senaye.gov.az/content/html/2281/attachments/State%20Program%20for%20the%20development%20of%20industry%20in%20Azerbaijan%20in%202015-2020.pdf

- Financial development of scientific research institutions by providing additional internal grants and support mechanisms (establishment of the Science Development Foundation under the President of Azerbaijan Republic, even if it was not so successful due to management trap);
- Creation of the normative legal base of science;
- Enhancing the efficiency of scientific research in the establishment of innovation ecosystem;
- Development and strengthening of the country's scientific potential.

The National Strategy envisages the mutual development of science and technology and formation of a productive innovation ecosystem with the successful transformation of patents, inventions, and scientific-design works into innovative products that will eventually improve technical effectiveness and strengthen the scientific potential of the country. Moreover, the state policy on science and technology development covers modernization and introduction of novel scientific and technical infrastructure, equipment, and laboratory facilities of research institutions of ANAS as well as other scientific institutions and organizations to ensure the development of highly qualified scientific human resources.

Obviously, the integration of science, technology, and innovation will ensure the development of the non-oil sector, especially fundamental and applied sciences. As one of the results of the Strategy, the e-science platform AzScienceNet was reconstructed, and the overall infrastructure was renovated for obtaining high-quality technological base and significant functional capabilities using BigData technologies[78]. In addition, several important measures and programs had been implemented to popularize science among the youth in order to ensure the further sustainable development of scientific fields and integration of innovations and ICT technologies in everyday life.

Last but not least, the establishment of ANAS Hi-Tech Park (HTP) was another

[78] ANAS, "Academician Akif Alizadeh makes a final summary report on the implementation of the "National Strategy on Science Development in the Republic of Azerbaijan in 2009-2015," http://www.science.gov.az/news/open/3577

achievement of the national scientific strategy[79] for strengthening the application and promotion of innovation as well as ensuring the transfer of scientific technologies to the industry and development of specific non-oil sectors in the main cities and regions of the country.

On May 10, 2019, the Cabinet of Ministers approved a new regulation for the government procurement rules of ANAS-HTP products. The government agencies declare their needs for procurement to ANAS-HTP, which in turn announces such offer to its tenants. The tenants then prepare their proposals without participating and contesting in tenders. If an ANAS-HTP (its residents' offers) proposal is not selected by the government agency, such agency should justify its decision of buying another offer and rejecting the ANAS-HTP products to the President's Administration. This rule may be criticized, but it is a great opportunity for ANAS-HTP residents to supply products for government agencies.[80]

The President of Azerbaijan has signed an Order on January 10, 2019 to increase the effectiveness and provide coordination for innovative development. It includes central government entities, ANAS, universities, state-owned big enterprises, and many other stakeholders. Agencies will be responsible for innovation issues based on scientific successes, knowledge, and digital advancements to develop, introduce, and use goods, services, technologies, process, and solutions. The Order provides an invitation to leading foreign consulting companies to develop jointly an "innovation strategy" of Azerbaijan (draft to be submitted in June 2019). Coordination working group members have to propose the assigned top managers, certain share of annual expenditures for innovation activities, and relevant rules and procedures for registry, laboratory use and tests, and other infrastructure-related issues in each agency involved. It also aims to encourage startups and human resources to support innovative activities.[81]

The Director of ANAS-HTP is a member of the working group on "Science and Education."

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[79] Ibid.

[80] https://www.docdroid.net/3ubNu9q/nk-qerar.pdf#page=2

[81] President's Order on increasing the effectiveness and coordination for innovative development in Azerbaijan. https://president.az/articles/31491

The next step was adoption in December 2016 by Azerbaijan's president of the "Strategic Roadmap on the Production of Consumer Goods of Small and Medium-sized Enterprises in the Republic of Azerbaijan." This roadmap has been developed to ensure the competitiveness, inclusiveness, and sustainability of the economy of the country.[82] The goals of the Strategic Roadmap are:

- Improve the business environment in the country to encourage the expansion of SMEs;
- Ensure SMEs' access to funding resources for the sustainable development of SME businesses;
- Increase SMEs' access to foreign markets for the international development of local SME businesses;
- Encourage the introduction of innovations and ICT technologies to increase the competitiveness of SMEs' products as well as strengthen R&D activities in this area;
- Strengthen the legislative base and stimulation of financial and tax policy for the development of SME businesses.

Moreover, the Decree on the "Further Improvement of Management of the Small and Medium-sized Entrepreneurship in Azerbaijan" (signed in December 2017) ensures further improvements of the investment framework and legal protection of the business environment and enhances SMEs' access to financial sources to upgrade the level of competitiveness of local businesses by creating favorable conditions for development.[83]

A vital achievement of the Decree was the establishment of the "Small and Medium Business Development Agency of the Azerbaijan Republic" under the Ministry of Economy of Azerbaijan. Financial support and development of SMEs in the country are the key goals

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[82] President's Administration, "Strategic Roadmap on the Production of Consumer Goods of Small and Medium-sized Enterprises in the Republic of Azerbaijan," https://mida.gov.az/documents/Ki%C3%A7ik_v%C9%99_orta_sahibkarl%C4%B1q_s%C9%99viyy%C9%99sind%C9%99_istehlak_mallar%C4%B1n%C4%B1n_istehsal%C4%B1na_dair.pdf

[83] President's Administration, "Decree of the President on the Further Improvement of Management of Small and Medium-sized Entrepreneurship in Azerbaijan," https://az.president.az/articles/26657

of the Agency. Thus, the state government has obviously been interested in actively participating in development projects and activities to maintain the progress of SMEs in Azerbaijan via state funding and legislative programs.

To sum up, Azerbaijani state programs did not fully achieve the mid- and long-term objectives of innovation policy in the context of economic development with exact applied and quantitative key performance indicators (KPI) for further monitoring and evaluation of the results. Strategic government stakeholders for innovation development (Ministry of Economy, Ministry of Education, Ministry of Transport, Communication & High Technologies, and ANAS) play a major role in the legitimate support for the establishment of an innovative ecosystem.

Furthermore, the introduction of incentives for and investments in R&D for innovations and technologies should not be the only part of innovation policy. SMEs and startup businesses are especially interested in the mutually beneficial progress of the process but do not have sufficient power to influence somehow the adoption and transfer of technology through accessible tools.

3. Status of Technology Transfer Ecosystem in Azerbaijan

3.1 Technology Parks and Incubation Centers

Technology parks and incubation centers have started emerging in Azerbaijan in recent years. Initial steps aimed to follow the ongoing global trends for transferring the knowledge and technologies into industries. Later, the government prepared an exemplary charter for technology park as substitute legal documents pending the adoption of clear vision, legislation, and management infrastructure. Even before the process got started, there have been several challenges and improvement in intellectual property rights, patenting, and commercialization.

The availability of technoparks is pivotal in the establishment of an STI ecosystem in Azerbaijan. This will help formulate and develop a specified competitive and innovative environment in the country. Obviously, there are three main operating mechanisms of technoparks:

- Support for independent research and project teams wishing to implement their own innovation projects;
- Support for the commercialization of scientific knowledge collected in university and scientific-research teams;
- Solving the technological issues of the producers by concluding technological decisions from the independent market of innovation[84].

Technoparks or technological scientific parks are recognized for the organization of innovative-favorable environment for the development of SMEs and startups. The key aim of technoparks is the transformation of the results of scientific and technological activity,

[84] Kenan Aslanli (2017), "Ways of improving the innovation policy in Azerbaijan," Entrepreneurship Development Foundation (EDF), p.10, http://edf.az/ts_general/eng/layihe/IIED/policy-papers/kenan-aslanli_innovation.pdf

mostly R&D accomplishments, into competitive product for entry into new and existing local and international markets. To be more precise, there are 11 major technological parks and innovation centers today in Azerbaijan. The list can be found below:

[Table 2-1] List of Major Technoparks and Innovation Centers

#	Organization (alphabetically)	Webpage
1	ADA Innovation Center	http://economy.gov.az/article/ada-innovasiya-merkezi/29030
2	Azerbaijan National Academy of Sciences (ANAS) Hi-Tech Park	http://www.ameaytp.az/content.aspx?id=2&lng=az
3	ASAN (State Agency for Citizen Service and Social Innovation) Innovation Center	https://icenter.az/en/main/
4	BARAMA Innovation and Entrepreneurship Center	https://barama.az/en
5	Bakcell APPLab Innovative Mobile Development Center	https://applab.bakcell.com/
6	Baku Engineering University Technopark	http://qutechnopark.az/en
7	INNOLAND Innovation Center	http://innoland.az/
8	Ministry of Transport, Communication & High Technologies Hi-Tech Park	http://www.hightech.az/ http://www.mincom.gov.az/en/view/organization/2/
9	Mingachevir High Technology Park under the Ministry of Communication & High Technologies	http://hightech.az/mhtp/
10	NEXT STEP Innovation Center	http://icnextstep.com/
11	Shamkir Agropark Logistics Center (Shamkir Agropark LLC)	http://shamkiragropark.az/home
12	Sumgait Chemical Industry Park	http://www.scip.az/index.php?lang=en
13	Sumgait Technology Park (STP)	https://www.stp.az/?l=en

Nevertheless, the availability of dozens of technoparks or innovation centers is not sufficient for the formulation of an STI Development Strategy along with an Innovation Ecosystem in Azerbaijan. The reasons for the current unsatisfactory situation are: (1) lack of clear vision and strategy for STI development; (2) lack of legal infrastructure and law base; (3) lack of managerial capacity for technoparks and innovation centers; (4) lack of internal funding as well as investment limitations; and (5) lack of applied innovative ideas.

In this context, to strengthen innovation and incubation base, first, a legal system should be adopted for the development of STI in the country. Currently, to address the lack of legal base for the establishment of technological parks and innovation & incubation centers, state technoparks were founded based on the Decree on the "Creation of Special Economic Zones in the Republic of Azerbaijan" and "Exemplary Statute on the Technology Park." Thus, the adopted legislation is not sufficient for the establishment and management of technoparks in the Republic of Azerbaijan, which will be facing challenges in the development of innovative entrepreneurial activity in the country.

At the same time, according to the Tax Code of Azerbaijan Republic, all technoparks and innovation centers established by a presidential decree are exempted from income tax on the profits, VAT, and some other fiscal payment deductions along with the construction and maintenance of the infrastructure of STPs. Moreover, the legal entities, as a resident of industrial and technology parks, are free from paying property tax for using the property of the industrial and technology parks as well as from land tax for using the land of the industrial and technology parks for 7 years in accordance with the laws of the Azerbaijan Republic.

According to the European Commission and its guidance report titled "Setting up, Managing, and Evaluating EU Science and Technology Parks" (2013), "an STP matures between years five and ten… This may be a time to moderate public sector investment in favor of private funding (particularly for capital expenditure)."[85] In order to check the track of development of STP, experts highlight performance evaluation and monitoring as

[85] European Commission, "Setting up, Managing, and Evaluating EU Science and Technology Parks," http://ec.europa.eu/regional_policy/sources/docgener/studies/pdf/stp_report_en.pdf

beneficial tools to analyze the activity and development strategy and improve general and specific governance and performance of STPs.

To be specific, below is a comparative analysis of Azerbaijani technoparks and innovation centers by using certain performance indicators:

[Table 2-2] Comparative Analysis of Major Technoparks and Innovation Centers by Indicators

	ADA Innovation Center	ANAS-HTP	ASAN Innovation Center	Baku Engineering University Technopark	BARAMA Innovation and Entrepreneurship Center	Mingachevir High Technology Park under MoTCHT	Innoland	MoTCHT Hi-Tech Park	NEXT STEP Innovation Center	Shamkir Agropark Logistics Center	Sumgait Chemical Industry Park	Sumgait Technology Park
Area	~100 sq.m	25.6 ha	~1000 sq.m	~1000 sq.m	n/a (sq.m)	n/a	n/a	50 ha (planned)	n/a	240 ha	505,64 ha	250 ha
Way of establishment	By University	By President	By state agency	By University	By private entity	By President	By public organization	By President	By private entity	By President	By President	By President
Tax incentives	-	+	-	-	-	+	-	+	-	+	+	+
Effective office environment	+	+	+	+	+	+	+	+	+	+	+	+
Internal funding opportunities	-	+	+	+	+	n/a	+	n/a	+	n/a	n/a	n/a
Ability to attract external funding	+	n/a	n/a	+	+	n/a	+	n/a	+	+	+	+
Science and research capacity	+	+	-	+	-	-	-	-	-	+	+	+
Availability of specific startup programs	+	+	+	+	+	n/a	+	+	+	-	-	-

To sum up, Azerbaijan is a country with lack of experience in the management of technoparks and innovation & incubation centers. Note, however, that the Azerbaijani government is willing to establish an innovation ecosystem with the further reorganization of the national economy and diversification of non-oil sectors focusing on STI. Moreover, the availability of youth generation full of new innovative ideas and readiness to realize innovative projects wields a positive impact on the whole situation.

3.2 Patenting and Commercialization

As mentioned above, science and research have been largely conducted at research institutes since the Soviet Union period. As a command economy center, those research institutes were assigned with planned programs in accordance with overall country strategies. Their aim was to produce quality research outcomes. Implementation of those findings was completely held by the government. Therefore, researchers and their affiliated institutes had no commercial interest due to the absence of market economy and private property rights. Such mindset still exists among certain academe environments even though the economic system and property rights have been totally reformed into market mechanisms. Understanding of members of the academe has historically led to their role of conducting a research project, and then publishing it to strengthen their academic status without considering the commercial opportunities from such.

Before the last administrative government reforms, there were two government bodies on intellectual property and copyright protection: the State Committee on Standardization, Metrology and Patent, and State Agency for Copyrights. In April 2018, President I. Aliyev merged the State Agency for Copyrights with the Center for Patents and Trademarks, and the State Committee on Standardization, Metrology, and Patent was formally put to an end. The new agency is called Intellectual Property Agency.[86]

[86] Presidential Decree (April 20, 2018), "İstehlak bazarına nəzarət, standartlaşdırma, metrologiya və əqli mülkiyyət hüquqları obyektlərinin mühafizəsi sahəsində idarəetmənin təkmilləşdirilməsi ilə bağlı tədbirlər haqqında Azərbaycan Respublikası Prezidentinin Sərəncamı" (in Azerbaijani), https://www.president.az/articles/28139

Intellectual properties are regulated by the Law on Science, Law on Intellectual Property Protection, Law on Patents, Law on Copyrights and related rights, and Law on the Provision of intellectual property rights and countermeasures against pirates.

An internal, not publicly disseminated research has been conducted to explore the exploitation opportunities of intellectual properties in Azerbaijan. The findings showed that there is substantial potential for intellectual property, especially within the ANAS system. Note, however, that this potential is mostly underexploited, underestimated, and even poorly protected. Because of the introduction of innovation and technology commercialization concepts within science audiences, research institutes can acknowledge their potential to bring about funding in addition to the state science budget. For this to be realized, the research institutes should adopt proper management staff and schemes for STI.[87]

At ANAS, all the research outcomes, i.e., intellectual properties, are collected in a dataset, all the outcomes belong to the assigned research institutes, and the author's rights of ownership are recognized by ANAS. Nonetheless, they do not have any standardized contract mechanisms to manage the ownership, revenue, and commercialization aspects between the authors and institutes assigned.

Before the establishment of ANAS-HTP, researchers and patent owners had seen no clear sign of commercialization probability for their own projects by the institutes. Some of them who had better knowledge and skills or better networks to commercialize the patents in the market had relatively good prospects and got financial advantages. Still, most of this kind of processes exceeded the ANAS boundaries. Now, the picture is comparatively different, and research institutes seek commercialization opportunities with the help of ANAS-HTP.

Recently, the ANAS Presidium decided to assign collecting all patents and copyrights of its research institutions under ANAS-HTP to assess and commercialize (if possible) such assets. On average, more than 500 projects are conducted annually, with approximately 20+

[87] Internal Report on Intellectual Property and Commercialization (2015)

having real commercialization potential.[88] There were 19 patents in 2017, which was very much lower than 54 in 2016. All in all, ANAS-HTP registered 457 active patents of the ANAS system[89].

Funding for the commercialization of patents and other intellectual property outcomes are lower than expected. Some public funds, e.g., ICT Development Fund, Science Fund of state oil company SOCAR, etc., as well as limited internal budgets are released for this purpose. Moreover, there are some startup competitions, e.g., ClimateLaunchpad[90], New Idea Competition[91], etc., and events that help new ideas get limited initial (but not sufficient) funding for their ideas. For instance, the Innovation Contest, organized jointly with 4 ministries and other stakeholders, rewards winning teams.[92]

Moreover, as already mentioned in the Report, different opportunities, such as innovation competitions and contests, are organized by local innovation and incubation centers and labs in order to push TC and TT in the country with further attraction of foreign investments in the local IT sector as well as the introduction and application of Azerbaijan innovations in foreign markets.

It is also necessary to mention the passion of Azerbaijan's innovative youth and young innovation project managers who successfully applied at and participated in the different foreign innovation contests and incubation programs with local startups. This gives additional opportunities in the TC and TT of local innovative products.

There are many competitions for innovation and hi-tech startups, such as "Yeni fikir" (New idea contest), "Startup Azerbaijan," I2B (Idea to Business), etc.

The commercialization of technologies is a pivotal part of the process of development of

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88 Internal Report on Intellectual Property and Commercialization (2015)

89 https://www.ameaytp.az/index.php/patent/patent-bank

90 Social Innovation Lab, http://socialinnovationlab.az/azerbaijan-is-now-part-of-the-worlds-largest-green-business-idea-competition-climatelaunchpad/

91 Baku Engineering University, http://yenifikir.az/en/

92 Azernews.az (2017), "First contest of innovative projects held in Baku," https://www.azernews.az/business/124447.html

innovations. In the real world, accelerating commercialization and ensuring an innovation's development require solving a number of essential issues, such as spurring of the development of startups and possibility of using innovations by small and medium-sized enterprises. Another point deals with the acceleration of university cooperation with state and private companies of the industrial sector. Note, however, that it is necessary to resolve the issues of copyright protection by determining who owns the results of scientific research and how the royalty should be distributed in the ecosystem. There is a need to define the legal and regulatory approaches regarding the share distribution and identify ways of stimulating commercialization, including the methods of forming a partnership between the state and the private sector as stated by the guidelines of the Board of Azerbaijan's Intellectual Property Agency.

Overall, the patenting process is established smoothly. Still, there is a long way to go in terms of the commercialization of those knowledge outputs. This fact is also evident in the "Knowledge & technology outputs" sub-indicator of the Global Innovation Index 2018, where Azerbaijan is ranked 89th.[93]

3.3 International TT and TC Cooperation

The commercialization of technologies is a pivotal part of the process of development of innovations. In the real world, accelerating commercialization and ensuring an innovation's development require solving a number of essential issues, such as spurring of the development of startups and possibility of using innovations by small and medium-sized enterprises. Another point deals with the acceleration of university cooperation with state and private companies of the industrial sector. Nonetheless, it is necessary to resolve the issues of copyright protection by determining who owns the results of scientific research and how the royalty should be distributed in the ecosystem. There is a need to define the legal and regulatory approaches regarding the share distribution and identify the ways to stimulate commercialization, including the methods of forming a partnership between the state and the private sector as stated in the guidelines of the Board of

[93] Global Innovation Index 2018, https://www.globalinnovationindex.org/gii-2018-report

Azerbaijan's Intellectual Property Agency.

As a public-private cooperation result, the Azerbaijan Innovations Export Consortium (AZINNEX) is "jointly established by 10 leading internationally recognized companies of Azerbaijan specializing in ICT innovations to become a reliable, flexible, and global partner in the world."[94] The key mission of AZINNEX is introducing innovative, competitive, and quality IT products and services from Azerbaijan to the global market, which will lead "to its becoming a trusted global partner in the provision of local innovative IT solutions."[95]

As the first local consortium that focuses on the commercialization of local IT products and research results in the global market, AZINNEX serves customers with the most innovative and cutting-edge technologies developed by Azerbaijan's IT companies. It provides a full range of solutions to deliver the highest-level sale and after-sale services. AZINNEX successfully exports to more than 30 countries around the world, including Russia, Ukraine, Estonia, Poland, Serbia, Belgium, Kyrgyzstan, Afghanistan[96], etc.

3.3.1 International Cooperation to Improve TT/TC Opportunities

At the macro-level, since Azerbaijan's economy had been fueled by high revenues from the oil resources for years, it was relatively easier for local big enterprises to adopt and acquire modern technologies from foreign sources, so it did not invest in domestic research institutes and/or in-house innovations. Today, however, the picture has changed with world prices of oil and gas dropping rapidly, leading many economies, including Azerbaijan's economy, to face domestic currency devaluation or diminishing revenue implications. In that regard, it can be observed, though very vaguely, that both government and private sector are thinking of investing more in and dedicating more resources to technology development.

In a rather micro-level, ANAS itself works on cooperating with various partners outside the

[94] Azerbaijan Innovations Export Consortium (AZINNEX) (2018), About, http://azinnex.com/en/home

[95] Ibid.

[96] Ibid.

country and in the domestic economy to improve TT/TC capacity. ANAS supports the participation of scientists in the European Union calls and programs and cooperation with post-Soviet countries. Nowadays, ANAS pays special attention to working with Asian partners, including Chinese and Korean TT/TC issues.

Recently, ANAS-HTP accepted proposals to become residents that are also affiliated or which closely cooperate with foreign companies. For example:

- “EPC GROUP” LTD is a resident that works on Azerbaijan’s gold mining and methanol productions as well as construction and industrial projects in Iran.
- “Millers Oils Azerbaijan” LTD is a joint venture by a British company and ANAS-HTP. The resident works on the production of innovative lubricants for the automobile industry.
- “Buta Armor” is a very new resident at ANAS-HTP, and it is also a joint project with Bulgaria’s “Mars Armor” company (#1 in Europe) in producing defense wear and anti-mine protections. It satisfies the NATO, API, and QOST international standards and certifications. In addition, ANAS-HTP continues to initiate cooperation with foreign companies and entities to promote its advantages and residents.
- The ANAS-HTP delegation met the British Petroleum (BP) Azerbaijan Office to advertise motor oil and lubricants and experimentation opportunities for toxic elements in petrochemical products.
- ANAS-HTP also works with the national oil company of Azerbaijan (SOCAR) in jointly developing solutions for the oil and gas industry.
- The RUSAL (Russian Aluminum) enterprise representative also visited ANAS-HTP to learn its products in mining and producing aluminum as well as its fiscal incentives for residents.
- In addition, delegations from Italian, Russian, Iranian, Spanish, Israeli, Ukrainian, Egyptian, Belarusian, and other countries discuss technology commercialization opportunities in various industries ranging from lubricants to defense products.

Moreover, ANAS-HTP had been represented in the EU Eastern Partnership event in 2018 wherein the main discussions were about research-industry collaborations, EU4Innovation, and Technology Transfer as well as active participation in the HORIZON2020 framework.

The ANAS President underlined the importance of ANAS-HTP in cooperation with the "Caspian European Club," which includes more than 5k companies from 70+ countries, and signed a Memorandum of Understanding between the Club and ANAS-HTP.

"Millers Oil Azerbaijan, a resident of ANAS-HTP, won the Innovation Award by HVM Catapult in 2018 for production efficiency, competitiveness, and innovations in business growth. It was selected among lubricant producers based on "nano-drive technologies." The resident got an opportunity to represent a wider region in finals in London.

3.3.2 Cooperation with STEPI

ANAS-HTP is working on better cooperation with international partners and donors. Note, however, that there are few tangible results of such cooperation efforts. Among them, cooperation and mutual understanding are very important for ANAS-HTP in cooperation with STEPI. The Korean experience and willingness to share knowledge and expertise are immensely valuable for Azerbaijani technoparks and research institutions to transfer/commercialize their outputs and products into markets. Since 2017, STEPI has been assisting ANAS-HTP to develop its capacity in building a better framework and the capacity for science commercialization.

So far, STEPI has focused on working with ANAS-HTP and has not yet reached a broader level of cooperation in Azerbaijan, especially with ANAS research institutes. If the ongoing project brings about the expected outcomes and impact, future broader cooperation would definitely help ANAS-HTP, and ANAS in general, improve its TT/TC opportunities and hopefully carry out profitable science commercialization with Korean partners.

ANAS-HTP has to work more on technology gap and source identification, technology selection, contracting and supply issues, R&D cooperation, licensing, FDI promotion, IPR issues, technology marketing, performance assessment, and technical assistance as well as the adoption/internalization of technologies by overcoming market, formal, and/or informal challenges. In this regard, the role of STEPI would help ANAS-HTP learn and follow benchmarks.

3.4 Structure & Function & Role of ANAS-HTP, TTO, and Organizations of TT & TC

ANAS-HTP is under -- and reports to -- the Presidium of ANAS; therefore, its 5 Supervisory Board members are the President (Chairman of the Board), Vice President, Scientific Secretary, Chief of Administration from the Presidium of ANAS, and a Director of one ANAS Institute (currently, a scholar who was the former minister of education).

The 7 Council of Experts members are responsible for the evaluation of tenant proposals, technology assessment, and support for feasibility studies during TT/TC processes. The Council members currently include the Vice President of ANAS (head), a Director of one ANAS Institute (secretary), another Director of an ANAS Institute, and the representatives from the Ministry of Economy, SOCAR, State Copyright Agency, and Armed Forces Military Academy.

The Management of ANAS-HTP is led by its Director, who has two deputies (one is currently vacant, and the other will probably be, too). Along with administrative divisions (legal, HR, accounting, etc.), there are departments on the following:

- Coordination and monitoring,
- Engineering infrastructure,
- Organizational issues and security,
- Patent, certification, marketing, and business incubation, and
- ANAS Strategic Scientific Analytic Experimental Center.

The bureaucracy and organizational structure of ANAS-HTP are apparently not compatible with TT/TC management and are not consistent with modern standards. High employee turnover and lowered salaries do not attract talents, creating inconsistent institutional memory that negatively affects the ANAS-HTP brand in the market. Organizational units are not assigned to clearly defined roles. For example, the patent, certification, marketing, and business incubation departments do not accept tenants for business incubation because of the new Director's vision. Or, the long name of the ANAS Strategic Scientific Analytic Experimental Center represents some vital roles, but it is strongly associated with the defense products of ANAS-HTP.

There are no appropriate and skilled staff members for technology marketing and sales, financing, project management, etc.

All these lead ANAS-HTP's activities on TT/TC to become a supply-oriented, passive operations unit in TT/TC issues.

3.5 Benchmark Analysis

Benchmark analysis was conducted by the authors to get a clear view of ANAS-HTP in comparison with Chungnam Technopark. The differences between both technoparks are shown at the table below. In detail, despite the relatively similar TTO model and availability of incentives for startups and residents, the pivotal differences are cover mission and vision as well as innovative approaches and overall mindset.

[Table 2-3] Benchmark Analysis of ANAS-HTP and Chungnam Technopark

	ANAS-HTP	Chungnam Technopark
Date of establishment	• November 2016 by the President's order	• 1999, funded by the Ministry of Trade, Industry, and Energy and Chungcheongnam-do Provincial Government
Vision	• To ensure national science and industry cooperation and to establish a national innovation ecosystem and knowledge economy through the commercialization of science	• To be a global technical innovation leader for creating future value
Mission	• Encourage innovation, research, and commercialization of technology; • Reduce the dependence of the economy on oil and gas using scientific and innovative potential; • Formulate an environment that shapes science and technology-based thinking that can lead to innovative products; • Encourage and promote potential selling inventions prepared by residents; • Exchange scientific and technological innovation experience with different countries of the world	• Enhanced competence for future growth • Upgraded systems for industrial development • Advanced infrastructure and support systems • Improved management system
Infrastructure	• Territory - 25.6 ha • Modern equipped laboratory • Strategic Scientific Experimental Center	• Fully equipped Cheonan Valley • KOLAS-accredited testing laboratory • Infrastructure, production hall, and venture hall • Information & Media Convergence Center • Display Center • Automobile Center • Bio Center • Creative Business Center • Clinical Support Center, etc.
Incentives	• Residents are exempted from property tax, land tax, and profit tax for 7 years according to the amendments to the Azerbaijan Republic law. • Residents are free from VAD and customs duty for technique, technological equipment, and installations.	• Support incentives for residents • Networking opportunities • 3 years for the initial residence, extension available 2 times for 2 years each

	ANAS-HTP	Chungnam Technopark
Portfolio	• Scientific spheres: - Physical, Mathematical, and Technical Sciences - Chemical Sciences - Earth Sciences - Biological and Medical Sciences	• Covers: - ICT areas including IT and software - Bio industry, maritime bio materials - Mechanics - Etc.
Approaches	• Passive TT operations • Focus on research rather than commercialization as an existing culture inhibits cooperation	• Proactive TT operations (TT processes are available) • Willingness to assess and develop technologies further before licensing • Mind communication and collaboration • Customer-oriented thinking
IP Licensing	• TTO office model	• TTO office model
Mindset	• Traditional cooperation process • Supply-oriented • Partial support to individuals • Weak initiative culture & bureaucracy • High level of staff turnover • Low level of specialized workforce	• New cooperation process • Demand-oriented • Commercialization of R&D • Full-scale support • Strong initiative culture • Skilled and dedicated professionals

Source: https://www.ctp.or.kr/global_site/en/sub.asp?menu=0101

On this basis, there are some effective tactics that are focused on the improvement of the current mindset:

- Being open to change (Be aware of, analyze where this transforming world is heading and curiously keep track of change and new phenomena)
- Embracing creativity (Application of a certain kind of cultural activities by organizations that enables creativity – e.g., Valve Software* -- stressing freedom wherein every employees’ responsibility is to be creative in everyday work)
- Thinking and acting fast (Innovation within an organization must be a fast-moving process to keep up with the change going on outside of the organization)

- Creating easy pathways for innovation (Treating failures negatively will limit employees' willingness to be innovative, so such should be avoided at all costs)
- Expecting failures (If you espouse innovation but fail to act on any of the ideas presented to you, employees will get discouraged)

Likewise, without really focusing on educational background, it will be difficult to achieve actual changes in thoughts, vision, and skills of society on innovations, technology transfer, and commercialization. Lack of vision, less relevant and ill-substantiated strategies, and incompetent management impede the performance of STI parks in the country. Note, however, that rapidly growing trends in the neighboring world and globalization had clearly affected mindsets. In all tiers, understanding of STI issues penetrates the daily agenda, but the process is proceeding very slowly and vaguely. It is necessary to:

- Provide education on entrepreneurship from school to PhD (Denmark's experience in the dissemination of entrepreneurial culture from kindergarten to graduate schools, including establishing an online education platform for knowledge distribution).
- Train staff to specialize in industry and market analysis (increasing and dissemination of knowledge as well as consultancy).
- Learn the best practices of TT/TC (knowledge-sharing experience from successful foreign practices).
- Proactively engage in networking (global and local) (collaboration and networking with foreign specialists in the area of innovations, startups, TT, and TC).
- Apply technology marketing tools for strengthening the marketability of products and companies for expanding commercialization practices).

4. Analysis of Challenges and Recommendations

Considering the current local circumstances, i.e., political will, socio-economic environment, and need for economic diversification, as well as global trends in making the world a significantly technological place within the Fourth Industrial Revolution, Azerbaijan should change the mindset of policymakers, academe, youth, and businesses to learn how to survive in harshly competitive markets with genuine innovations. In this process, STI as a cornerstone has no other choice but get reformed immediately. In order for STI development to be achieved, Azerbaijan needs to adopt and effectively implement certain relevant policies on related measures. Some of these measures can be implemented simultaneously, whereas others need prerequisite and sequential policy formulation and applications.

Legislative reforms

- Competiton Law should be adopted and effectively implemented soon. No extra bureaucracy to control competition
- Laws regulating venture capital and innovations should be adopted by learning foreign (e.g.Korean) experience
- Courts on economic-business cases should be strengthened
- Property rights, especially IPRs should be respected and the laws must be effectively enforced

Governance

- A highest-level government authority should be assigned to manage and coordinate STI issues and foster innovation ecosystem (kind of Ministry/Agency for STI development), otherwise the governance becomes chaotic and mobbish
- ANAS itself should be reformed for industry demands with closer ties to businesses and universities
- STI and research community should adopt strategy against aging leadership

Access to Finance

- Joint funding initiative, e.g. Innovation Fund, as fund of funds, could be established with PPP framework to fund scienctific research and R&D, and guarantee risky investments on STI
- Angel investors, venture capitalists and private funds should get incentives in accordance to their funding to STI projects
- Corporate exit and IPO strategies should also be developed for long-term for innovative unicorns
- Most importantly, increasing R&D spending and science budget upto 2% of GDP in mid-term period

Human Capital Development

- Management of STI parks and academia should be trained to learn more about park management, IP commercialization, etc. from international success stories
- STEM education, entrepreneurship edication, study abroad (science and technology fields) schemes, participation in international STI projects should be encouraged
- Redesigning school and university curricula to improve project-based life-long learnings

- Considering the ongoing preparation of the "innovation strategy" of Azerbaijan, ANAS and ANAS-HTP should be actively involved in the innovation strategy development process to contribute to technology commercialization elements. Although it is not the only player, it can acquire competence and recognition in the innovation ecosystem.

- Considering the special and designated advantages, ANAS-HTP should position itself in the market as a corporate technology transfer entity with large-scale TTO projects. Since there is no business incubation, it can specifically concentrate on and specialize in TTO.

- Considering weaker organizational capabilities, ANAS-HTP should take measures on building effective and appropriate organizational structure with clearly defined roles for each subunit. Those units should be trained and specialized in particular industries. Those trained staff members should be "recognized star" experts in their respective industries.

  In order to have such staff, ANAS-HTP should empower its human resources. Instead of low salaries, it can propose "salary plus bonuses" based on TTO outcomes and KPIs.

  Staff members can also be trained within the STEPI cooperation.

- ANAS-HTP should invite more foreign partners. This can help ANAS-HTP learn and adopt those technologies and industry skills for later "reverse engineering." An effective tool is attracting FDI with modern technologies (e.g., Chinese, Korean, Turkish, etc.) to establish (or become) local representatives.

- ANAS itself should assist ANAS-HTP by delivering quality research with commercialization in mind. Otherwise, research programs and outputs with little commercialization prospects will not contribute to the TTO of ANAS-HTP. The Presidium of ANAS should develop strategy and programs for commercialization-oriented research from the short-, medium-, and long-term perspectives.

CHAPTER 3

Sharing Korea's Experiences in Technology Transfer and Commercialization Policy and System

Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan

Chapter 3. Sharing Korea's Experiences in Technology Transfer and Commercialization Policy and System

1. Technology Transfer Promotion Policy and System in Korea

1.1 Introduction

There is a saying in Korean folktales that "nothing is complete unless you put it in final shape." With technology, too, it means that, no matter how much technology you have, we have to commercialize it to contribute to our competitiveness.

Since the late 1990s, Korea has begun preparing measures to ensure that technologies developed by public research institutes, such as universities and research institutes, were transferred smoothly to the private sector so that they could commercialized. The following describes Korea's technology transfer promotion policies and efforts over the past 20 years:

The institutional framework for promoting technology transfer and commercialization of R&D performance includes the Framework Act on Science and Technology, Act on the Promotion of Technology Transfer and Commercialization, Act on Performance Evaluation and Performance Management of National R&D Projects, Act on the Promotion of Industrial Technology Innovation, Act on the Promotion of Research and Development Specialization, and Framework Act on Knowledge Property.

The most representative of these is the Act on the Promotion of Technology Transfer enacted in 2000. The Act aims to lay the foundation for promoting technology transfer and commercialization of public development technologies.

The Act defines technology in a wide range, from patents registered or applied under the Patent Act and other laws related to intellectual property such as utility, new design, layout design, and software of semiconductor integrated circuits, capital goods concentrated with knowledge, and information on them.

The term "technology transfer" means that technology is transferred from its holder (including a person who has the right to dispose of the technology) to another person through assignment, grant of a license, technical guidance, joint research, joint venture, merger and acquisition, etc.

Meanwhile, commercialization is defined as “the process by which products are developed, produced, or sold using technology, or the improvement of technology related to such process.”

The Act included almost all matters necessary to promote the transfer and commercialization of technology, including the establishment and implementation of plans for promoting technology transfer and commercialization, expansion of bases for technology transfer and commercialization, promotion of technology transfer and commercialization, establishment of a technology evaluation system, and financial support.

1.2 Major policies by period

1.2.1 Before 2015

The policy directions and implementation strategies of the plans that have been established by MOTIE (Ministry of Trade, Industry, and Energy) before 2015 are as follows. [Table 3- 1] presents the “Technology Transfer and Commercialization Promotion Plans” (1^{st} to 4^{th}) made and implemented during this period.

[Table 3-1] Technology Transfer and Commercialization Promotion Plans (1st to 4th)

Technology Transfer and Commercialization Promotion Plans
<u>1st ('01~'05)</u> Vision: Creating and Activating a Technology Transaction Market Strategy: 1) Support for the vitalization of the technology trading market and readjustment of the s ystem 2) Foundation for Promoting Technology Transaction and Commercialization
<u>2nd (‘08~‘10)</u> Vision: Infrastructure for technology transfer and business expansion. Strategies 1) Public promotion of commercial application and technology transfer 2) Scaling your technical evaluation systems and technology financing 3) Base broadening of international cooperation
<u>3rd ('09~'11)</u> Vision: Promoting technology-based global enterprises Strategy: 1) Exploration and management of technology support 2) Support system for the commercialization of all cycles 3) Support for technology financing 4) Support for entry into the global market
<u>4th ('12~'14)</u> Vision: Creating a virtuous-cycle ecosystem of technology and market Strategy: 1) Reinforcing the linkage activities between the technology and the market 2) Enhancing the competency of the people involved in technology commercialization 3) Promoting convergence and open innovation 4) Enhancing the infrastructure for market mechanism

In order to promote technology transfer in the public sector and promote technology transactions in the private sector, the government (MOTIE, Ministry of Trade, Industry, and Energy) has decided to introduce the Act on the Promotion of Technology Transfer (January 2000), and performance standards for the designation of technology traders (2000 - October) and subregulations of the Act were prepared, including guidelines for the designation of technology transaction institutions and evaluation institutions (December 2001).

By amending the Technology Transfer Promotion Act and the Patent Act, a technology transfer consultation organization was established as a corporation in national and public schools.
The system was improved to enable faculty members to own, manage, and utilize their job inventions.

In December 2006, the Act on the Promotion of Technology Transfer was wholly amended to the Act on the Promotion of Technology Transfer and Commercialization, and policy support for technology evaluation and finance was strengthened. In addition, the R&BD program was started in 2006 to support the establishment of enterprises (independent corporations) among the outstanding tasks that have gone through the planning process for commercialization.

In June 2009, a new growth engine investment fund was first created with the private-public partnership to promote private investment in future new growth engine industries such as IT, BT, and robots.

In 2010, the Act on Technology Transfer and Promotion of Commercialization was amended to provide the basis for fostering private commercialized companies specializing in technology transaction, evaluation, consulting, and commercialization. In addition, the Public Research Institute Hi-tech Holding System was introduced to commercialize the technology directly in order to provide new solutions for utilizing public technologies.

Meanwhile, an Intellectual Property Management Company with private-public partnership was set up in 2010 to enhance the added value of IP by promoting the use of Intellectual Property and handling disputes regarding IP. Since 2010, with the goal of fostering experts of technology commercialization, Korea has supported the establishment and operation of three Special Graduate Schools of MOT (Management of Technology) and five other universities of technology management curricula.

In 2012, the 4th Technology Transfer and Commercialization Promotion Plan was set up, aiming to enhance the core competency of the performing entity for technology business, to strengthen the linkage activities between technology and market, and to promote converging and open innovation. In particular, with the emergence of a market-oriented business model as the core of its commercialization policy, the R&BD project was expanded, and a comprehensive blueprint -- which included expanding support for the convergence and integration of technologies, promoting open innovation, and enhancing entrepreneurship -- was presented.

In 2013, policies were announced and implemented to create a “boom in Idea Commercialization” in the market by promoting the commercialization of “Business Idea.” By designating 10 support institutions for commercialization for 10 industries, the government provided support services for commercialization (correction of ideas, planning of business models, R&BD, etc.) to businesses that have only ideas and lack the ability to commercialize them.

In addition, the government started providing support measures such as product certification and priority purchase by public institutions for superior products produced through the development of business idea merchandising.

The Initial Commercialization Fund was launched in 2013 to help small businesses with superior technology overcome the Death Valley period. Fund for initial commercialization was established with 25 billion KRW in 2013, and the portion of government investment was 20 billion won.

In order to finance low-interest technology financing for small and medium-sized companies with excellent technologies and to support various consulting services, Industrial Bank and Woori Bank were designated as banks exclusively for the commercialization of R&D. These banks provide loan incentives and consulting to small and medium-sized enterprises that have succeeded in developing technologies by utilizing operating profits from the ministry's R&D contribution (approximately 3.3 trillion won as of 2013).

In April 2014, the government established and announced the 5th Technology Transfer and Commercialization Promotion Plan (2014-2017) to improve the system so that the technology transaction market can become self-sustaining and expand in scale.

Furthermore, the R&D system was reorganized from a commercialization perspective to provide the growth conditions for technology service businesses, such as strengthening the operating mechanism of the technology transaction market, enhancing the technology marketing capabilities of public research institutes, and providing customized technologies for technology commercialization.

As for the main contents, first, the government decided to introduce proper guidelines for technology brokerage fees, which are not being paid properly by public research institutes, in order to improve the technology transaction market system. Second, it decided to reform the technology bank (NTB) system to focus on the demand for new business creation based on technology information by increasing the amount and quality of technology information database so that companies can find the necessary technologies more easily. Third, the government decided to introduce a technology transfer and commercialization index to induce self-rescue efforts and measures for the development and transfer of transferable technologies.

In May 2014, "Activation Measures for Technology Finance to Create Industrial Engines" was announced to develop new industries and provide support for technology-intensive businesses, including creating funds, expanding technology-based loans, and expanding the infrastructure for promoting technology investment and loans.

1.2.2 From 2015 till 2016

The effects of the technology transfer commercialization policy that had been supported were visible. This is a time when the government's policy stance to keep pace with the fourth industrial revolution -- which is rapidly changing and converging -- was changed. The technology transfer rate of public research institutes increased by 31.2% (2013)→31.7% (2014)→38.6% (2015)→38.0% (2016), and the number of technologies transferred also grew to 7,495 (2013)→8,524 (2014)→11,614 (2015)→12,357 (2016).

[Table 3-2] 5th Technology Transfer and Commercialization Promotion Plan

5th Technology Transfer and Commercialization Promotion Plan ('15~'17) Vision: Creating an Ecosystem for Technology Transfer and Commercialization of Creative Economy Strategy: 1) Activating the operation of the technology trading market 2) Improving the technology marketing capability of public research institutes 3) Providing customized technologies that are highly likely to be commercialized 4) Preparing the growth environment of the enterprise in the early commercialization phase

The amount of technology transfer imports also increased to 135.3 billion won (2013), 140.3 billion won (2014), 204.2 billion won (2015), and 177.1 billion won (2016).
New investment in venture capital, which is a private investment for technological commercialization by small and medium enterprises, has expanded to 1.91 trillion won in 2010)→ 2.38 trillion won (2017) to surpass 2 trillion won.

This period promoted the 5th technical transfer promotion plan (2014-2016), established the 6th promotion plan (2017-2019), and implemented 12 key strategic tasks. To strengthen the working mechanism of the technology trading market, the National Tech-Bank (NTB), a technology transaction infrastructure, was improved, and the registration of the government's R&D performance with the NTB was required.

To enhance the activity of private technology trading institutions that aid in technology transfer, the government also supported both technology trading consultants and companies by carrying out "the project to support the commercialization and consulting services of new growth engines."

In 2015, the Demand Technology Platform, which allows companies to explore and match the demand technologies required, was established. By 2017, more than 4,000 cases of corporate technology demand were found, and consulting and technology adoption were supported.

In addition, an investment technology evaluation model was developed for use in investing in companies, providing the basis for shifting the technology financial ecosystem from a credit-based, loan-oriented one to a technology-based one and helping innovative small- and medium-sized companies receive 152.8 billion won in investment and loans between 2015 and 2017.

1.2.3 After 2017

In the 6th promotion plan (2017-20197), which is currently in effect, the government set up key tasks such as “Creating an Ecosystem to Promote Open Innovation.” The following are the strategies that take into account the views of the demand, supply, infra, and system.

- Demand: Expand the demand base for “Buy R&D”
- Supply: Supply the technologies that the demand companies desire (buyable R&D)
- Infra: Clear gaps between suppliers and customers
- System: Establish a pan-government-level cooperative system

To implement it, the Korean government has been pushing for 12 detailed tasks as follows:

[Table 3-3] Main Tasks of the 6th Technology Transfer and Commercialization Promotion Plan

[Task 1] Introduction of the Open Innovation B&D System
[Task 2] Expanding the Competent Technology Demand Group
[Task 3] Improvement of the Technical Transaction Promotion System
[Task 4] New Corporate Support Model for Technology Commercialization Projects
[Task 5] Reinforcement of Public R&D Marketability
[Task 6] Promotion of High Value-added Technology startups
[Task 7] Building a technology market platform that converges online and offline
[Task 8] Improvement of private-centered technology transaction promotion system
[Task 9] Support to overcome the Death Valley of technology-based businesses
[Task 10] Improving the mindset toward Buy R&D
[Task 11] Strengthening the collaborative governance for technology commercialization
[Task 12] Establishment of a constant system for the discovery of regulations and difficulties of technology commercialization

The 6th Technology Transfer Promotion Plan (2017-2019) aimed to provide support for the companies' adoption of external R&D (Buy R&D), and it has been pushing for and supporting detailed tasks aggressively since 2017.

First, to expand the demand base for B&D (Buy and Develop)-oriented R&D, the government introduced at the end of 2016 an open-innovation "B&D" system that increases the productivity of public R&D through the use of external technologies, and it plans to activate it in the future. In addition, the government will expand the basis of finding demand through the introduction of coaching systems for small and medium-sized companies that introduce technologies and vitalize technology exports.

Second, the government expanded support for private sector-led innovation models for commercialization, which are more suitable for commercialization compared to existing R&D systems, and strengthen the marketability of public R&D, including improvement of systems for expanding the market use of research results, expansion of technology in the product portfolio, and improvement of types of implementation of public R&D performance.

Third, the online technology transaction information network (NTB) was to be reorganized from simple technology DB service to a platform linking online and offline. In addition, the technology brokerage fees and technology transfer contributor compensation system have been improved and settled in order to improve the profit structure of private technology trading consulting companies and make them self-reliant. To reduce the risk of failure of a company with insufficient funds in the Death Valley section, the government planned to increase the size of its policy funds and the reliability of technology evaluation.

Fourth, the government strengthens the cooperative system of each ministry in the field of technology commercialization. In particular, the government plans to form a consultative meeting on technology commercialization, which will discuss ways of establishing major policies related to technology transfer and commercialization, and to coordinate and support projects in order to help Korean companies transfer and commercialize the technology effectively and continuously.

So far, we have looked at the policies of the past 20 years in Korea. We could tell that, thanks to the Korean government's policy efforts, Korea has set up the framework of the support system for the infrastructure of the technology trading market and innovative M&As.

As a result, the transfer performance of public technology has been improved. Still, the culture of cross-enterprise technology transactions has not been activated. The companies still prefer the closed self-development of technology.

At this point, we are going to move on to the output and outcome of technology transfer and commercialization of public R&D organization in Korea.

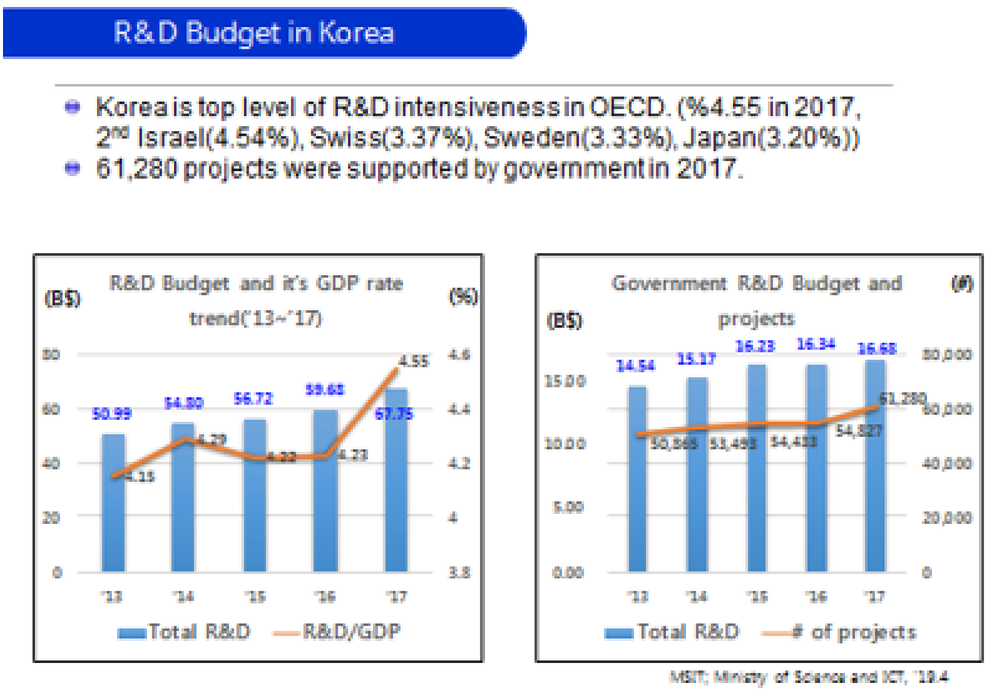
2. Technology Transfer and Commercialization of Public R&D Organization in Korea: Status and Limitations

2.1 Current Status of R&D Investment in Korea

2.1.1 Proportion of R&D investment to GDP

The proportion of R&D investment to GDP in 2017 was 4.55%, a 0.22% increase from the previous year's 4.23% and the highest in the world. The next highest was Israel (4.59%), followed by Switzerland (3.20%) and Sweden (3.33%t).

[Figure 3-1] Proportion of R&D Expenses to Total R&D and GDP in Korea



Source: Ministry of Science, Technology, Information, and Communication press release (2019)

Annual R&D investment is 67.75 billion dollars, the fifth largest in the world next to the United States, China, Japan, and Germany.

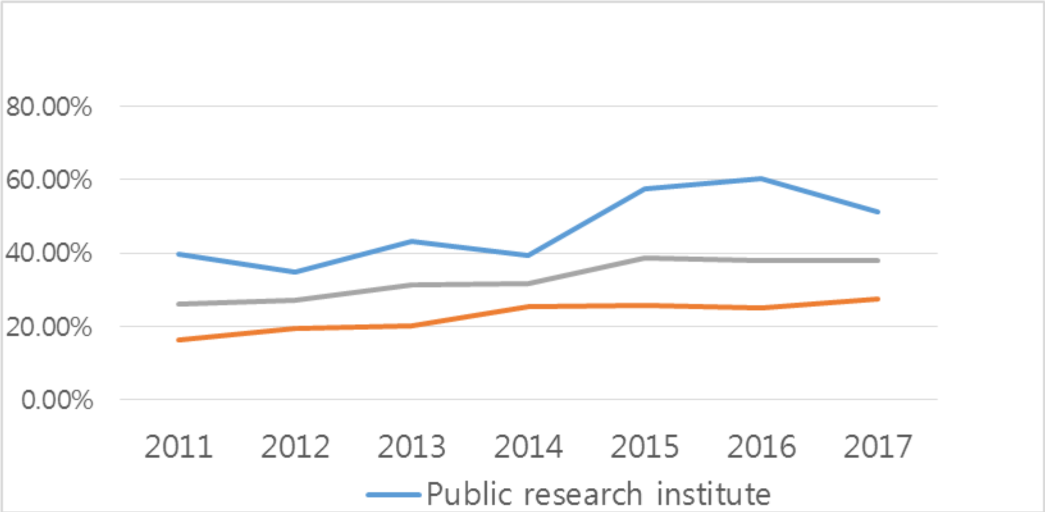
When looking at the research and development costs by resource, private funds amounted to 76.3% or 51.07 trillion won, with the ratio of private sector dependence. Government and public funds accounted for 24.6%, or 16.68 billion dollars, whereas foreign funds accounted for only 1.0%.

2.1.2 Technology Transfer Rate

Looking at the trends for each indicator through data, the technology transfer rate of all public research institutes, including universities, was 38.6% in 2015, up 17.8% from 20.8% in 2004. The technology transfer rate in both public research institutes and universities went up to 60.3% in 2016 but dropped slightly in 2017.

If we look at the trend, the figure has been on a steady rise since 2008 when it increased from 20.8% in 2004 to 27.4% in 2007. The technology transfer rate in public research institutes increased to 42.4% in 2007 but dropped to 29.7% in 2008 before rising again. In particular, it rose 46.3% year-on-year to a record high of 57.5% in 2015. Throughout the entire analysis period, the technology transfer rate of public research institutes was higher than that of universities. The technology transfer rate of universities increased from 10.1 dollars in 2004 to 17.2% in 2006 and then declined; since 2010, however, the rate has shifted to 13.8% and increased to 25.8% in 2015.

[Figure 3-2] Trends in Technology Transfer Rates of Public Research Institutes (2004-2017)



		Public research institute	University	Total
Technology transfer rate by year	2017	51.0%	27.4%	37.9%
	2016	60.3%	25.0%	38.0%
	2015	57.5%	25.8%	38.6%
	2014	39.3%	25.4%	31.7%
	2013	43.0%	20.3%	31.2%
	2012	34.9%	19.5%	27.1%
	2011	39.6%	16.4%	26.0%

Source: Ministry of Trade, Industry, and Energy (2018), A Survey Report on Technology Transfer and Commercialization

As for the overall technology transfer rate of public research institutes, the gap between the US and Korea is narrowing due to the continuous increase in Korea's performance, but the gap in efficiency of technology transfer is still wide.

2.2 Technology Royalty Income & Technology Transfer Efficiency

2.2.1 Technology Royalty Income

Over the period '07 -'15, the number of technology transfers by public research institutes increased three-fold, and technology fees increased four-fold. In particular, university and public research institutes' revenue from technology royalty jumped 45.5% in 2007 to 204.2 billion won in 2015. Among them, imports of technology fees by public research institutes rose 58.6% year-on-year, a sharp improvement in public research institutes' 2015 commercialization performance in contrast to a drop in technology fee imports over the cited period in 2014.

In 2016, the technology royalty considerably decreased in public research institutes but increased back in 2017. Meanwhile, the figure in the university has been on a steady rise since 2007.

[Table 3-4] Trends in the Import and Technical Loyalty of Public Research Institutes

	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
대학	150	265	277	327	426	482	369	505	618	683	700
공공연구소	893	1,023	740	918	832	1,170	985	898	1,424	1.088	1.127
전체	1,044	1,288	1,017	1,245	1,258	1,651	1,354	1,403	2,042	1,771	1.827

(Unit: KRW 100 million)

Source: Ministry of Trade, Industry, and Energy (2018), A Survey Report on Technology Transfer and Commercialization

2.2.2 Technology Transfer Efficiency

We can see in the table of trends in technology transfer efficiency (formerly research productivity) of public research institutes that the efficiency of technology transfer among all public research institutes decreased slightly in 2011 to 1.32% after increasing from 2008 to 2010; then in 2015, the ratio increased 27.4% to 1.72%. Throughout the entire analysis period, technology transfer efficiency was also higher in public research institutes than in universities. The efficiency of technology transfer in public research institutes increased from 2008 to 2010 but declined to 1.69% in 2011. Afterward, the ratio increased 39.4% year-on-year to 2.08% in 2015. These trend of up-down appeared from 2016 to 2017. Technology transfer efficiency in large universities increased from 2008 to 2012. The figure dropped slightly in 2013 to 0.94%, and then rebounded to 1.23% in 2015, remaining about the same from 2016 to 2017.

[Table 3-5] Trends in Technology Transfer Efficiency (formerly research productivity) of Public Research Institutes (2008-2017)

	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
대학	0.6	0.83	0.85	0.92	1.05	0.94	1.16	1.23	1.24	1.09
공공연구소	1.9	1.84	2.02	1.69	1.75	1.63	1.49	2.08	1.54	1.37
전체	1.3	1.38	1.48	1.32	1.47	1.36	1.35	1.72	1.24	1.09

Source: Ministry of Trade, Industry, and Energy (2018), A Survey Report on Technology Transfer and Commercialization

As shown above, the latest technology transfer rates and efficiency of technology transfer among public research institutes in Korea are gradually improving, especially compared to the US, increasing significantly in 2015 compared to the previous year.

2.3 Obstacles to Technology Transfer and Commercialization: from the Viewpoint of Technology Providers

From the technology supplier's perspective, the main obstacle to technology transfer and commercialization of public research institutes is the lack of internal and external resources (labor, budget, etc.) for technology transfer and commercialization activities. Items with slightly lower response rate include:

- Difficulty in finding entrepreneurs or enterprise who/that will adopt or commercialize the technology and their business feasibility

- Lack of highly business-worthy technology
- Low understanding or commitment of researchers' technology transfer and commercialization activities, etc.

The following is the response ratio for each item:

1. Lack of internal and external resources (labor, budget, etc.) available for technology transfer and commercialization activities: 19.7%
2. Difficulty in finding companies and entrepreneurs to introduce or commercialize technology: 18.0%
3. Lack of highly business-worthy technology: 15.6%
4. Low participation in technology transfer and commercialization activities of researchers (technology developers): 12.2%
5. Customer’s low (negative) understanding of business feasibility and value of public research organizations’ technologies: 10.3%
6. Lack of system or institutional-level incentives to facilitate technology transfer and commercialization activities: 10.0%.
7. Lack of government-level institutional incentives to promote technology transfer and commercialization activities of individual institutions: 7.6%
8. Difficulty in managing performance after technology transfer and commercialization: 4.3%
9. Existence of restrictions, such as restrictions on the transfer of performance of state R&D projects: 2.2%

Source: Ministry of Trade, Industry, and Energy (2018), A Survey Report on Technology Transfer and Commercialization

So far, we have examined the current status of R&D investment, royalty income & technology transfer efficiency and obstacles in Korea. In the next chapter, a new paradigm of technology transfer will be presented to improve the outcome.

3. Suggestions of a New Paradigm of Technology Transfer and Commercialization

3.1 Limitations of Traditional Technology Transfer and Rise of Technology Marketing

A new paradigm of technology transfer will be introduced to overcome the barriers in the technology transfer process. In this regard, all the major elements -- from the definition & characteristics of technology marketing, examination & selection of technologies to sell strategy and implementation of technology marketing to prospecting and contacting of the customers -- should normally be addressed.

Due to the limitation of space, however, the presentation in this paper will be confined to the definition & characteristics and composition to be addressed. Only these will be helpful to change the perception of the people in charge of technology transfer.

The limitations of technology transfer performance in Korea have been described. It is now time to look into the problems from a more systematic perspective. These problems could be grouped into 8 stages in the process of technology transfer, although there might be a lot of reasons that limit the performance, from lack of pre-evaluation of technology to low competitiveness.

3.1.1 Limitations of Traditional Technology Transfer

The abovementioned stages can be summarized into three big steps such as (1) insufficient mindset toward technology performance, (2) lack of recognition about technology marketing as well as the technology process, and (3) weak commercialization. The following are the typical problems or weak points at each stage:

The first problem is that there was not enough technology evaluation prior to technology development. The assessment of the technology of this stage requires real emphasis on marketability, not only on technicality. The excessive emphasis on the evaluation of technology results in a wide gap between the R&D results and the market in the process of technology commercialization. In this regard, it is worth referring to ETRI's process including the survey of pre-technology demand for small business, which leads to excellent technology transfer performance.

The second problem is the lack of suitable technology from the consumer's perspective, which is often attributed to the lack of sufficiently proven marketability in the preceding step. Public technology refers to the technology whose introduction is desired from the perspective of the consumer, as it has undergone an evaluation and selection process consisting mostly of technical experts.

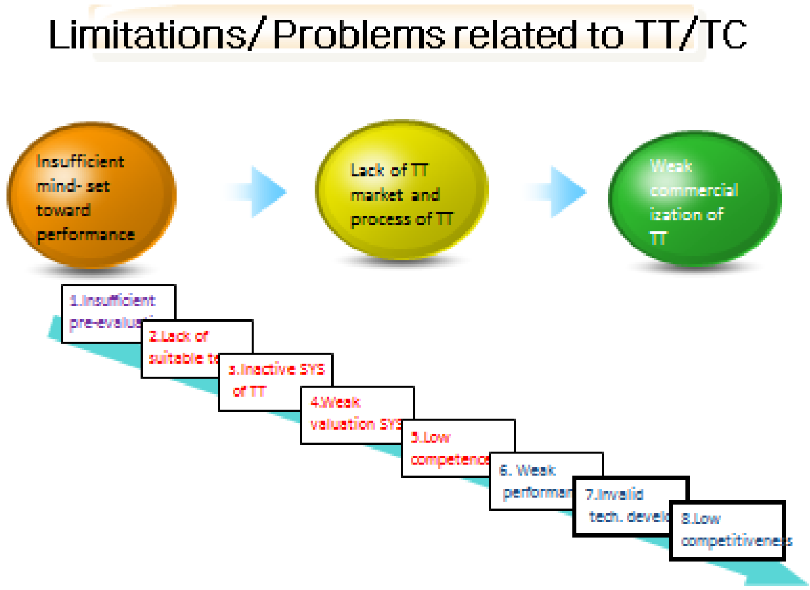
Third is an inefficient technology transfer system. technology transfer systems refer to a comprehensive group of people and activities involved in technology transfer, such as technology provision, technology adoption, technology transfer support policies and institutions, etc. The third problem refers to the state wherein these systems are not functioning properly.

Fourth is the lack of valuation. Technology valuation is an indispensable condition for the transfer of technology. Just as it would be difficult for both parties to negotiate on an apartment to be traded if they do not know its proper value, the trade of the technology will be difficult if the two sides have too different views on the proper value of the technology to be traded.

Fifth is poor competitiveness. This is particularly true in the case of developing countries, where either the technology provider, the technology adopters, or the technology itself is not competitive enough or where the support agencies are less efficient.

The sixth and the next two steps relate to the technology transfer performance and the resulting lack of competitiveness. These are due to the preceding steps and are an inevitable phenomenon caused by the failure of various conditions to establish the necessary infrastructure or ecosystem in order to achieve direct and indirect outcome or impact, such as strengthening the external competitiveness of a single institution or economic development and job creation in a country.

[Figure 3-3] Problems in the Process of Technology Transfer



3.1.2 New paradigm of TT/TC: Technology marketing

Technology transfer can be largely divided into three generations. If we look at the basic philosophy, first is technology diffusion, second is the technology channel, and third is technology marketing. The following is a detailed explanation of each generation:

3.1.2.1 1st Generation

The premise of the first generation is that the technologies will be sold on their own if their quality is good enough. Thus, the technology transfer agency's efforts are very limited and passive, and, with the exception of very few technologies, technology transfer performance is weak.

The technology flow is one-way, and the main medium of technology transfer is seminars, etc. The technology market at this stage consists of a small number of innovators. Thus, the main role of this generation is only a handful of innovators.

3.1.2.2 2nd Generation

The basic concept of this second generation is matchmaking. Thus, technology transfer is deemed to take place through matchmaking between the technology supply and demand.

The government policy is to establish technology transfer networking, such as technology transfer channel, Techno-mart, etc.

Technology transfer performance is relatively poor as most technologies are not ready to be applied by the companies. The technology flow is mostly bilateral, and the main medium of technology transfer is of course matchmaking. The technology market in this generation is considered to grow according to the matchmaking competence. Thus, the major player of this generation is the matchmakers.

[Table 3-6] New Paradigm of Technology Transfer

	1st Generation	2nd Generation	3th Generation
Basic concept	Tech diffusion	Match-making	Tech. Marketing, Tech commercialization
Basic assumtion	There are sufficient demand for good tech.	TT takes place by a match-making between the tech. supply and demand	A proactive effort to search the customers of the tech is important
Government policy	Any special policy is not needed	Establish of TT networking, such as TT channel, Techno-mart..is required	A new tech commercialization program is needed
Performance	Not works well except few exellent tech.	Most tech. are not ready to be applied by the companies	TT/TC is accelerated by Tech. MKT and commercialization efforts
Characteristics of TT sys.	Institutional (role of tech suppliers is limited, such as to IPR management)	Bureaucratic(If there is a demand, the supplier responds to it)	Marketing(evaluate the tech., actively search for the customer)
Direction of knowledge flow	one-way	Mostly bilateral	Multilateral
Channel	Seminar, journal, internet..	Match-making	Customized
Supplier of tech.	passive	-	Acttive
Market of tech.	TT/TC realized by the leading innovative enterprises	Market can grow by the match-making competency	Market can grow by the activities of Marketing
Major player	Leading innovative enterprises	Match-makers	Tech. marketer

3.1.2.3 3rd Generation

The basic concept of this generation involves technology marketing and commercialization. Thus, proactive effort to search the customers of the tech is important in this generation. When it comes to government policy, a new tech commercialization program is needed.

Technology transfer performance is accelerated by technology marketing and commercialization efforts. The characteristics of the system include marketing (i.e., evaluate the technology, actively search for the customer).

The technology flow is mostly multilateral, and the main medium of technology transfer is customized. The technology market in this generation is considered to grow according to the marketing activities. Thus, the major player of this generation is the technology marketers.

3.2 Then, Why Technology Marketing?

Then, why should we learn about technology marketing? That is because it includes the principles and practices for facilitating technology transfer in an effective way by shortening the distance between the research lab and market with the help of market-oriented R&D and transfer activity.

Technology marketing helps effectively find customers and enhance the marketability of technology. It also modifies and/or complements the technology by the demand and develops the appeal through additional R&D for the immature technology to meet the demand.

With the help of STP (segmentation, targeting, and positioning) of technology marketing, the marketer can identify the best use of technology and determine the value in the market segment. It develops the concept, positioning, and search for a new market through a convergence of technologies.

Technology marketing strengthens the power of persuasion in the potential market through advertising, publicity, etc.

Technology marketing analysis means to define all possible applications of the technology and determine the total market that the technology addresses. It also assesses the penetration volume and rate within each application.

In addition, technology marketing assesses the probability of success in each of the markets or applications and prioritizes the highest return with the highest probability of success while eliminating the lowest return with the lowest probability of success.

Why is it possible? The answer is simple. Because it is the function of marketing!

3.3 Definition and Characteristics of Technology Marketing 1)

In the previous chapter, the background of emergence of technology marketing was introduced. In this chapter, the concept and characteristics of technology marketing will be presented. Due to space limitations, the presentation will be limited only to the concept and contents of technology marketing.

3.3.1 Concept of Technology Marketing[97]

If marketing is regarded as an act of exchange, technology is the subject of exchange in technology marketing. Nowadays, together with the term technology management, technology marketing is also more frequently becoming the subject of discussion more widely.

In technology marketing, there are many cases wherein the involvement of transfer parties and the level of risk are not commonplace due to the characteristics, complexity, and intangibility of the technology itself. As a result, the general marketing principles are

[97] Most of the contents of this chapter are extracted from the following books: Sang Hyuk Suh (2004), "Guidebook of Technology Marketing" Bringing New Technology to Market," APCTT (Asia Pacific Center of Technology Transfer), a public organization under the umbrella of the UN.

not applied well to technology. Therefore, a separate research domain of technology marketing has been established.

Like all other faculties, the concept of the term "technology marketing" is diverse depending on the scholar or user. Note, however, NTTC (National Technology Transfer Center) defines technology marketing as "the effort to transfer & transact technology efficiently." The research results and literature related to technology marketing are still scarce, and there is a tendency to define the concept very narrowly. As a matter of fact, technology marketing is regarded as technology transfer, technology sales, etc.; often, there are cases wherein it partly overlaps with the commercialization of technology.

The technology marketing concept in this text sees technology as a product and defines it as the various efforts for its efficient and effective marketing. Originally, marketing is widely known to have the following concept structure (American Marketing Association, 1985):

Marketing is a process involving the planning/ execution of conceptualization/pricing/distribution/promotion of products/services/ideas for the accomplishment of individual/organization goals so that exchange can take place.

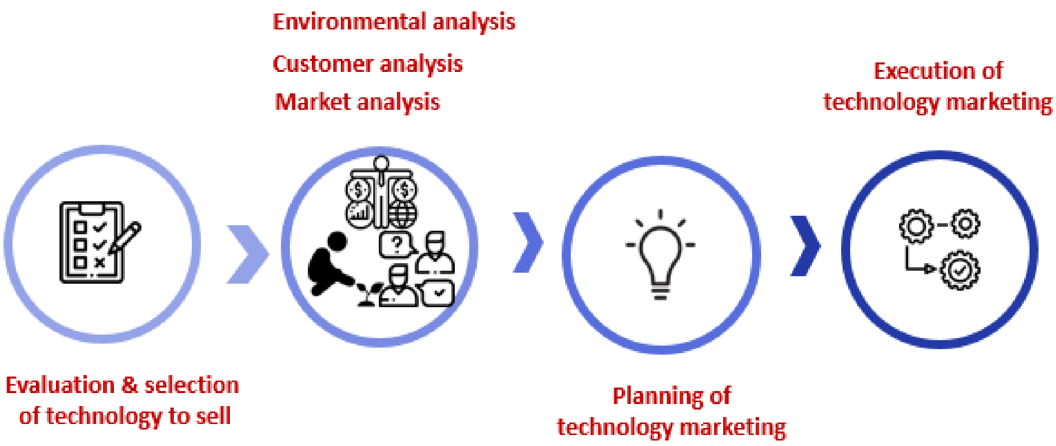
If this is applied to technology marketing, the subjects of exchange such as "products, service, ideas" etc. are compressed into technology. Thus, technology marketing can be defined as follows:

A Process involving the planning & execution of conceptualization, pricing, distribution, and promotion of technology so that exchange activity can take place

In this text, the technology devising process -- that is, the activities involving demand analysis, technology forecast, new technology prospects, etc. for the development of the technology -- is omitted due to lack of space.

Thus, the main contents of this text can be summarized as follows:

[Figure 3-4] Main Contents of Technology Marketing

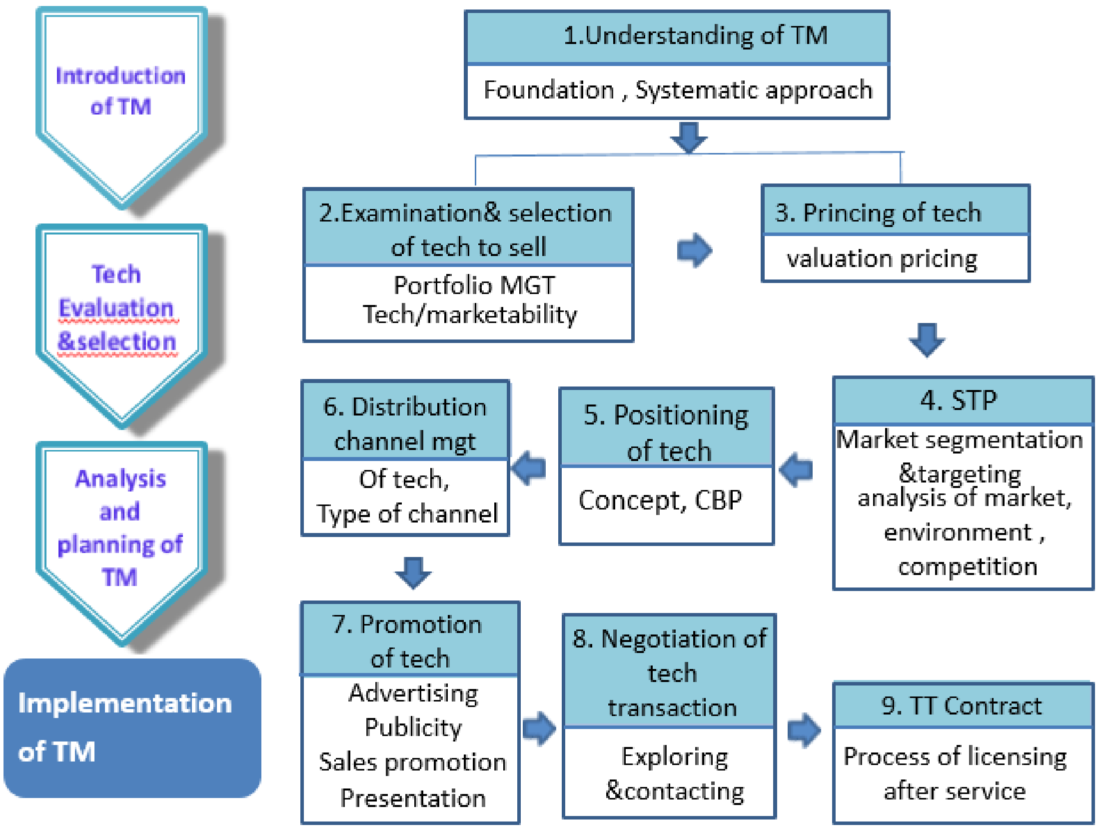


The technology marketing research contents can be further specified as follows. The next figure describes the contents to be addressed to understand the theory and practice of technology marketing. The first stage relates the basic understanding of technology marketing, which covers the foundation and systematic approach.

The second and third stages deal with the evaluation and selection of technology to sell. The next stage addresses the strategy of technology marketing, which mainly addresses STP and positioning.

The final stage describes the implementation of technology marketing, which includes promotion, negotiation, and contract.

[Figure 3-5] Detailed Contents of Technology Marketing



3.3.2 Characteristics of Technology Marketing[98]

With marketing activities targeting technology, the first priority in terms of establishing and executing the marketing strategy is to examine what characteristics it has compared to general marketing. To do this, the features of technology will be examined first.

[98] Sang-Hyuk Suh, "A study on the actual situation & promotion plan for technology utilization by advanced venture enterprises," Science & Technology Policy Institute

3.3.2.1 Characteristics of the Technology as the Transfer Subject

Technologies subject to transfer are mostly new technologies and advanced technologies. The characteristics of this kind of technology include ingenuity, complexity, innovativeness, non-standardization, etc.

- Uncertainty
- Innovativeness
- Reduced development period
- Short life cycle
- Somplexity

- Uncertainty: Technology cannot be touched or tapped, and its usefulness is difficult to predict precisely. From the viewpoint of the technology purchaser, whether the technology can be properly utilized cannot be known.
- Ingenuity: As there are many advanced science research results, it is ingenious, complex, and in some cases still being developed.
- Innovativeness: There are cases wherein technology or products, which are the results of technology, drive out existing technology or other products from the market. The emergence of new technology speeds up the decline of the existing technology; in cases wherein the scope is wide, it can suddenly transform the condition of a particular industry.
- Reduced development period & short life cycle: The speeding up of development and shortened life cycle of today's technology is the general trend. New technology is quickly developed and replaced.
- Complexity: In accordance with the fusion with other technologies and advancement, technology development has high complexity with diverse technological factors and technology solutions in relation to buyers.

3.3.2.2 Characteristics of Technology Marketing

In accordance with the technology characteristics above, technology marketing also has characteristics that are different from marketing in other areas as stated below.

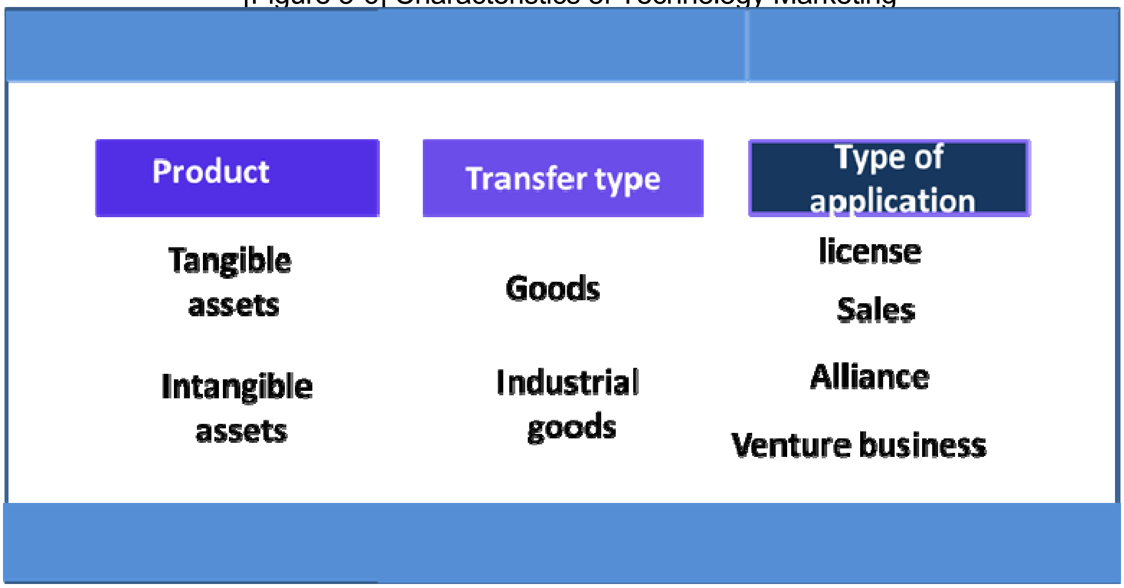
- High risk
- Long transfer period & procedural complexity
- Importance of time management
- Importance of opinion adjustments between transfer parties

- High risk: The burden of high risk is felt by the decision maker in relation to technology marketing. With technology itself being intangible, the risk becomes greater the more complex and elaborate the technology is and the shorter the life cycle.
- Long transfer period & procedural complexity: Technology transfer is more complex than the transfer procedure of general products, and it often takes a long time.
- Importance of time management: With the shortening of technology life cycles and rapid increase in development speed, efficient time management is more important. This means that, if there is hesitation after technology development, it can soon be weeded out from the market.
- Importance of opinion adjustments between transfer parties: technology buyers and sellers are all composed of many sections, and there are many cases wherein their attitudes differ. Differences in opinion often occur between the research personnel and the marketing manager within the selling organization and between the high-level management and the technology director within the purchasing organization. This is an obstacle in technology transfer and transactions.

The characteristics of technology and technology marketing have been explained above. As technology is an intangible asset, it is closer to a service than a product, with the transfer type more industrial rather than consumable. Thus, the application of a marketing strategy for services and industrial property would be most favorable.

On the other hand, however, there is a need to devise an advanced venture product marketing technique in accordance with the complex, high-risk, innovative characteristics of the technology. Finally, the technology can be simply transferred but can also be utilized through licensing, strategic alliance, venture enterprise, etc.

[Figure 3-6] Characteristics of Technology Marketing



4. Technology Entrepreneurship, Startup, and Incubation

4.1 Technology Entrepreneurship

4.1.1 Definition and Significance

In today's turbulent business environment, companies often seek to develop entrepreneurial thinking. At the same time, universities and nations around the world are increasing entrepreneurial training across a wide array of majors.

Large, mature, and capital- and technology-intensive organizations struggle to keep up with the global competition, changing demographics, and rapid technology advancements. Even more challenging is the ever-increasing complexity of business models, partnerships, and value propositions as well as a host of other business innovations. Organizational members that are not accustomed to these dynamic developments find it difficult to assimilate this bewildering array of issues to find new competitive directions. These drivers are addressed in entrepreneurial thinking and education.

Entrepreneurial thinking combines the existing resources and methods with new developments and creates new competitive advantages. By blending assets and capabilities as well as the installed base with market and technology trends and competitive landscape and profound insights into customer needs, entrepreneurial thinking generates innovative solutions to the most pressing challenges faced today. Therefore, to get people to think entrepreneurially, they need to be taught how entrepreneurial instruction takes place, and this should be of great importance to large companies. Few students start new companies right out of college, but they and the companies they eventually work for greatly benefit from entrepreneurial thinking. Universities offer a wide array of entrepreneurial programs that range from art entrepreneurship to non-profit companies as well as family-owned businesses. These are all worthy approaches to entrepreneurial education but do not prepare the people for entrepreneurial activities in large companies.

Entrepreneurial education in the United States and Korea is growing rapidly in a variety of directions. For large companies to benefit from these innovative concepts, the entrepreneurial education models must include understanding the industry challenges and opportunities, working in highly constrained environments, and working across organizational boundaries as well as traditional entrepreneurship topics such as early-stage market research, business plan writing, and development and production of winning products. Ideally, the industry should be deeply involved with university education to help educators understand the industry needs to prepare corporate entrepreneurs (Choi and Markham, 2019).

4.1.2 Entrepreneurship Education Plan

Entrepreneurship education — the teaching of skills and cultivation of talents that students need to start businesses, identify opportunities, manage risk, and innovate in the course of their careers — is now a staple in American higher education (Kauffman, 2013). The current entrepreneurship education pedagogy appears to be quite diverse and eclectic. Entrepreneurship education initiatives employ a traditional small business management approach and a more recent high-growth venture creation approach. Although lecture and case study approaches are very much evident, there seems to be a growing consensus that technology entrepreneurial education has different content, and that the pedagogical focus of experiential learning via games, simulations, or even actual venture creation may improve learning outcomes (Forest and Peterson, 2006; Neck and Greene, 2011; Rideout and Gray, 2013).

At the core of the entrepreneurship education curriculum, one finds courses essential to the starting and running of a new business. For example:

- Introduction to entrepreneurship
- Developing a business plan
- Financing a small business

- Legal environments for new businesses
- Cases in small business
- Managing fast-growing companies

[Table 3-7] University Entrepreneurship Education Programs

University entrepreneurial education	Program/Contents	Authors
New Venture, Startup	Small business management and high-growth venture creation Mindset	Rideout and Gray (2013), KCI (2013)
Family business		Zhang, et al (2013)
Small business	Financing a small business, cases in small business	Gordon, et al (2012), Kauffman (2013)
Ranking index	Availability of internships, externships, experiential and cooperative learning, and consulting opportunities for small business owners Companies started by graduates over the last five and ten years Annual business plan or new venture competition	princetonreview.com

Source: Choi and Markham (2019)

There are now many entrepreneurship programs around the world. The Princeton Review surveyed more than 300 undergraduate and graduate business schools on their entrepreneurship offerings and ranked them according to three criteria: First is the availability of internships, externships, experiential and cooperative learning, and consulting opportunities with small businesses; Second, the number of students enrolled in their entrepreneurship offerings and the number of companies started by graduates over the last five and ten years, including how many of those companies are still in business; Third, whether they have partnerships with other schools that allow access to their entrepreneurship programs. Universities host numerous business plan or new venture competitions, hackathon, pitch deck, startup weekend, etc. The Princeton index and these activities are focused on startup and venture creation and are exclusively oriented toward small business, not large companies.

In trying to broaden the definition of entrepreneurship, the Ewing Marion Kauffman Foundation launched the Kauffman Campuses Initiative (KCI) in December 2003 to encourage new, interdisciplinary entrepreneurship education programs throughout American colleges and universities. The Foundation sought to make entrepreneurship a campus-wide experience to help schools become more entrepreneurial and to ensure that thousands of students at diverse campuses would begin to see their own knowledge and resources from a more entrepreneurial perspective (Torrance, 2013).

[Table 3-8] Key Success Factors of the Kauffman Campus Initiative (KCI)

Type of Activity	Number of Universities
Curriculum Development	8
New or revised programs of study for non-business students	6
New courses in entrepreneurship	8
Modified courses in entrepreneurship	6
Course development grants for faculty	6
Co-curricular Activities	8
Entrepreneurship clubs	8
Lectures and speakers	7
Workshops and seminars	6
Business plan competitions	7
Internships	6
Venture incubators	5
Research Opportunities	7
Grants for faculty research in entrepreneurship	5
Grants for student research in entrepreneurship	4
Research conferences and visiting scholars	5
Databases of entrepreneurship literature	2
Other Types of Faculty Involvement	8
Advocating for entrepreneurship in school	6
Leading program components	6
Serving on entrepreneurship committees or boards	5
Participating in on-site faculty development training workshops or meetings	5
Attending outside training in entrepreneurship	6
Community Involvement	8
Serving on boards	5
Raising funds	7
Teaching courses	8
Speaking	7
Judging competitions	6
Providing internships	6
Serving as entrepreneur-in-residence	3
Attending open events	7
Receiving training, education, and consulting services	5

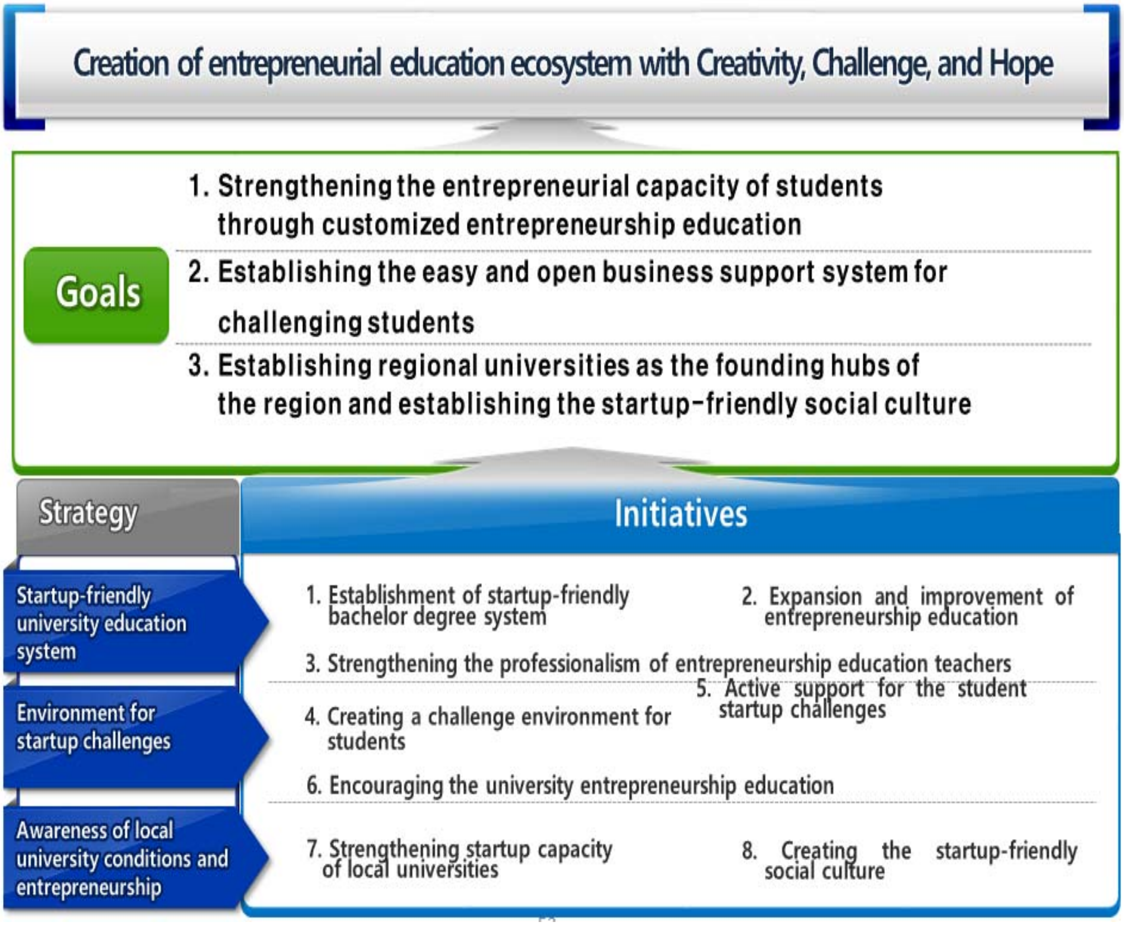
Source: Lara Hulsey, et al (2006)

For instance, Babson College is one of the first academic institutions in the world to offer a course in entrepreneurship. Since then, Babson has been internationally recognized as the leader in entrepreneurship education. "Entrepreneurship isn't just an academic discipline at Babson; it is a way of life. Our faculty and staff recognize the interdisciplinary value of entrepreneurship and weave it throughout our curricular and co-curricular programs. The skills learned in our entrepreneurship classes are vital for the success of any business -- large or small, public or private, corporate or not-for-profit, local or global," they said. Nevertheless, Babson is generally focused on new venture creation and problem solving in the community (Choi and Markham, 2019).

4.1.3 Korean graduate school of entrepreneurship

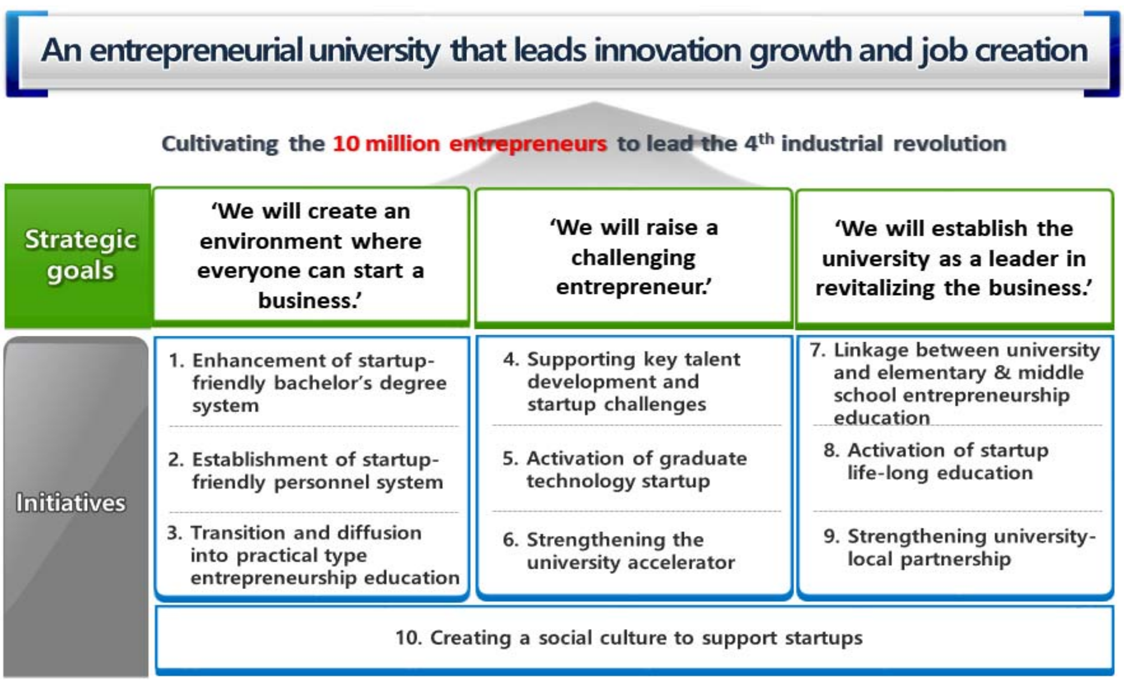
In Korea, a graduate school of entrepreneurship has been run since 2004 through government-level support to provide systematic, professional entrepreneurship education. The purpose of the professional graduate school is to nurture startup experts who can adapt well to uncertain circumstances. In order to achieve this goal, the schools have provided practical training programs for each startup stage to differentiate them from programs in existing business schools. The background and purpose of the students who enroll in the entrepreneurship graduate school vary. For example, students go on to entrepreneurship graduate school for reasons such as the use of information and infrastructure for incubation, construction of a human network, preparation for startup, and acquisition of a degree. Accordingly, formal education programs that do not consider students' entrepreneurship motivation are ineffective. On the other hand, well-designed curricula and programs that take into account the motivations and goals of students can enhance the effectiveness of entrepreneurial education. Entrepreneurship curriculum and programs are highly evaluated in terms of encouraging entrepreneurial attitudes and awareness of students, increasing intent to start a business. Thus, in many studies on entrepreneurship education, the effectiveness of entrepreneurship education is often measured by the degree of entrepreneurship intention. In addition, in order to narrow the gap between educators and students with regard to entrepreneurship education, it is necessary to analyze the importance of and satisfaction with entrepreneurship curriculum and programs as perceived by students. This generally allows students to improve their knowledge and skills through well-established entrepreneurship curriculum and experience. Note, however, that most entrepreneurship education research focuses on program design and implementation, and objective evaluation practices by experts and students are still poorly addressed in this field. Moreover, there is lack of evaluation on the effectiveness of entrepreneurship education in higher education such as graduate-level courses and analysis of its importance, and satisfaction is insufficient (Byun, Chung-Gyu, et al, 2019).

[Figure 3-7] First University Entrepreneurship Education Plan in Korea (2013-2018)



[Figure 3-8] Background of the Second University Entrepreneurship Education Plan in Korea (2018-2023)

[Figure 3-9] Second University Entrepreneurship Education Plan in Korea (2018-2023)



[Figure 3-10] Second University Entrepreneurship Education Plan in Korea (2018-2023): 10 Initiatives

Initiative 1 | **Enhancement of startup-friendly bachelor' s degree system**

① Entrepreneurship Entrance ② Establishment of a minor or department (program)
③ Expansion of substitute credits for startup ④ Establishment of Credit Exchange Center
⑤ Alternative startup system ⑥ Establish linkage network of institutions related to university startup

Initiative 2 | **Establishment of startup-friendly HRM system**

① Strengthening the evaluation of achievement related to start-up challenge and business support
② Expansion of startup research program ③ Expansion of operation of core business teachers
④ Creation of teacher startup manual
⑤ Startup research / enhancement of educational capacity

Initiative 3 | **Transition and diffusion into practical type entrepreneurship education**

① Established online entrepreneurial education platform ② Selected as 'Startup Class 100'
③ Activation of entrepreneurial education utilizing maker space and coding education
④ Activation of market-oriented and field-oriented entrepreneurial education
⑤ Student Entrepreneurial Commitment Support

Initiative 4 | **Supporting key talent development and startup challenges**

① Finding and cultivating 100 famous clubs ② Production and utilization of prestigious business club list ③ Expanding entrepreneurial access opportunities through startup contests
④ Introducing 300 prospective teams ⑤ Domestic and Overseas Business Practices

Initiative 5 | **Activation of graduate-student technology startup**

① Survey of graduate student startups ② Graduate Student startup activation program design
③ Establishment of post Doc. and post graduate startup support system: Promotion of 'team startup' system for graduate students and faculty ④ Establishment graduate student training
⑤ Teacher start-up support operation: Startup Lab. ⑥ Introducing entrepreneurship studies abroad post Doc.

Initiative 6 | **Strengthening the university accelerator**

① Expansion of university startup fund ② Strengthening university accreditation capacity: Bridge business, etc. ③ Expansion of stakes in startup companies ④ Relaxing the limitation for technology holding company subsidiary investment

Initiative 7 | **Linkage between university and elementary & middle school entrepreneurship education**

① Institutional reinforcement for school-university linkage support
② Activated entrepreneur competition contest sponsored by the university
③ Elementary and junior high school teachers and principal training courses

Initiative 8 | **Activation of startup life-long education**

① Senior-youth collaboration startup aimed at senior market
② Establishment support system for veterans
③ To induce women to accumulate human capital based on engineering

Initiative 9 | **Strengthening university-region's partnership**

① Local problem-solving startup training
② Developing the around of campus as an entrepreneurial space
③ Establishment of business-oriented settlement space in which universities and regions are linked: Activation of old cities

Initiative 10 | **Creating a social culture to support startups**

Universities that are successful in creating entrepreneurially minded students for the corporate world must have an administration that is aware of this need and embed it in their mission and vision. The interviewees recognize the need for faculty attuned to the entrepreneurial needs of corporations. The interviewees strongly recognized that there must be focus on student development that goes beyond classroom settings and actively engages with a variety of companies in the business community. Finally, while these activities must be encouraged and supported by the administration, programs and faculty are evaluated, and they live in their home units. This means that academic units must each establish their own vision, reward faculty for participation, create a student focus, and promote community engagement at the department level from the perspective of open innovation. Therefore, to be successful campus-wide, Corporate Entrepreneurship Education (CEE) must include (1) entrepreneurial leadership; (2) faculty champions; (3) student-focused policies; (4) engagement with the community; and (5) decentralized, autonomous structure of entrepreneurship programs.

[Table 3-9] Key Factors in Entrepreneurial Education

CEE key factors	Contents
1) Entrepreneurial Leadership in Universities	Vision of Industry-University collaboration for corporate entrepreneurship. Entrepreneurship vision tied to the university vision/mission. Administrative support to start and sustain funding strategies. This can come from tenured faculty, administrators, or non-tenured track specialists.
2) Faculty champion	Although leadership can come from anywhere, a strong program will have a faculty champion, ideally with strong STEM background, academic and industry experience, and entrepreneurial experience. These are exceptional faculty, so faculty and program development in multiple colleges is necessary.
3) Focus on student	Student-friendly policies and engaging content and programs need to be focused on involving a wide variety of students. Connect students with community, teamwork, competitions, and coworking accessibility, process, and hands-on program provided.
4) Community engagement	Industry-University collaboration culture; identify key stakeholders and partners, corporate participation, angel and investor networks, industry sponsors, mentor in residence, and internship opportunity
5) Decentralized/ Coordinated	Diverse autonomous discipline-based program. Each academic unit may sponsor their own program to meet specific industry needs associated with each discipline. Nevertheless, there is also a need for coordinated and collaborated structures.

Source: Choi and Markham (2019)

Applying the corporate entrepreneurship education (CEE) principles to define a campus-wide entrepreneurship education program, we make the following suggestions for a university-based corporate entrepreneurship program (Choi and Markham, 2019). The program should contain the following:

- Educational process that covers all topics in corporate entrepreneurship
- Deep connection and cooperation with multiple corporations
- Corporate entrepreneurship projects that allow students to get firsthand experience trying to be entrepreneurial in a large company
- Corporate mentors who can guide students to develop a compelling business plan for use within existing corporations
- Internship and job connection so that students can gain experience and have a view of job opportunities in large companies
- Competitions for corporate entrepreneurship projects to call attention and increase competition and innovation in corporate entrepreneurial thinking
- Establish a Corporate Entrepreneurship Clinic that works with existing companies and matches students with projects and mentors connected to their interests
- Executive education programs and short courses to help existing managers and executives understand what entrepreneurial thinking is and how to promote it within the organization

These features may be contained in wide variety of academic programs; for example, MBA programs may include a corporate entrepreneurship course as part of an existing entrepreneurship program. We also identified successful entrepreneurial programs across the university that could include corporate entrepreneurship. A separate concentration may be offered as well with support from area corporations. An even more ambitious approach would be to start a Master of Science degree focused on corporate entrepreneurship. At the undergraduate level, concentrated courses may be offered similar to the MBA level. A minor in corporate entrepreneurship may be offered for non-business majors to expand job opportunities (Choi and Markham, 2019).

The benefits for corporations to participate in corporate entrepreneurship programs will include more thought and creativity on the part of faculty to solve pressing corporate innovation and competitive issues. Students from both business schools and other majors will be more able to understand the challenges and to formulate solutions for corporate challenges. Student projects will result in actual project work being done and new thinking applied to old problems as well as companies getting a close look at potential new hires. Executive and other professional courses will help existing managers develop and manage in a more entrepreneurial way. Finally, participation in these types of programs gives companies a positive venue to present themselves to potential employees and contribute to the hiring and retention of an entrepreneurially thinking workforce.

Universities also benefit from providing corporate entrepreneurship programs by offering programs of interest to students, job placement, internship opportunities, connection with area corporations, potential research opportunities, and positive venue to present the university as one that helps the local economy by responding to the educational needs of area employers.

4.2 Technology Incubation and Startup

"Business incubation is a unique and highly flexible combination of business development processes, infrastructure, and people designed to nurture new and small businesses by helping them survive and grow through the difficult and vulnerable early stages of development." Technology incubation provides SMEs and startups with the nurturing environment needed to develop and grow their businesses, offering everything from virtual support and rent-a-desk to state-of-the-art laboratories and everything in between. They provide direct access to hands-on intensive business support, finance, and experts as well as other entrepreneurs and suppliers to make businesses and entrepreneurs really grow (www.nbia.org).

Korean incubators and residents are as follows:

[Table 3-10] Number of Business Incubators

	2011	2012	2013	2014	2015	2016
BI	255	266	264	282	275	267
Companies	4,764	5,145	5,646	6,333	6,655	6,365
Budget (Billion Won)	38.0	32.2	25.7	20.4	22.7	23.8

[Table 3-11] Business Incubator Classifications in Korea

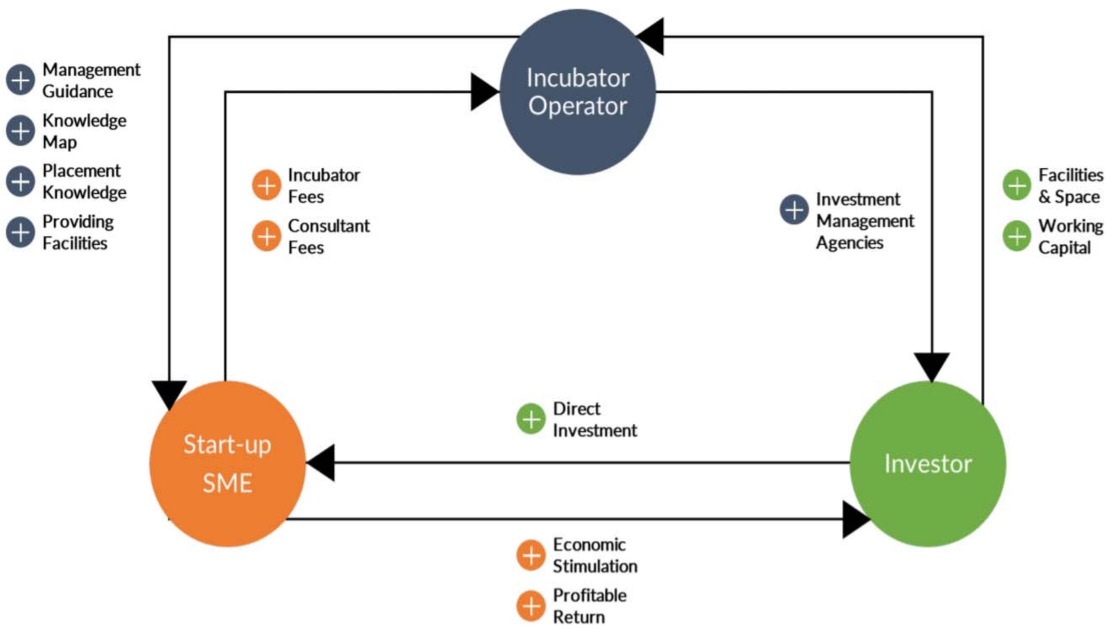
Structure	Category	Characteristics and Profile
Management	Public	Founded by the central government / local governments and public institutions
		Focused on job creation and economic diversification of the economy
		Primary target is expanding the tax base
	Universities and Research Institutions	Established by Universities and Research Institutions
		Focused on the transfer of technology and commercialization of research
		Target is to strengthen and commercialize University revenues
		Contribute to the enhancement of the institution's image in the community
		Finding research grant opportunities in cooperation
		Providing entrepreneurship opportunities for graduates and faculty
	Private	Operated by Individuals and Private Companies as a business investment
		Seeking profit and the structuring of companies to produce profit
		Investments in technologies and to secure technology, real estate development purposes
Specialization	Specialized Center	More than 70% of the resident enterprises have a specific area or specific target
		Secured through Small Business Administration for a specific area of incubation
		The use of specialized services, depending on the growth stage of the enterprise
		Industry professionals needed to ensure proper support
	Non-Specialized Center	Targeting a wide range of industries and specialties
		Professional managers need to have general management capacity for a wide range
Geographical Location	University-related Center	The accumulation of human capital, technology and equipment advantage of the university
		Suitable for the cultivation of high-tech industry
		Earliest possible commercialization of research results
	Urban Center	Easy access to amenities and proper facilities
		Market proximity and marketing advantage
	Rural-area Center	Proximity to indigenous resources nearby
		Expenses are relatively inexpensive, such as space occupancy fees
Operational Support	Self-Reliant Center	Faithful to the established objectives and policies of the operations center
		Specialized support services, eligible to focus on specific sectors
	Non-Self-Reliant Center	Policies reflect the operational support agency
		Public interest can be excluded to the operating profit of the institution

Source: Park, G and Kim, Jay (2016), Manual of the Korean Business Incubator Model, Rehoboth. p. 7.

[Figure 3-11] Objective of Business Incubator in Korea

Source: Park, G and Kim, Jay (2016)

[Figure 3-12] Basic Model of Incubating Operating System in Korea

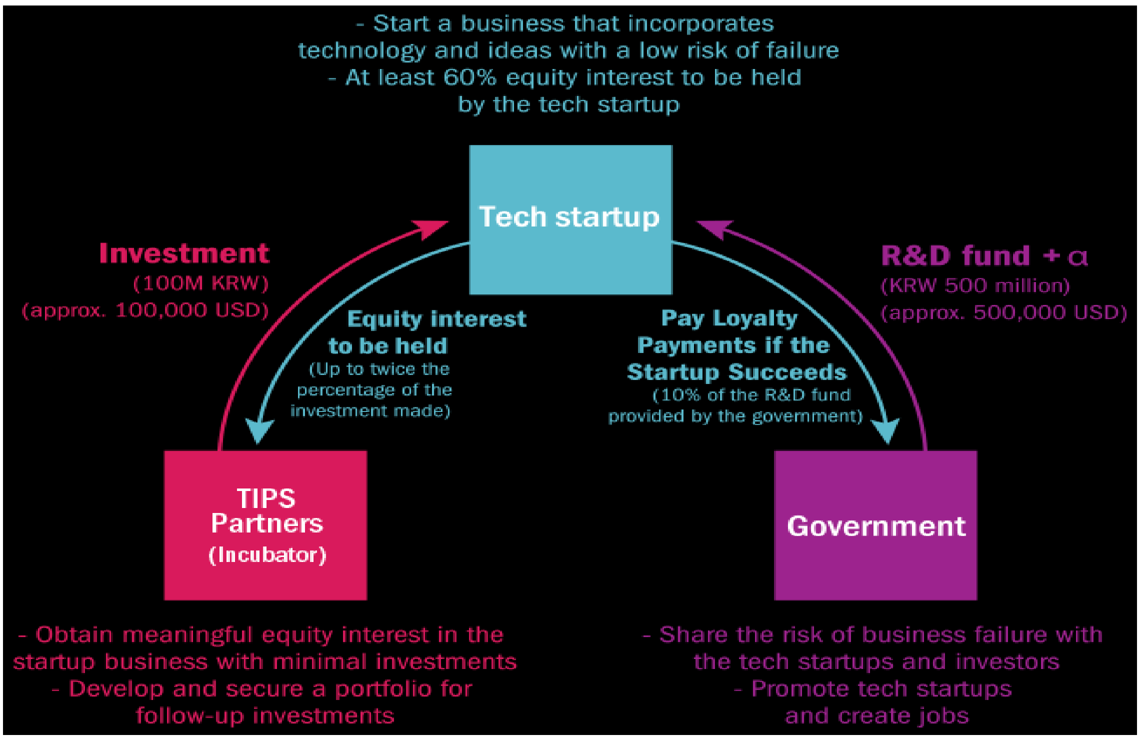


TIPS: Technology Incubator Program for Startups

The South Korean government has been running the Accelerator Investment-Driven “Tech Incubator Program for Startups” (TIPS) with public- and private-sector partners to discover and nurture promising startups.

TIPS (Accelerator Investment-Driven Tech Incubator Program for Startup) is designed to identify and nurture the most promising startups with innovative ideas and groundbreaking technologies. In order to support them in entering the global marketplace, it appoints and designates successful venture founders who are now angel investors and leaders of technological enterprises as their incubators/accelerators. It then offers seamless service encompassing angel investor networking, incubating, mentoring/professional support, and matching R&D funds.

Startups below three years old are eligible to apply for the program led by the Ministry of SMEs and Startups. Each selected team is eligible to receive up to 1 billion won ($838,000) in funding for use as seed capital, research and development expenses, or marketing fees. For years, the program has helped propel many startups to the next stage in their business development by providing them with the necessary capital and resources needed to bring their products to market and scale up their business (Sohn, 2019).



Source: http://www.jointips.or.kr/about_en.php

Korean unicorns

Would you rather work at an established Korean startup? Below is a list of Korean startups valued at more than one billion US dollars (Source: CBInsights). It consists of Unicorn companies, valuation (B$), category, investors etc. The unicorn companies of Korea are as follows: Coupang, Bluehole, Yello Mobile, Woowa Brothers (Baedal Minjok), L&P Cosmetic, Viva Republica (Toss), and Yanolja

[Table 3-12] Korean Unicorn Companies

Unicorn	Valuation (B$)	Category	Investors
Coupang	$9	eCommerce / Marketplace	Sequoia Capital, Founder Collective, Wellington Management
Bluehole*	$5	Travel Tech (sic)	Tencent Holdings, Stonebridge Capital, IMM Investment
Yello Mobile	$4	Mobile Software & Services	Formation 8
Woowa Brothers*	$2.6	On Demand	Hillhouse Capital Management, Altos Ventures, Sequoia Capital
L&P Cosmetic	$1.78	Beauty & grooming	CDIB Capital
We make price	$1.33	eCommerce / Marketplace	IMM Investment, NXC
Viva Republica (Toss)	$1.2	Fintech	Bessemer Venture Partners, Qualcomm Ventures, Kleiner Perkins Caufield & Byers
Yanolja	$1	Travel Tech	SBI Investment Korea, Partners Investment, GIC

* Bluehole is the company behind PlayerUnknown's game Battlegrounds.

* Woowa Brothers is the company behind food delivery app Baedal Minjok.

5. Technopark Status and Introduction to Chungnam Technopark

5.1 Status of Technoparks in Korea

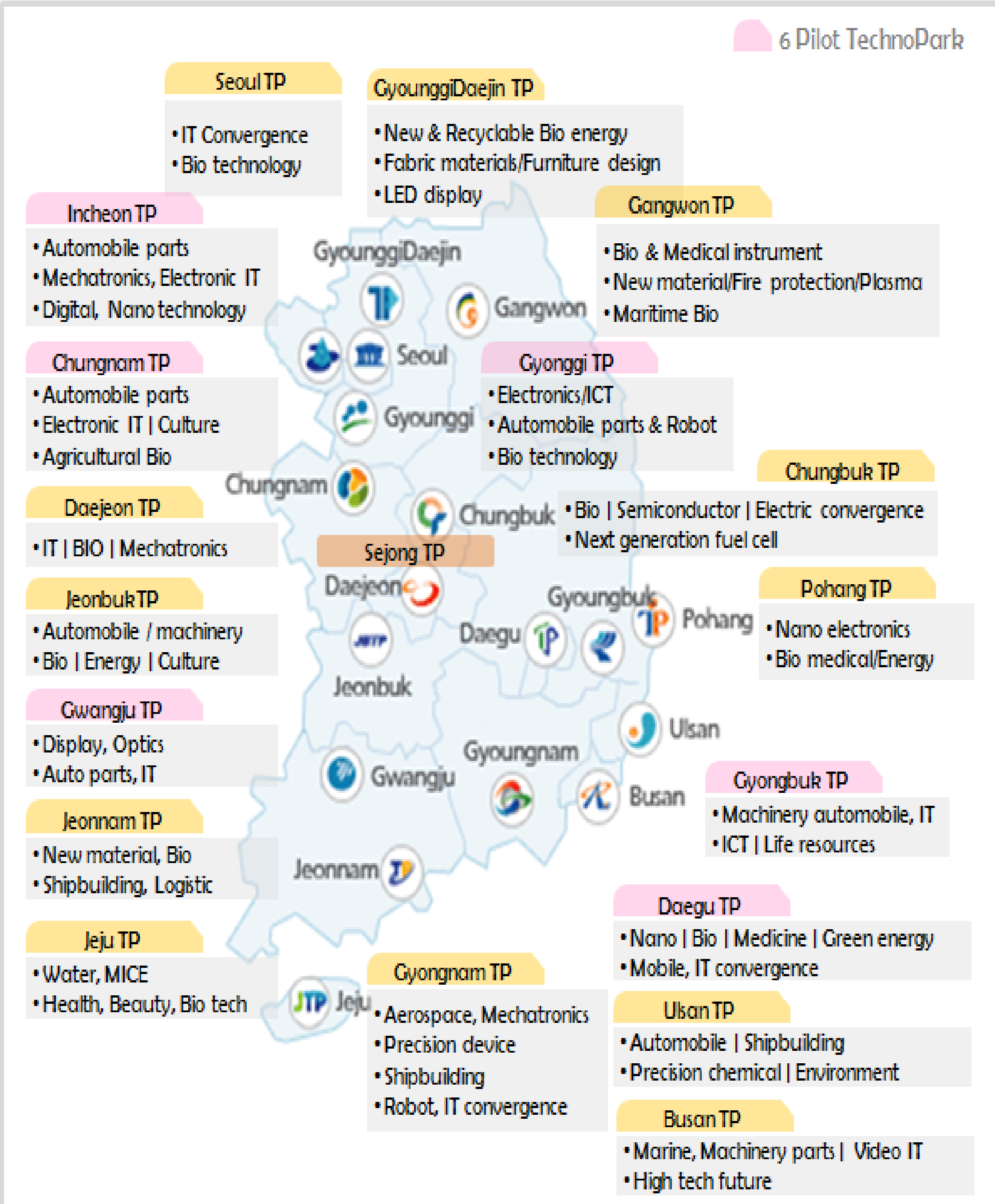
Technoparks in Korea are established/operated for balanced regional development, and there are a total of 19 technoparks in Korea (one each in metropolitan cities/provinces and special cities/provinces (Seoul, Seong, and Jeju)).

Most of the technoparks have secured the budget needed for establishment/operation with funds from the central government and local governments, with the budget from the central government taking up the highest portion.

Each technopark plays a role in establishing regional industry development policies, fostering technology-based companies, nurturing talents, and establishing technology innovation infrastructures, focusing on the key industries in the region.

Among the 19 technoparks, Chungnam Technopark is considered a leading technopark along with Incheon Technopark, Gyeonggi Technopark, Gyeongbuk Technopark, Daegu Technopark, and Gwangju Technopark. (Leading technopark: A technopark established in the early stage, i.e., late 1990s)

[Figure 3-13] Status of Technoparks in Korea



Source: Korea Technopark Promotion Association

5.2 Introduction to Chungnam Technopark

5.2.1 Industrial/Regional Environment

Chungcheongnam-do Province, where Chungnam Technopark is located, is adjacent to the Yellow Sea Free Economic Zone and is home to large corporations such as Samsung Electronics and Hyundai Motor.

Chungnam's key industries are steel, petrochemical, automotive, and display, which account for a large share in the region (47.1% of local manufacturing businesses, 59% of workers, 58.8% of production value, and 64.6% of added value).

In relation to local industries, Chungnam houses 4 specialized centers, 5 research institutes, and 12 universities equipped with R&D capacity and infrastructure to support businesses centered on existing products.

As of 2016, the size of the economy expanded with local economic growth rate of 3.83% (national: 2.85%) as well as GRDP of 7.11%.

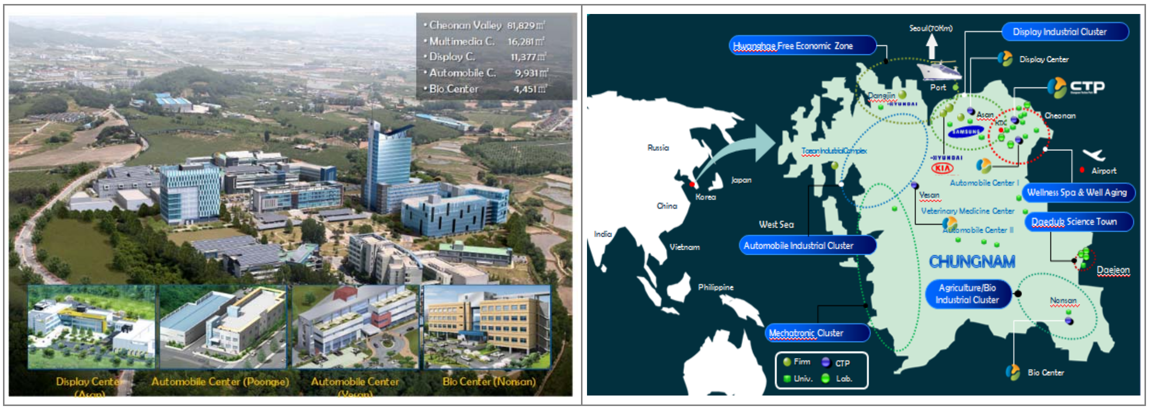
Chungnam is driving national growth with the nation's third largest growth contribution rate (13.6%) after Seoul and Gyeonggi.

This is attributable to the growth of manufacturing, which accounts for more than 50% of local production (12.1% of national production, 2nd highest in Korea). As such, the region's industrial structure is concentrated on manufacturing, on which the local economy is highly dependent.

Mainly composed of the country's major export industries, Chungnam's industries account for more than 10% of national export output (4th largest in Korea), and they are sensitive to changes in the external environment.

Chungnam-based businesses account for 4.09% of the country and grow at an annual average rate of 2.43%. This is attributable to the increase in new corporations, with the average growth rate (9.69%) exceeding the national average (6.55%).

[Figure 3-14] Chungnam TP Complex Composition and Chungnam’s Regional Industry Status



Source: Chungnam Technopark

5.2.2 History of Chungnam Technopark

After being selected as a pilot technopark in 1997, Chungnam TP was approved for establishment in 1993. Since its opening in January 1999, Chungnam TP has expanded its infrastructure, including the completion of Cheonan Valley, opening of the Multimedia Center in 2005, and opening of the Display Center in 2006.

Chungnam TP has received numerous awards for its contribution to regional development, including the Prime Minister's Award, Future Minister's Award, and Industry Minister's Award.

In particular, Chungnam TP has maintained its outstanding status, having been selected as the best technopark in the country in 7 out of 10 government evaluations performed since 2008.

1997. 12	Designated as one of 6 Techno Park Projects
1999. 01	CTP GRAND OPENING
2002. 09	Completion of BI and Post BI at Cheonan Valley
2005. 11	Opened Multimedia Center at Cheonan Valley
2006. 10	Opened Display Center (Asan)
2007. 12	Opened Automobile Center (Poongse, Cheonan)
2010. 11	Opened Automobile Center (Yesan)
2011. 02	Consolidated Bio center (Nonsan)
2013. 08	Opened 'Secondary Battery Center'
2014. 08	MOU with KTL for secondary battery certification center
2016. 07	Veterinary Medicinal Center (Yesan)
2017. 05	Wellness Spa Clinical Center
2017. 08	Well-aging Functional Food Development Center
2018. 12	Selected as 'Best Corporate Training Center'

Best Techno Park

Awarded by Ministry of Trade, Industry and Energy(MOTIE) (Excellent Organization)

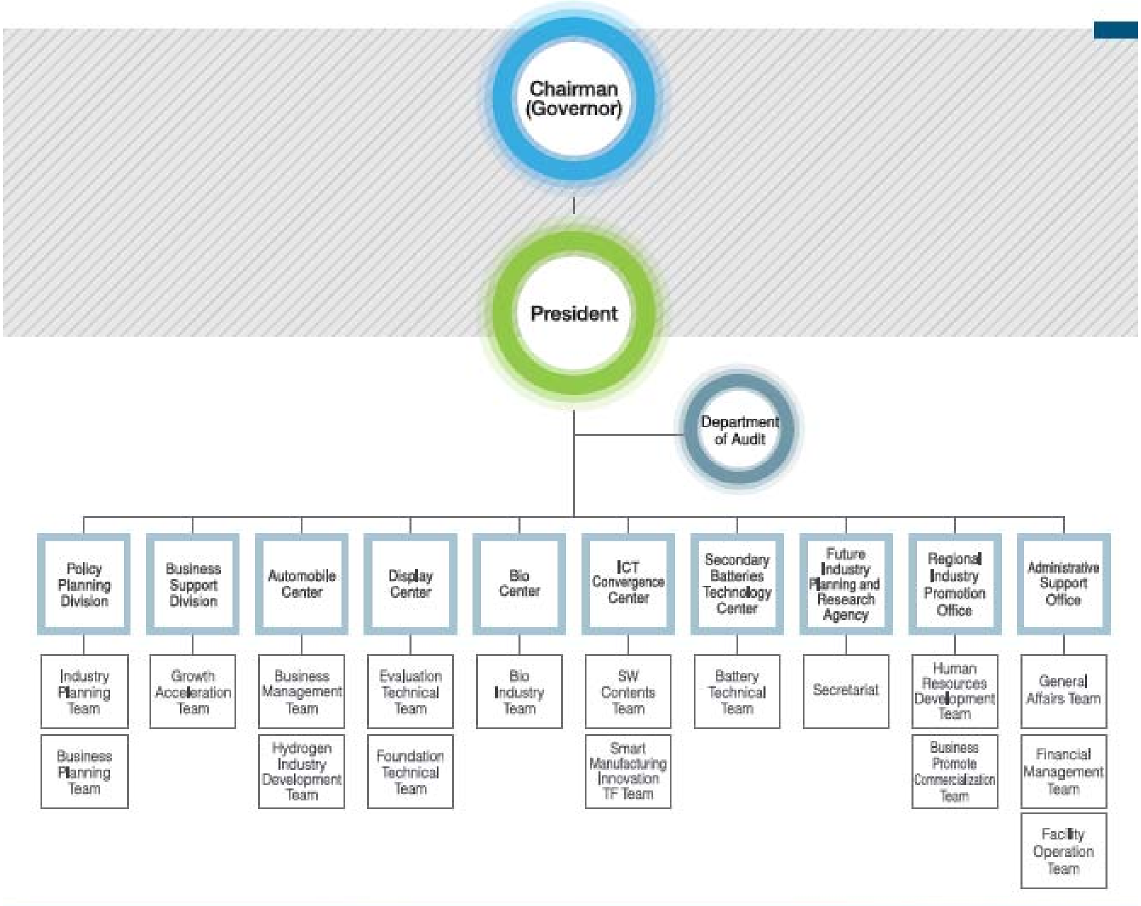
- A(Top Ranking) for 7 times of total 10 times evaluations since 2008

5.2.3 Chungnam Technopark Organization & Main Roles

Chungnam TP consists of specialized centers in five industries: Policy Planning Division, Business Support Division, Regional Industry Promotion Office, Future Industry Planning Research Agency, and Administrative Support Office.

As a technological innovation base, Chungnam TP plays key roles in 1) supporting the technological innovation and commercialization of small and medium enterprises (SMEs), 2) supporting the acceleration of startups according to the stage of corporate growth, 3) fostering promising SMEs in Chungnam, 4) establishing SME development strategies and securing future growth engines, and 5) linking jobs and nurturing technical professionals.

[Figure 3-15] Chungnam Technopark Organization Chart



Source: Chungnam Technopark Website

5.3 Goals and Strategies for Promoting Technology Transfer Commercialization (Chungnam TP)

Chungnam TP, which started supporting technology transfer in 2003, set four goals aimed at promoting technology transfer and commercialization: "establishment of basis for market invigoration"; "improvement of outcome/success rate of commercialization"; "enhancement of business operation efficiency," and; "specialization/differentiation of organizations."

Considering the related problems and environment, Chungnam TP has established strategies for each goal. The details are as follows:

[Figure 3-16] Chungnam Technopark's Goals and Strategies for Promoting Technology Transfer and Commercialization

Goal		Grand Strategy
Establishment of basis for market invigoration (quantitative/qualitative levels of demand)	*Importance of demand information in the technology trading market*	Turning "demand study into demand generation" and clarification
		Reflection and sophistication of demand features/characteristics
Improvement of outcome/success rate of commercialization (consistent management/scope of support/level)	*Seamless support and efficient problem solving*	Expansion and sophistication of scope of support
		Establishment of a performance-oriented monitoring/support system
Enhancement of business operation efficiency (cooperation resources/framework/complementation)	*Utilization and organic system*	Utilization of external resources
		Network consolidation and clear role assignment
Specialization/Differentiation of organizations (specialized positions/capabilities)	*Firm position and high competence*	Establishment of specialized positions/roles
		Continuous capacity building of institutional/human resources

5.3.1 Turning Demand Study into Demand Generation and Clarification

5.3.1.1 Background

In order to invigorate the technology trading market, securing sufficient quantitative and qualitative levels of technology demand information is key, but there is low inflow of technology demand information from the market.

Most SMEs are unable to recognize or derive demand technology by themselves due to their limited capacities in technology planning, external environment analysis, and strategy establishment.

This has caused extremely low response rates in technical demand surveys.

The response rate of simple technical demand surveys is usually less than 1%, and the rate of valid demand information among the responses is also very low.

Ultimately, it is necessary to change the mindset and think that technology demand information should be actively created, not surveyed.

Even if companies derived technology demand on their own (very few companies), they tend to explore by themselves instead of consulting an external party.

When companies can clearly derive and recognize technology demand, the ability to search for technology is also high, so the role of intermediary becomes unnecessary.

After all, the introduction of technology is a means of securing technology, and the desire to secure technology arises in the process of developing new products (differentiated products) or promoting new businesses (new item development).

In other words, the demand for items (new businesses) or ideas (product differentiation) tends to be more evident and exposed than the demand for technology.

5.3.1.2 Content

Seek the conversion of "demand study into demand generation'' by actively providing technical ideas on difficulties.

Introduce the concept of demand creation idea (DCI) as a way of triggering demand and use it to expand the quantity of demand.

Actively secure technology demand from new item discovery and needs for differentiated products.

Target mid-sized companies in the mature stage and firms with stagnant growth to find basic demand information intensively.

Concentrate support capacity on finding new items for companies in the mature stage to make a new leap so that they can successfully select items and derive technology demand as required for commercialization.

Specify vague demand in the early stage with the help of experts in each field.

Process demand information to clarify the field of technology, purpose of acquisition, quantification of the required performance, and restrictions.

5.3.2 Reflection and Sophistication of Demand Features/Characteristics

5.3.2.1 Background

There are many limitations in the existing method of collectively recognizing/processing all technical demand information at the same time.

Although there is something in common in terms of wishing to secure technology, when each technology demand information's features and characteristics are not reflected, it is difficult to utilize the supplied technology.

In other words, it is necessary to specify ambiguous demand and, at the same time, identify the unique features and characteristics of technology demand information in order to provide optimal support.

5.3.2.2 Content

Determine the existence and feasibility of related technology based on the significance, current status, and same field of a product's unique function and identify the features/characteristics in terms of demand.

Reflect the type of demand technology from a technology perspective and make the final classification. Sophisticate technology demand information by establishing/adding new values and pursuing universal application.

Establish the best intermediary strategy by identifying priorities in advance as necessary for the final stage.

5.3.3 Expansion and Sophistication of Scope of Support

5.3.3.1 Background

From a policy perspective, the final purpose of technology transfer is technology commercialization, but low commercialization rates are emerging as a problem.

One of the factors that hinder successful commercialization compared to the results of technology transfer is discontinued support.

It is necessary to prepare a system that can continuously support not only technology transfer but also commercialization.

5.3.3.2 Content

It is necessary to expand the scope of technology transfer support to cover the brokerage/intermediary roles prior to technology transfer as well as commercialization.

When a company pursues commercialization after adopting technology, there are needs for additional development, certification, and other commercialization resources and funding, and it is important to support these effectively.

5.3.4 Establishment of a Performance-oriented Monitoring/Support System

5.3.4.1 Background

Another reason for the low rate of successful commercialization is the failure to provide continuous monitoring and support.

There are many cases wherein companies that acquired technology and benefitted from R&D are not consistently managed after receiving primary support.

Each agency has a DB for companies that received support, but items essential for managing commercialization performance are missing, often leading to inefficiency.

5.3.4.2 Content

Establish a DB that can be utilized for network cooperation support.

Implement information management to identify the progress and difficulties of each commercialization project.

Consistently manage related information on companies that acquired technology and benefitted from R&D to enable timely and proactive support.

Collate commercialization information on companies related to technology transfer into a DB, conduct continuous monitoring, and solve problems with the help of various experts by identifying the factors that delay or hinder commercialization.

5.3.5 Active Utilization of External Resources

5.3.5.1 Background

In order to solve diverse and complex problems and difficulties of companies, cooperative support from various experts along with comprehensive support services are needed, but they are not fully implemented in many cases.

Individual entities and supporting projects have limitations in providing all the services necessary for companies, so it is necessary to use experts and link multiple support projects actively to improve performance compared to the injected budget.

5.3.5.2 Content

Maximize the outcome of support relative to the budget by actively utilizing other projects and external experts.

Provide customized business information to the relevant companies for the timely use of related business.

Enable organic connections by actively implementing joint programs and projects with external experts and institutions.

5.3.6 Network Consolidation and Clear Role Assignment

5.3.6.1 Background

When forming a cooperative network to improve business operation efficiency, most cases tend to be a formal network.

Practical cooperation is not possible without a clear sense of common goal among the members.

Factors that hinder cooperation include ambiguous and vague cooperation topics, system that leads to a decrease in commitment and responsibility, and formal, one-off cooperation.

5.3.6.2 Content

Provide strong cooperation motivation and consolidate the network by forming a cooperative system based on needs and demands.

Allocate clear roles through project/program-based cooperation.

Increase the commitment and responsibility of network members by motivating them to establish and sustain a sense of common goals.

5.3.7 Establishment of Specialized Positions/Roles & Continuous Capacity Building of Institutional/Human Resources

5.3.7.1 Background

There are various supporting players in the domestic technology transfer and commercialization market.

The similar functions of the players make cooperation rather ineffective.

Although the functions supporting technology transfer could be the same in a broader sense, the characterization and differentiation of each organization are most important. This requires the establishment of specialized positions and roles as well as enhanced capabilities.

5.3.7.2 Content

Establish specialized positions and roles considering the competitive advantage over other supporting players, growth potential, and demand (needs) in the market.

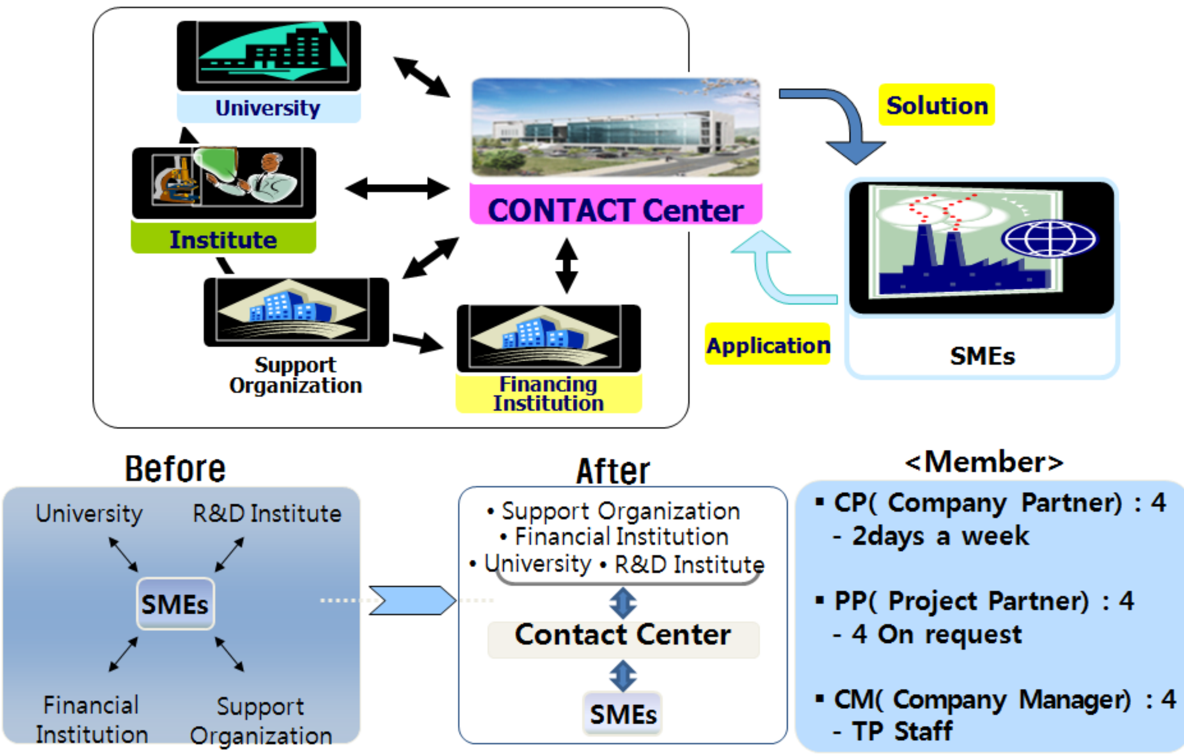
Continue to strengthen the capabilities of institutional/human resources according to specialized/differentiated goals.

Realize organizational specialization and differentiation through specialized positions and continuous capacity building.

The organization supporting technology transfer requires expertise in various fields such as technology, management, and law, and the expertise of dedicated personnel is closely related to the outcome.

5.4 Program Specializing in Technology Commercialization (Chungnam TP)

[Figure 3-17] Overview Diagram of the Contact Center Operation Program



5.4.1 Contact Center Operation Program

Difficulties facing new technology-based companies are very diverse, and companies need support from various experts and related organizations to resolve such difficulties.

There is a need for customized corporate support services to meet the needs of complex, diversified companies. Therefore, a measure was prepared to support companies effectively through a single window by integrating the scattered information and capabilities of innovators.

Companies previously inquired with and contacted universities, research institutes, and support institutions individually, but they can now receive integrated guidance and support through the contact center.

General companies have been able to receive guidance and support services for handling difficulties, but member companies are entitled to receive consistent mentoring and more professional services.

The operating organization consists of Technopark's experts and external specialists. For each project, a cooperation team is formed to solve corporate problems.

CP (Company Partner)	• Works 2 days a week; acts as a mentor dedicated to each company (regular visits to the company); selected from those who have insight into corporate support • Experienced corporate executives (SME/conglomerate) • Master's or PhD degree holders with experience in corporate technology development
PP (Project Partner)	• Experts in each field provide advice and consultation if needed • Consists of technical, financial, and consulting experts, including university professors, researchers, and corporate employees
CM (Company Manager)	• Technopark's corporate support staff • Regularly visits companies and manages member companies together with CP

Application of a member company	• Receive an application from a member company • Decide whether to visit the company (operation meeting)
Site visit	• Dedicated CP/CM visits the site • Write a contact chart • Submit data to the strategy table
Strategy table	• Review the approval of membership • Notify membership fees • Decide on a dedicated partner
Partner's visit	• The partner visits the site • Corporate demand survey (matching point consultation) • Prepare for the signing of an agreement (1-year agreement)
Signing an agreement	• Sign a 3rd-party agreement (signing ceremony) • Draw up a customized company growth roadmap
Project execution	• Regular support activities • Customized support activities (coupon provision) • Regular exchange meetings (learning activities)
Consistent management	• Matching support depending on the situation • Learning activities to manage changes
Measurement of outcomes	• Comprehensive measurement of support activities; promotion • Satisfaction survey (member company) • Reflect the results to the agreement the following year (degree of cooperation)

5.4.1.1 Comprehensive Support Program for Technology Commercialization

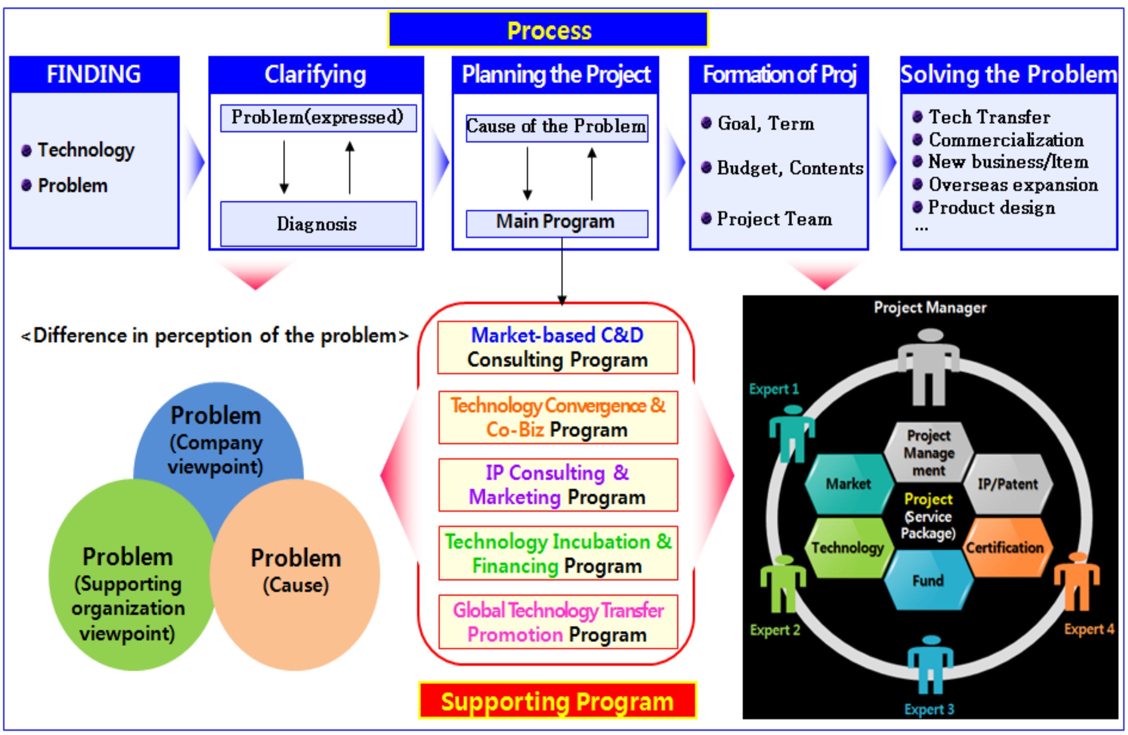
The program consists of five stages (finding → clarifying → project planning → project formation → problem solving) and provides project-based support.

A project is a collection of services designed to solve a customer's problem; a project team is a group of professionals or specialized organizations providing these services.

It is assumed that the problems of complex, diversified companies cannot be solved by a single expert with a single service.

Five regular subprograms are provided to solve problems, and each program contains professional support content.

[Figure 3-18] Overview Diagram of the Comprehensive Technology Commercialization Support Program



The overview of the program and key contents in each stage are as follows:

Stage	Main content
S1) Finding	• Find support subjects based on the problems of technology or the company
S2) Clarifying	• The stage when problems are clarified and specified assuming that the problems expressed by the company, the problems recognized by the supporting agency, and the actual causes are different
S3) Planning the Project	• Match the finally derived problems with the content of five programs • If necessary, provide comprehensive support with multiple programs
S4) Formation of Project	• Stage for formalizing project goals, setting terms, and calculating budgets • Stage for forming a project team and concluding a contract
S5) Solving the Problem	• Stage when the formed project team solves problems by providing professional services • Supports various areas such as technology transfer, technology commercialization, investment attraction, marketing, and intellectual property

Program	Overview
P1) Market-based C&D Consulting Program	• For companies requiring new business (item) discovery or product differentiation, analyze external environments such as the market/industry as well as internal capabilities to set items and establish goals for differentiation
P2) Technology Convergence & Co-Biz Program	• For technologies that can be applied universally such as components/materials or companies that need cooperation with other firms, support new market entry and cooperation between companies
P3) IP Consulting & Marketing Program	• Support intellectual property management, patent consulting, and technology marketing for technology transfer
P4) Technology Incubation & Financing Program	• For tech startups and companies that want to attract investment, provide startup consulting, fund consulting, IR and investor matching, M&A support, etc.
P5) Global Technology Transfer Promotion Program	• Support technology transfer overseas and introduction of overseas technology, provide support for global partner discovery, negotiation, promotion, marketing, etc.

5.5 Technology Transfer Commercialization Promotion Plan

5.5.1 Supply → Demand

5.5.1.1 Key Issues

The differences in the perspectives of technology owners and technology adopters (companies) often lead to failure to deliver the full potential of technology.

First of all, in terms of differences in language and expressions, technology owners explain technology from a perspective limited to technology (e.g., technological performance, technological sophistication, experimental results, etc.).

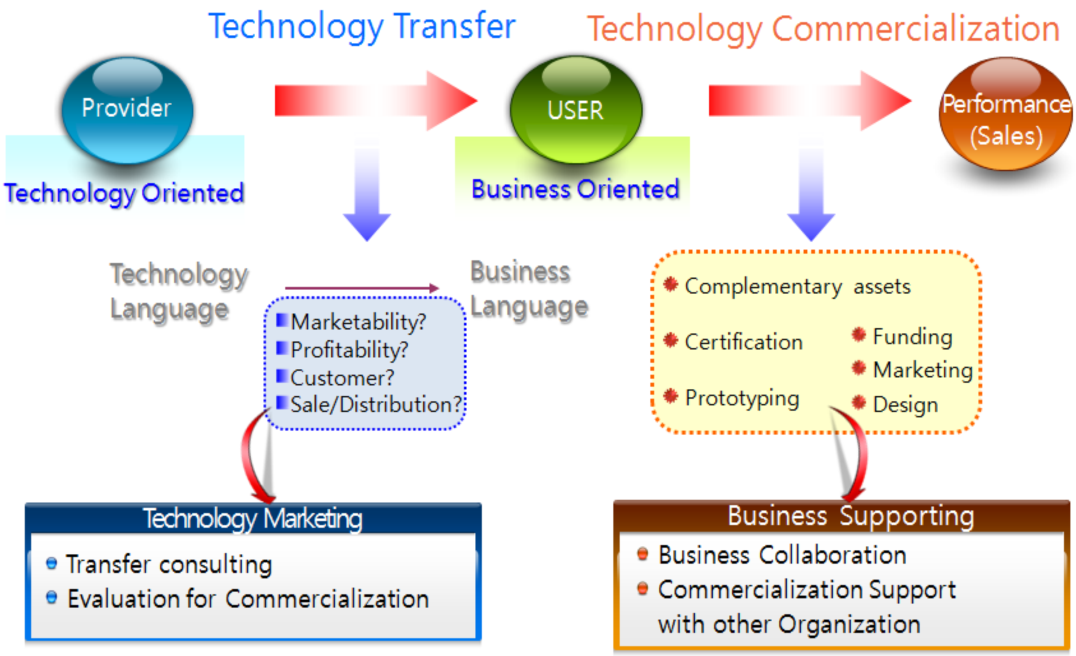
On the other hand, technology adopters (companies) are more interested in what effects the technology will bring to their business.

Therefore, it is necessary to produce appropriate technical promotional materials with the technical language converted into business language. At this time, a business model, including marketability, profitability, customers, and sales/distribution of the applied product/business, should be created together and provided to the technology adopters (companies).

It is also important to calculate the value of technology in addition to expressing/delivering the technology or to provide technology adopters with an objective evaluation of marketability, technology, and feasibility.

Even when a company is pushing for commercialization after the technology is transferred, there are various demands for support such as complementary assets for commercialization, certification for market entry, prototype production, fundraising, marketing, and design; effective support for these determines the success of commercialization.

[Figure 3-19] Key Points in the Technology Transfer Commercialization Process



5.5.1.2 Promotional Activities of Technology

Promotional activities of technology are largely divided into online and offline activities.

Online activities include posting technical promotional material or listing technical information on an online platform, sending regular newsletters to companies, and carrying out promotion through media.

Recently, active online marketing using YouTube or social media has been highly effective.

The highest delivery effect was found to come from modeling and videos of the applied product, followed by detailed data including visual images and brief information.

[Figure 3-20] Examples of Online Technology Promotion Channels

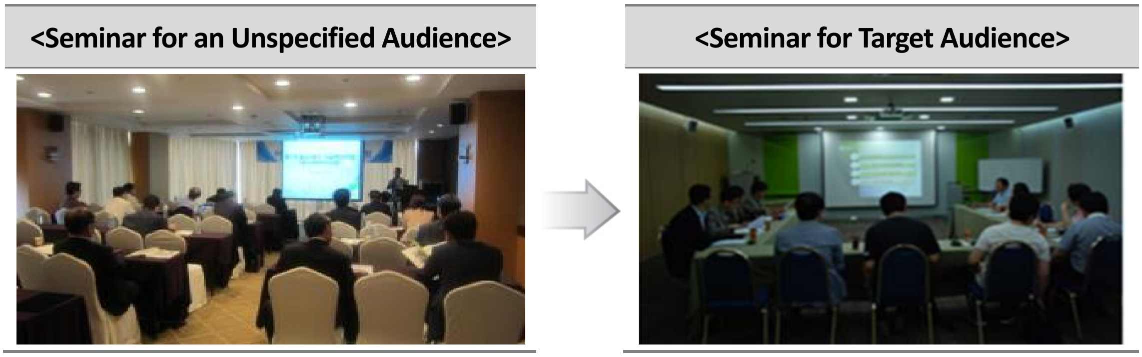
On the other hand, offline activities include technology transfer seminars and consulting sessions.

These events have the advantages of enabling the direct monitoring of companies' reactions as well as immediate contacts with companies of interest, but seminars for a number of unspecified companies may not effectively attract attention.

In other words, participants in such seminar can perceive the technology to have low value, thinking that it can be transferred to anyone.

Therefore, instead of holding a seminar for a large number of unspecified companies, it is important to create an atmosphere wherein target companies would want to adopt the technology competitively.

[Figure 3-21] Examples of Offline Technology Promotion



5.5.2 Brokerage/Negotiation

5.5.2.1 Technical Matching

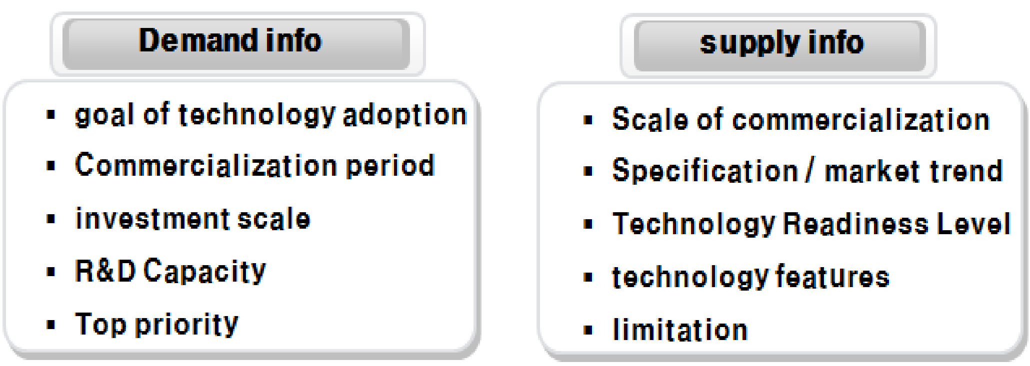
It is important to understand the features and characteristics of each information in the matching stage together with the initial technological relevance based on technical information and demand information.

First of all, in terms of demand information, the purpose of technology introduction, period of commercialization (expected), size of the company and investment, R&D capacity and experience, ability to absorb the technology, and matters that should be considered first when introducing the technology should be identified.

In terms of supply information, it is necessary to understand the scale of commercialization (to be pursued with technology), characteristics and trends of the industry/market, technology readiness level, features of the technology, possible additional support, trade restrictions, etc.

To prevent mismatching, these considerations should be made first; if a technology transfer contract is concluded without such considerations, problems may arise during the commercialization process.

[Figure 3-22] Key Considerations of Supply/Demand Information



5.5.2.2 Negotiation and Contracting

The technology transfer intermediary should coordinate the opinions and conditions or support negotiation between the technology owner and the technology adopter considering the various issues in the technology transfer negotiation process.

Common issues involved in the technology transfer negotiation process include:

- Definition of technology: In the case of technologies such as know-how, which have no legal property rights like patents, the definition of the technology to be transferred should be clarified because it could give rise to controversy.
- Contract period: The longer the contract period is, the higher the royalty, and setting the term for an excessively long period is not desirable.
- Scope/Method/Information and material index and time: The method, items, and timing of technology transfer also need to be defined clearly. Thus, the schedule needs to reflect the items/list and timing.
- Advanced invention: If additional technology is developed from the transferred technology, both the technology owner and the technology adopter can claim ownership of this technology, so a related clause is needed.
- Type of transfer: The technology adopter wants to obtain transfer or exclusive license (sole license) to secure the exclusive right, and the technology owner wants to provide its technology to as many users as possible. In Korea, regardless of the intention of the technology owner, granting an exclusive right is restricted for technologies developed through a national R&D project.
- Contract termination: The contract needs to specify the reasons for contract termination, e.g., when the technology adopter did not faithfully pursue commercialization or when the technology owner did not faithfully transfer the technology.
- Guarantee: There are also frequent disputes as to how far the performance and effectiveness of the technology should be guaranteed. Since it is unfair for the

technology owner to assume all technical responsibilities, clear guidelines need to be set.

- Royalty: As the most controversial element in a technology transfer contract, royalty can be paid in cash, in kind, or in the form of equity. It is difficult to pay a large amount of royalty at the contract stage when the effects of the technology are expected but not yet realized. Therefore, in many cases, royalty is set based on sales and operating profit (running royalty).

5.5.3 Demand → Supply

5.5.3.1 Key Issues

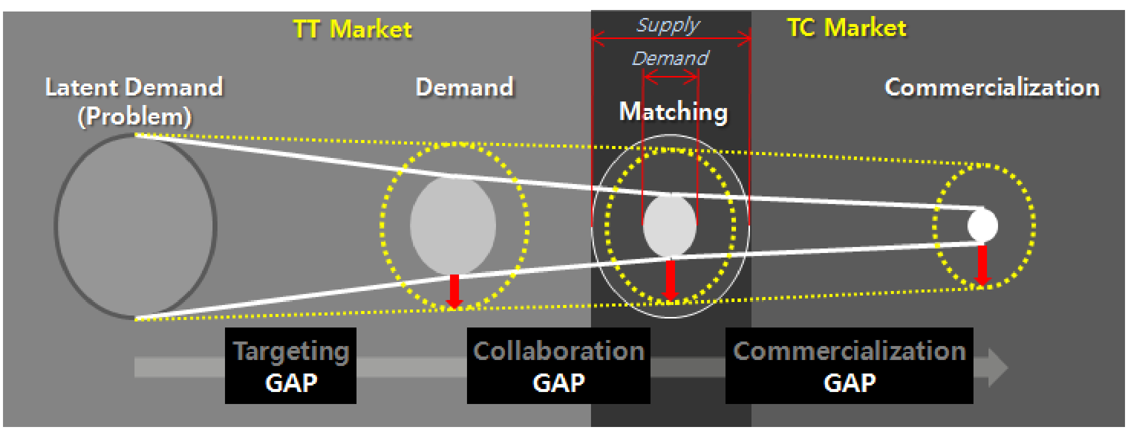
The key is to raise the level of technology demand quantitatively and qualitatively in order to increase the success rate of technology transfer and commercialization.

In general, the demand for technology is far lower than the technology supplied in the technology trading market.

Balancing supply and demand by deriving true demand from latent demand can increase the success rate of technology transfer matching.

Unlike a simple quantitative increase, deriving true demand can also increase the success rate of technology commercialization.

[Figure 3-23] How to Improve the Success Rate of Technology Transfer Commercialization



In the past, Korea also relied on technology demand surveys as a way of finding demand for technology.

The problem was that sufficient demand was not secured in technical demand surveys because the response rates were extremely low (around 0.5% of response rate).

Even the responses lacked sufficient information, or they were vague and ambiguous in most cases. Initially, this problem was perceived as a matter of corporate security (trade secrets).

From a fresh perspective, however, asking companies what technology they need is like asking patients what medicines they need.

[Figure 3-24] Concept of Finding Demand

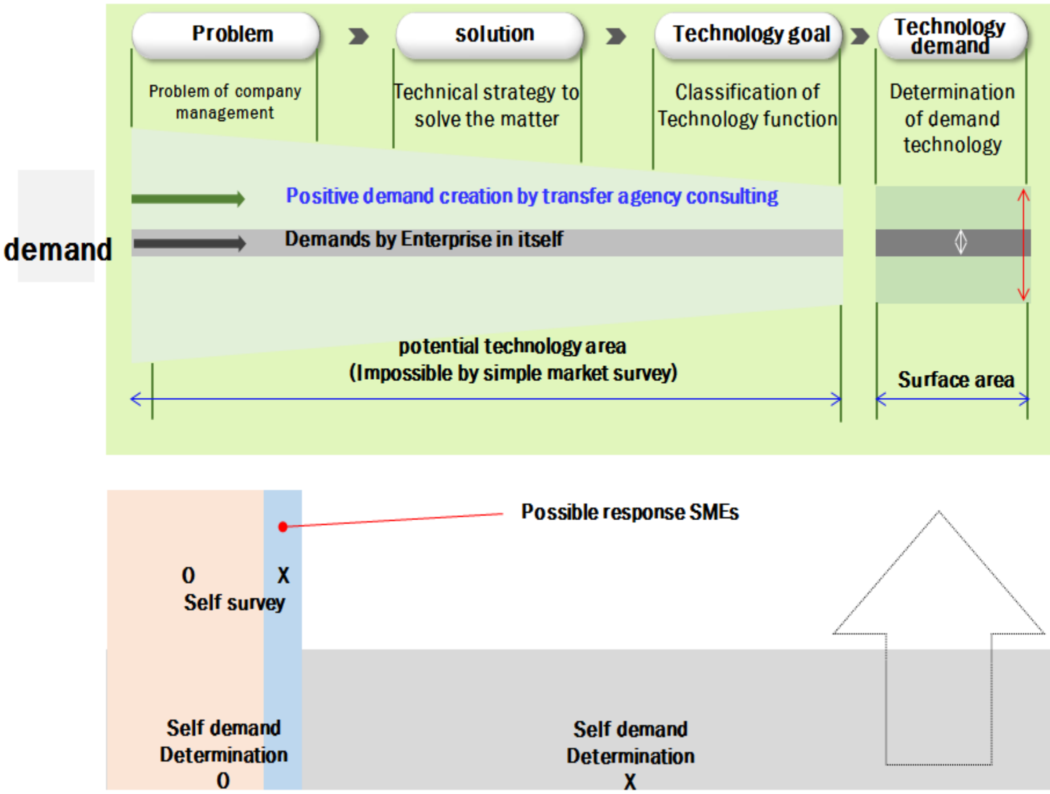
Most SMEs are not well-aware of what technology they need.

If firms are clearly aware of technology demand, it is easy to push for technology trade, but this applies to a small number of companies only. Ultimately, many firms fail to derive clear demand despite the existence of latent demand.

Even though there is a desire for innovation, there are frequent cases of mismatch with the technology transfer support service due to the ambiguity of demand.

Therefore, the commercialization of technology demand is also a challenge, as is the death of the developed technology in the technology trading market; the key is to secure enough advanced technology demand information by actively materializing latent demand.

[Figure 3-25] Difficulty in Finding Demand



5.5.3.2 How to Specify Demand

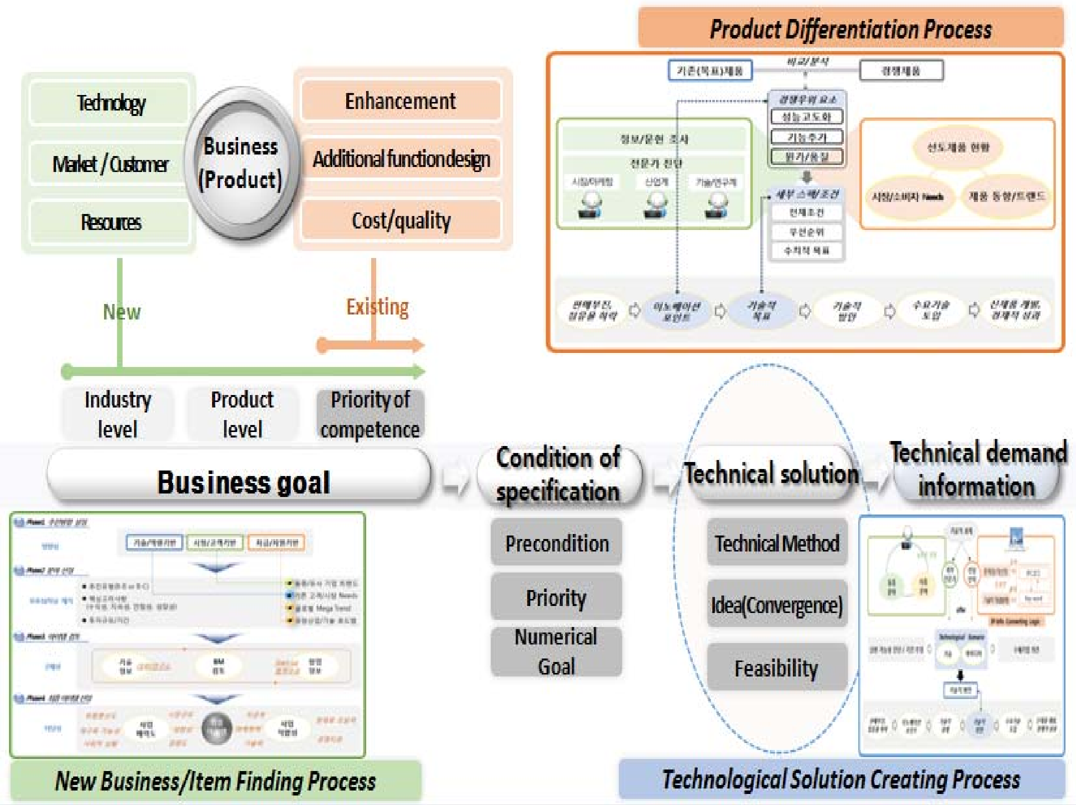
As discussed above, technology needs to be found and created rather than surveyed.

The demand specification process promoted by Chungnam TP derives the technology demand from problems related to the management of a company.

It consists of four steps: Setting/Establishment of innovation goals → Derivation of detailed specifications/conditions (product unit) → Derivation of technical solutions → Derivation of final demand information. It also includes three processes that complement each step of the main process.

Below is the detailed description:

[Figure 3-26] Demand Specification Process



<Main Process>

Set/Establish innovation goals.

- Establish innovation direction and goals to address corporate management difficulties and establish the final goals by modifying/complementing existing goals.
- Divide them largely into two categories — new business (item) discovery and product differentiation — and subdivide them into three categories considering detailed directions.
- Use the new business/item-finding process and the product differentiation process.

Derive detailed specifications/conditions (product unit).

- Derive the technical goals (detailed specifications/conditions) from the innovation goals.
- Derive quantitative objectives but review the preconditions for the selected goals and prioritize in detail in case of multiple goals.
- Actively use the product differentiation process.

Derive technical solutions.

- Derive technical solutions to meet the detailed specifications/conditions (technical goals).
- Actively consider convergence ideas in addition to the technical solutions in the same field.
- Derive two or more solutions, but select/determine the best one by diagnosing/reviewing the feasibility and duration of each solution.
- Actively use the technological solution creation process.

Derive the final demand information.

- Confirm the final technical demand based on the optimal technical solution that was prioritized (selected).
- The final technical demand should be derived to have clear functional and quantitative objectives for the technology.

<Detailed Process>

1) New Business/Item Finding Process

Implementation direction setting

- Establish new business directions (focusing on clear directions) considering the company's capabilities, resources, and management policies.
- Set the optimal direction among technology/competence-based, market/customer-based, and fund/resource-based directions.

Area selection

- Select the desired area in terms of industry/technology and provide information focusing on the trends of companies in similar/same category, needs of the existing customers/market, global megatrends, and prospective industry/technology roadmaps according to the selected direction.
- Select the final area considering the type of implementation, key considerations, and investment size/period (focusing on eliminating ambiguities and contradictions between standards).

Item review

- Provide information focusing on detailed items according to the selected area with item information held by startups as well as technology held by research institutes.
- Review the provided items together with the business model (BM) and prioritize (focusing on securing concreteness).

Final item selection

- Review/Evaluate the prospective item candidates based on business attractiveness and suitability.
- Select the final target item based on opinions from experts/company (focusing on feasibility review).

2) Product Differentiation Process

Derivation of competitive advantages

- Identify the competitive advantages through comparison/analysis between the existing (target) and competing products.
- Identify differentiation points such as product performance enhancement, addition of new functions, cost reduction, and quality improvement.
- Avoid vague or abstract differentiation strategies, actively consider the market/ competitive environment, and exclude cases that are not realistic or feasible.

Derivation of detailed specifications/conditions

- Prioritize sub-elements to secure competitive advantages.
- Set numerical goals for sub-elements considering the preconditions for goals for individual elements.
- Set the final technical goal by reviewing the feasibility.

Implementation plans

- Conduct information/documentary research in related areas in parallel with offline expert diagnosis.
- Conduct intensive research/analysis on the status of leading products, market/ consumer needs, and product trends.
- Participating organizations in the network conduct information/documentary research, with external organizations to be commissioned if needed.
- Expert diagnosis will be carried out by those specializing in market/marketing, industry, and technology/research.

3) Technological Solution Creation Process

Suggestions for technical solutions

- Dedicated personnel in the network and external experts each create and suggest a technological scenario containing technology and ideas.
- With the participation of external experts in both the same industry and cross-industry, consider the possibility of finding new solutions (ideas) by cross-industry experts.

Review and selection

- Conduct a comprehensive review (evaluation) of the suggested technological scenario to select the best technical solution.
- Include external experts in the feasibility diagnosis and period estimation and reflect the opinions of the adopting company during the review (evaluation) process.

5.5.3.3 Demand Information Analysis

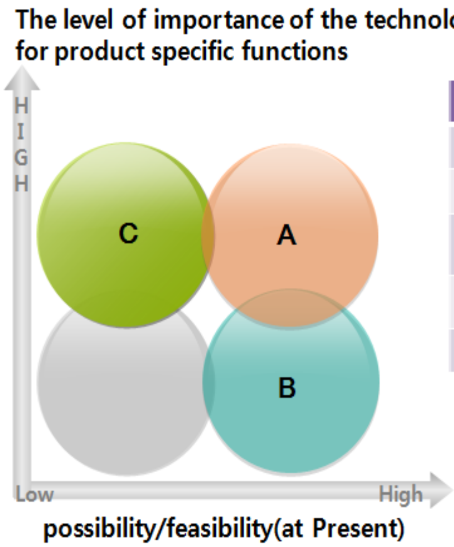
In addition to the demand specification process to specify ambiguous demand, it is also useful to analyze demand information and classify it according to the features and characteristics.

Through classification according to the features and characteristics of demand, the possibility of successful technology transfer can be increased, and it is possible to enhance the value of technology and the concept of products.

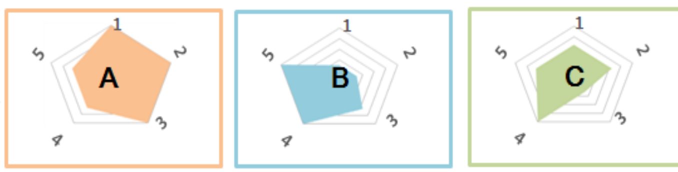
In the figure above, the X-axis means the possibility of existence and feasibility of technology based on the present time and the same field; the Y-axis means the percentage (or importance) of the demand technology in the unique function of the product.

The demand information located in the four areas each has different features and characteristics, showing different characteristics from five perspectives: requiring maximum performance, demand for rapid processing, expectations for completeness of technology, hope for appropriateness, and technical path dependence.

[Figure 3-27] Concept of Demand Information Analysis



		A	B	C
1	Requires maximum performance	H	L	M
2	Hope for appropriability	H	L	M
3	path dependence (Traditional technical method)	H	M	L
4	Expectations for completeness of technology	M	H	H
5	Degree of demand for rapid processing	M	H	M



Among them, the features of the three (A, B, C) areas except the area that shows low levels in all aspects are as follows:

[Table 3-13] Features/Characteristics by Demand Type]

Demand technology type	Content summary
Type A	・This demand technology is commonly recognized as a strategic/core technology by companies due to the high importance of the demand technology in the unique function of the product and high possibility of existence and feasibility of technology. ・Since the performance of technology determines the competitiveness of the product, the demand company tends to pursue high performance and has high motivation to own (protect) the right upon securing the technology. ・Technology in the same field should be explored, but mediation is necessary as the possibility of securing ownership and exclusive right to the technology are likely to become an issue in trading. ・There is less tendency to expect rapid processing and high completeness, and the demand company hopes to secure its unique technology even through additional R&D.
Type B	・This demand technology is commonly seen as a cumbersome technology and is recognized as one of several technologies required to develop/commercialize a product. ・The demand company does not recognize this as a core technology area to secure a competitive edge. It is a technology that does not require a sophisticated level but is simply not owned by the company. ・The company is satisfied with the performance level that meets the purpose, preferring quick processing and simple method. ・In some cases, it may become a key area for product differentiation in the future, so this demand could be sophisticated through consulting (adding new value). ・For simple, cumbersome technologies that need not be sophisticated, quick matching is needed centering on industry technologies.
Type C	・This demand technology is recognized by the demand company as a technical challenge or a mid- ~ long-term R&D goal. ・There is high technical flexibility as it is recognized as a task that cannot be easily solved based on the related field's domestic/overseas level and experience of the demand company. ・Application of technology in other fields should be actively reviewed, and methodologies such as TRIZ and OPIS may be effective. ・When reviewing the introduction of technology from other fields, high completeness can be demanded (expected), and there could be a dilemma in the ownership and problem solving of the technology.

CHAPTER 4

Capacity Development Process

Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan

Chapter 4. Capacity Development Process

1. Kick-off Meeting and Field Study

1.1 Background

The Azerbaijan National Science Academy (ANAS) has submitted the demand survey paper in 2017.

They requested the Korean government to support the capacity development of the Azerbaijan National Science Academy-High Technology Park (ANAS-HTP), a new center for technology transfer and commercialization in Azerbaijan.

To meet the demand from ANAS, STEPI has conducted a local workshop in 2017 and a training workshop for ANAS-HTP in 2018. STEPI trained more than 70 managers and staff of ANAS-HTP and Technology Transfer-related organizations.

As a follow-up of the training, ANAS requested the Korean government to support further the promotion of technology transfer and commercialization in Azerbaijan through policy consultation. This consultation will be composed of a kick-off workshop in Baku, an invitational training, and a dissemination workshop.

STEPI and ANAS jointly conduct the diagnosis of technology transfer ecosystem and share Korea’s experience relevant to Azerbaijan in order to support the promotion of technology transfer and commercialization in Azerbaijan.

1.2 Overview of the Kick-off Meeting and Field Study

- Name of Field Study
  - Field study for the assessment of framework conditions of the technology transfer ecosystem in Azerbaijan

- Period of Field Study
  - 2019.2.18 – 22 (Mon. – Fri.)

- Place of Field Study
  - Baku, Azerbaijan

- Purpose of Field Study
  - Review of the framework conditions of Azerbaijan’s technology transfer ecosystem and current status
  - Identify the specific demand for the 2019-2020 consultation project
  - Interviews and meetings with stakeholders

- Purpose of Kick-off Meeting
  - STEPI will set the detailed promotion direction of the project through meetings and interviews with related parties of the industry-university-institute collaboration for technical transfer in Azerbaijan and through field survey.

<Scope of New Projects in 2019-2020 (Draft)*>

- Support the improvement of operation and management with regard to technical transfer and commercialization in Azerbaijan*

- Summary of Korean experiences (policies, systems, successful cases)
- Institutional framework, governance, and operational structure analysis by ANAS-HTP
- Consultation on the development of an executable master plan for technical transfer and commercialization

▪ Support the development of the technical transfer and commercialization program in Azerbaijan*

- Support policy program design for experts/related parties of ANAS-HTP and private companies in Azerbaijan
- Share the Korean technical transfer and commercialization program and methodology

* The project scope will be confirmed in the local workshop for disseminating the outcome.

▪ Meeting Agenda

- Interview with the new director of ANAS-HTP to share the collaborative results
  - Meeting with related parties to organize the local response team and to identify the demand for invitational training
  - Discuss the 2019 project scope and promotion method with ANAS-HTP, Economic Development Center affiliated with ANAS, ADA University, Innoland, and Baku Engineering University research team
  - Discuss plans for organizing the local response team (max. of 10 people) and invitational training

▪ Hold the Kick-off Workshop

  - Presentation on the issues and status related to technical transfer and commercialization in Azerbaijan

- Introduction on the science & technology policy collaboration project of STEPI for developing countries
- Presentation on the concept, process, policy, and implications of technical transfer and commercialization
- Presentation on the Korean experience of technical incubation, TPM (Technology-Product-Market) connection, entrepreneurship training and success factors, etc.
- Roundtable meeting with Korean experts on technical transfer and commercialization (3 experts)

• Meeting with the main related parties on technical transfer and commercialization in Azerbaijan

- Interview with the main parties of the relevant departments and organizations on technical transfer and commercialization, sharing project promotion methods
- Share experiences in Korean technical transfer, commercialization, and technical incubating support projects

• Local fact finding related to technical transfer and commercialization, inspection of the main organizations related to the industry-university-institute collaboration

• Visit the Korean Embassy in Azerbaijan and KOICA Azerbaijan Office for discussion

- Share the difficulties related to the collaboration performance and project promotion with ANAS
- Introduce the project in year 2019 to discuss the methods of reinforcing science & technology collaboration between the two countries in the future
- Introduce new projects in year 2020 and request continuous interest and collaboration from the embassy and KOICA office

1.3 Research Team

▪ STEPI Research Team (total of 4)

Name	Position/Organization	Role/Area of Expertise
Eun Joo Kim	Head of External Cooperation and Public Relations, STEPI	Project Manager (PM)
So Hyun Kwon	Researcher, STEPI	Project Coordinator (PC)
Sang Hyuk Suh	Professor, Hoseo University	Technology Transfer, Tech Marketing
Jong In Choi	Professor, Hanbat University	Technology Incubation, Entrepreneurship

▪ Azerbaijan Research Team (total of 7)

Name	Position/Organization	Role/Area of Expertise
Mammad Baghir-zada	Deputy Director for Strategic Development, ANAS-HTP	Strategic Planning on HTP
Bunyamin Seyidov	Chief, Department of Relations with Eastern Countries, ANAS Presidium	Global STI Partnership Program Development
Zaur Akhmedzadeh	Head, Technology Transfer and Business Incubator Department, ANAS-HTP	Law in the field of TT
Isa Gasimov	Senior Specialist, Entrepreneurship Development Foundation of the Republic of Azerbaijan	Innovation and Entrepreneurship/ TT and Incubation in Univ.
Samir Veliev	Economist at the Economic Research Department under the Board of Directors, International Bank of Azerbaijan (OJSC)	Economic Analysis on Innovation and Industry

Name	Position/Organization	Role/Area of Expertise
Aysel Abdullayeva	Lecturer in Innovation Risk Management at UNEC Business School, Azerbaijan State University of Economics (UNEC)	Research on Innovation, Entrepreneurship, and Marketing
Vusal Suleymanli	Chief Innovation Officer, Innova Ventures Group	Tech Innovation & Entrepreneurship

1.4 Main Activities and Agendas

- Identify Azerbaijan's specific demand and expectation on the 2019-2020 project
- Interviews with policymakers, practitioners, researchers of TT-related ministries, institutions, research centers
  - An interview was conducted to diagnose the Azerbaijani technology transfer ecosystem and review the current policies and programs.
- Through the wrap-up meeting with Azerbaijan's counterpart team (to be organized by ANAS, Innoland, and Baku Engineering Univ.) the scope of research and activities for the year 2019 will be confirmed.
- The period, beneficiary, and specific topics of the capacity building workshop (in Korea) will be decided.

1.5 Schedule

DATE	TIME	DESCRIPTION	VENUE
Monday, February 18	10:00 – 12:00	• Opening Session - Welcome by the host organization - Introduction of STEPI's activities - Review of the framework conditions of technology - Transfer ecosystem in Azerbaijan (Azerbaijan expert) (1) - Korean experts' presentation (2) - Q&A	ANAS (Central Scientific Library, H. Javid avenue 129)
	12:00 – 14:00	• Lunch	Central Scientific Library
	14:00 – 15:30	• Meeting of Research Team of the 2019 Project (5-10 persons from ANAS, ANAS-HTP, Innoland, BEU, etc.) - - Discussion on the scope of activities in 2019	
	16:00 – 18:00	• Meeting at KOICA and Korean embassy	KOICA and Korean embassy
Tuesday, February 19	11:00	• Sumgait Technologies Park	H.Z. Tağıyev Dist., Sumgait
	19:00	• Dinner (3 from STEPI, 1 interpreter, 10 from the counterpart team)	Hosted by STEPI
Wednesday, February 20	10:00 – 11:30	• ADA University	ADA University (11 Ahmadbay Agha-Oglu Street, Baku)
	11:30-13:00	• Visit to Baku Engineering Univ. • (http://yenifikir.az/en/)	Baku Engineering Univ.

DATE	TIME	DESCRIPTION	VENUE
			(Khirdalan City, Hasan Aliyev str. 120, Baku, Absheron)
	14:00 – 17:00	• Innovation Center LLC	
Thursday, February 21	10:00 – 12:00	• Visit to ANAS-HTP	ANAS-HTP (Baku, Khatai District, Old Ahmedli Settlement, 622nd Street, 9)
	12:00 – 13:30	• Lunch	ANAS-HTP (Khatai District, Old Ahmedli Settlement, 622nd Street, 9)
	14:00 – 17:00	• Closing Session (Stakeholders' Meeting) a. Results of field study (STEPI, ANAS, Technical University, Baku Engineering University, State Oil and Industry University, and Science Technology Park) & Scope of the project for 2019-2020 b. Capacity building workshop in Seoul, Korea c. Final remarks	ANAS (Central Scientific Library, H. Javid avenue 129)
Friday, February 22		Wrap-up Meeting	

1.6 Official Letter for Kick-off Workshop

Official Letter of STEPI	Official Letter for ANAS
• **Fact-finding trips from Feb. 13 to Feb. 14 (Day 3-4, a flexible schedule)** - Site Visits: Technology Transfer Centers, Universities, Technology Parks, etc. - ANAS-HTP is requested to arrange the visits and meetings for fact-finding mission We strongly believe the project will build a closer relationship between our institutes, and furthermore, between two countries. Yours Sincerely, Eun Joo Kim Head, Global Training Program Team Project Manager, STEPI-ANAS-HTP Cooperation Science and Technology Policy Institute (STEPI) January 14th, 2019 Dear President Dr. Akif Alizadeh, On behalf of Science and Technology Policy Institute (STEPI), I would like to express my sincere appreciation to you and your staff at ANAS for your kind cooperation toward the *K-Innovation ODA Program of STEPI*. In 2017 and 2018, STEPI has organized two training workshops in Baku for technology transfer capacity development of ANAS-HTP managers. Based upon our cooperative relationship, STEPI and ANAS-HTP proposed a Korean Official Development Assistance (ODA) project titled *"Policy Consultation for Promotion of Technology Transfer and Commercialization in Azerbaijan"*. To our pleasure, STEPI got an approval from the Korean government for implementation of this year's cooperation project. The Policy Consultation Project consists of a kick-off meeting and capacity building workshop in Baku, a one-week training program in Seoul and a dissemination workshop in Baku. As the first activity for this year, we suggest that we should organize a kick-off meeting and one-day capacity building workshop in Baku in February. During this visit, we would like to meet with our counterpart experts of Azerbaijan and interview relevant stakeholders to have a better understanding of the status and challenges of technology transfer and commercialization in Azerbaijan. To effectively assist capacity building for promotion of technology transfer and commercialization in Azerbaijan, we should like to ask for your continued interest and support. The tentative schedule would be as follows: • **Stakeholders' Meeting on Feb. 11 (Day 1)** - Stakeholders' Meeting at ANAS-HTP with experts from ANAS Research Institutes, Universities, government agency representatives in the field of STI - Meetings with Korean Embassy and KOICA, Baku • **Capacity Building Workshop on Feb. 12 (Day 2)** - National Technology Transfer System, Technology Incubation - ANAS-HTP should identify the targeted group for the workshop and confirm the topics preferred by the group	AZƏRBAYCAN MİLLİ ELMLƏR AKADEMİYASININ PREZİDENTİ THE PRESIDENT OF AZERBAIJAN NATIONAL ACADEMY OF SCIENCES AZ 1001, Bakı şəhəri, İstiqlaliyyət küçəsi, 30 Telefon: (994 12) 492 35 29, Faks: (994 12) 492 56 99 E-mail: president@science.az AZ 1001, Baku, Istiglaliyyat st., 30 Phone: (994 12) 492 35 29, Fax: (994 12) 492 56 99 E-mail: president@science.az № 18-2/31 "06" february 2019 il/y. To: Ms. Eun Joo Kim Head, Global Training Program Team Science and Technology Policy Institute (STEPI) Project Manager, STEPI and ANAS-HTP Cooperation Dear Ms. Eun Joo Kim, Training workshops organized in 2017 and 2018 in Azerbaijan in the frame of cooperation between Azerbaijan National Academy of Sciences (ANAS) and Science and Technology Policy Institute (STEPI) were very significant for the learning of experience of Korean Republic – one of the leading countries of the world in the field of technology and innovation. We highly appreciate approval from the Korean government the project titled "Policy Consultation for Promotion of Technology Transfer and Commercialization in Azerbaijan" of STEPI and ANAS-HTP and support organization of the seminar in Azerbaijan on February 16-22, 2019 within the project. We believe that this step will open wide perspectives for the strengthening the cooperation between our organizations. Sincerely, Academician Akif Alizadeh

1.7 Meeting Minutes

1) Kick-off Meeting

- Opening Address (Nazim Shukurov, Former Director of ANAS-HTP)
  - Among the CIS countries, Azerbaijan established academies such as ANAS for the national economy, and efforts were made to maintain and protect ANAS and academies.
  - The Korean economic development experience and international experiences are learned and applied through the technical transfer and commercialization project between ANAS-HTP and STEPI, so ANAS-HTP is expected to be acknowledged internationally as well.

- Introduction on the Status of ANAS-HTP in Azerbaijan (Vusal Suleymanli, Former Deputy Head of Patent, Certification, Marketing, and Business Incubation of ANAS-HTP)
  - ANAS was established in 1945 and has been operated centrally with regard to scientific policies since 2013.
  - ANAS-HTP was initially operated as an innovation center for almost a year to be established as ANAS-HTP by a Presidential Decree of Azerbaijan in 2016 and has been officially operated since 2017. The slogan of ANAS-HTP is social and business collaboration.
  - In ANAS are a special zone and various residents located in this zone. ANAS-HTP has two buildings, including one library and another located in Ahmedli.
  - The official portal website of ANAS-HTP (www.ameaytp.az) carries information on not only the residents but also on technical commercialization from national requests.
  - Currently, there are ten residents in ANAS-HTP, and they are provided with promotions and other various benefits (place and facilities provided). In addition, ANAS-HTP performs consultation with international companies, and the best example is the Science Genetic Robotics Company.
  - 10 residents perform projects focusing on petrochemistry, HTP, and military, and there are various projects in the simulation stage.
    - ANAS Experimental – Industrial Plant LLC: This oil production company produces over 70 products; especially, naphthalene oil is produced.
    - Millers Oils Azerbaijan LLC: The company was established in partnership with a UK company and should be maintained for 130 years. Especially, the company is famous for producing premium oil.
    - ETP: ETP conducts research for the improvement of innovative products and cutting-edge technology, and military supplies are mostly produced. The products manufactured by ETP are mostly used in agriculture.
    - EPC Group: The company develops heavy industry technology; especially, alunite/aluminum extraction technology is developed.
    - Alquoritmika LLC: The company provides students with education on program

coding.

- DN Technologies LLC: 3D modeling & simulation program development and application technology is developed for the military sector.
- Azeltech LLC: The company develops radiation detectors, and patent rights are registered through ANAS-HTP.
- Alximik-H1 LLC: High-quality detergents and chemical products are manufactured, and production is enabled in partnership with Germany.
- AZMONBAT: The first battery and battery accumulator in Azerbaijan were produced by AZMONBAT.
- BUTA ARMOR LLC: The company is a global company that is famous for producing military uniforms.

• The plan for ANAS-HTP is to carry out development by putting emphasis on the patent right, and information related to the patent right is published in the Patent Bank Section of the official ANAS-HTP portal website (www.ameaytp.az). Consultations are held with about 50 global patent agencies to develop the Patent Bank, and advice from STEPI in Korea is also requested.

• In addition, the ANAS Strategic Scientific Analytic Testing Center is established within ANAS for the purpose of implementing the petroleum-refining process to develop safe toys and dietary products.

▪ Introduction of the 2019 Project (Eun Joo Kim, Head of External Cooperation and Public Relations, STEPI)

• Introduction of STEPI

• Introduction of the 2019 project (background and purpose) and Korean research team

• This project is an ODA project approved by the Korean government. It is not a large-scale project like KOICA but is a policy consulting project with a small budget. Nonetheless, a larger-scale project is expected to be created to contribute to the society and economy of Azerbaijan after 2 years based on strong mutual collaboration and commitment.

- Presentation on Technical Transfer and Commercialization (Professor of Hoseo Graduate School of Global Entrepreneurship)
  - Concept and process of TT/TC
  - Status and policy of TT/TC in Korea
  - Success factors of TT/TC

- Presentation on Technology Commercialization and Entrepreneurship (Professor of Hanbat University)
  - Purpose of HTP
  - Entrepreneurship Ecosystem
  - INC Model
  - T-P-M Linkage
  - Entrepreneurial Education Center

2) Research Team Meeting: Discussion on the project scope activities and work plan

STEPI emphasized that the most important factor in promoting the project is to clarify the project purpose, time, resources, and scope.

STEPI requested the Azerbaijan research team to clarify the project scope from the institutional, organizational, and program perspectives; clear awareness of the problem was requested as to what results will be obtained from the project as well as the program target and beneficiary of the program.

STEPI explained the overall work plan (draft) of the project and requested the following matters to Azerbaijan's research team:

- Selection of contact person in Azerbaijan
- Organization of the local response team in Azerbaijan:
  - Selection of local expert to prepare the draft report on the status of the TT ecosystem in Azerbaijan until April

- Request the selection of candidates for invitational training in Korea
  - Selected as candidates are those who can share and disseminate the contents and experiences of Korea to Azerbaijan through the 1-week program (working hours: 5 days) (max. of 10 people: ANAS-HTP manager, manager of the relevant organization, and policymakers)
  - Since it is a two-year project, candidates are selected for 2019 (1^{st}) and 2020 (2^{nd})

[Table 4-1] Work Plan on Policy Advice to Azerbaijan in 2019 (draft)

Schedule	Plan
Kick-off Meeting & Field Survey (February)	• Materialization of demand regarding the 2019-2020 project • Interview with related parties and field survey for identifying the status of the technical transfer ecosystem in Azerbaijan
Azerbaijan's TT Ecosystem and Report on the Korean Experience (March~April)	• Report on the status of Azerbaijan by Azerbaijan's research team • Report on Korea's experiences in TT by Korean experts
Invitational Training on Empowerment (June)	• Invite Azerbaijan TT experts (max. of 10 people) to Korea for on-site inspection and matching lecture
Workshop on the Dissemination of Korean Experience (October)	• Workshop on sharing and disseminating the Korean experience in Azerbaijan
Final Report (November)	• Publication of final report on the project

In 2018, this project was approved as a two-year (2019-2020) project, but only this year's budget was approved for the activity budget. Therefore, ANAS-HTP was reminded that an official letter from PCP and the relevant organizations (official document from the Ministry of Foreign Affairs) must be submitted until the end of February to approve the project budget for 2020.

Accordingly, ANAS-HTP revealed that the procedure above is currently in progress.

Zaur Akhmedzadeh (Team Leader) from ANAS-HTP explained that ANAS currently entrusted all work related to the patent right to ANAS-HTP, and that there are 450 patent rights registered internationally. Also mentioned was the great interest in the Korean experiences regarding the transfer of almost 500 patented technologies.

There is a technological assessment mechanism called TPM (Technology Portfolio Management), which tests the technicality, marketability, introduction, and project feasibility to provide matching with the company according to the results.

The Recommendation Factors in the Selection of Project Scope are as follows:

- Pursuit of convergence project (Education & Training + Consulting: Including investigation, analysis, and research)
- Determination of Main Function Required: Technical Transfer? Spin-off Startup? Techno Entrepreneurship? Hi-Tech Marketing?
- For technical transfer and commercialization, introduction of foreign technology is recommended along with domestic technology
- Will only the exclusive organization be included in the process of education & training for the technical project? Will researchers be included? Will public research institutes be included?

3) Visit to the Korean Embassy in Azerbaijan

(Eun Joo Kim, STEPI Team Leader) The technical transfer and commercialization project by STEPI-ANAS-HTP in 2019-2020 was introduced, along with the activities performed by STEPI. Difficulties (attitude and will of ANAS-HTP, limitations of the newly established organization, etc.) were mentioned regarding the project with ANAS-HTP in the current project initiation stage.

(Tong-op Kim, Ambassador) Azerbaijan is a country that still maintains the socialism of the Soviet Union. It is a nation rich in resources and is easily influenced by the change in oil price, so it has difficulty in recognizing problems independently. It can be somewhat slow in performing a project, but it was emphasized that time is required for the understanding of others.

Azerbaijan is similar to Korea in terms of tendency and temperament, traditionally showing conservative tendency. Strictly speaking, Azerbaijan is located in Europe but is more intimate with Central Asia that it has established emotional connection. In addition, Azerbaijan and Korea share the consensus of suffering from the tragedy of national division due to the ceasefire state with Armenia that people consider Korea to be special.

Currently, there are no issues regarding residence, private education, land and school (university tuition is also almost free), and housing in Azerbaijan. Azerbaijan is a country rich in resources and without any worries regarding livelihood, so the people do not work as much as we do in Korea. Since the people participating in the innovation program are special, however, we recommend that they perform the project by putting in more hours.

4) Visit to the Sumgait Technologies Park (STP)

STP was operated beginning 2009 and is the first technical complex in the Sumgait region to have manufacturing factories in various fields.

The total area of STP is 250 hectares, and the indoor production area is 50 hectares. Currently, more than 30 products are being produced in 12 factories, and there are a total of 2000 employees working in the factories.

The main products include electrical device, cable, machinery, high-polymer products, steel manufacture, overhead crane, zinc galvanizing, prefabricated panel and PVC window, etc.

Products manufactured in STP are being exported to Russia, UK, Denmark, Germany, Turkey, Georgia, Kazakhstan, Tajikistan, and Uzbekistan.

There is no incubation center within STP, but it is currently under planning.

5) Visit to ADA University

ADA University was established in 2006, and there are total of 4 schools (Public and International Affairs, Business, Education, IT, and Engineering).

The School of IT and Engineering was established in 2014, offering both undergraduate and graduate degrees.

ADA University aims for practical education through courses on Big Data study, AI application program and technology entrepreneurship, etc.

Nonetheless, there are difficulties associated with lack of faculty, and exchange and cooperation of research team and education & training program are required.

There is an incubation center, but it is still in the initial stage; funding was recently received from BP Company to operate the research center.

Advisers Lee Soo-hyun and Choi Jin-hyuk were dispatched from Korea through the NIPA Program.

There was an exchange of IT engineering program with KAIST in 2014.

The Information Olympiad will be held in Japan this year jointly with the School of Education and School of IT.

STEPI is also interested in ADA University for the activation of technical transfer and commercialization from the national perspective of Azerbaijan.

6) Visit to BEU

BEU (formerly Qafqaz University) is a national university established by a Presidential Decree two years ago in 2016.

BEU has 3 departments (Economics and Policy Making, Engineering, and Pedagogy); as the university is currently seeking to specialize in Engineering, the Department of Pedagogy will be closed after 2 years. There are a total of 16 majors in the Department of Engineering.

70% of the class is in English, followed by 25% in Azerbaijani language and 5% in Russian.

There is great interest in the internationalization of the technology sector as well in the national level, and emphasis is placed on the necessity of technical transfer and HTP; thus, BEU is aiming to be a technology-specialized university centrally in English. More focus will be given on mechanical engineering, computer science, and IT.

Currently, BEU has a total of 4,700 students and almost 300 faculty members.

The current Rector of BEU was the former Rector of Technical University for 17 years, having participated in the 3-year project (Construction of the Electronic Management System in Technical University) with KOICA. During this project, the CEO of Korea Polytechnics Colleges also visited the Technical University.

BEU has great interest in the collaboration with HTP on a national level. HTP experts are invited as lecturers in BEU.

BEU is operating a project referred to as NYC (New Idea Competition).

New Idea Competition (NYC) is a project hosted by BEU Technopark (Baku Engineering University Technopark), and support is provided to cultivate innovative entrepreneurship.

In addition, NYC is contributing to the improvement of entrepreneurship of the youth and qualitative improvement of scientific research and manpower training.

NYC plays the role of a bridge between the youth and entrepreneurs. Held once a year, this contest can be participated in by not only the main group of BEU but also by the youth all around Azerbaijan.

The main missions of NYC are as follows:

- Support innovative entrepreneurship
- Contribute to providing the infrastructure for the ecosystem to enable the development of cutting-edge technology by BEU Technopark
- Support socio-economic improvement
- Support ICT development
- Support youth startup
- Promote the entrepreneurial approach

The purpose of NYC is to contribute to discovering innovative ideas in the technology area. These areas include startup idea, mobile APP, Internet, program provision, game, service program, corporate software, new business model, service robotics, information technology, consumer electronics, medical information technology, etc.

7) Visit to Innoland (Incubation and Acceleration Center)

The Incubation and Acceleration Center, or “INNOLAND,” is an agency affiliated with the State Agency for Public Service and Social Innovations*.

* As a presidential institution established in 2012, Innoland is responsible for operating the public service center and promoting government computerization; integrated management is provided to the “ASAN Service” Center, the government service provided to the citizens of Azerbaijan.

“INNOLAND” is an innovation center that supports the establishment of the startup ecosystem, and innovation and development of the private sector inside and outside the borders of Azerbaijan are promoted. In addition, tools and community are provided to the entrepreneurs to develop and acquire innovative business.

The main activities are as follows;

- Startup Accelerator: Mentoring support, startup network, access to investors, legal guideline, development of skills, and consistent support after finishing the program
- Incubator: Technical development, administrative support, startup network, facility, linkage, access to capital, mentoring and specialized service
- Coworking Space: Open 24/7 all year round, locker, high-speed Wi-Fi, coffee/tea/water, technical equipment, max. of 4 members per team, 2.5 m^2 per person, 150AZN per month of personal desk per person (about 88 dollars), 125AZN (about 73 dollars) for 2 additional members per team
- Tech Academy: Full-stack Python coding, data science, AI, mentoring support, 24/7 lab use, intensive coding boot camp, real-life project experience

The main services are as follows:

- Services for enterprises: Networking and mentoring, meeting with investors, technical transfer and commercialization support, social contribution activities, business plan preparation, and cooperative acceleration programs
- Service for startups: Commercialization, business consulting, legal service, mentoring, marketing and sales service, search for financial opportunities, Microsoft BizSpark service, media, PR, communication, prototype lab
- Promotion of mechanism: Tax refund, finance rebates, custom rebates, creation of special type of visa

8) Visit to ANAS-HTP

As an organization established in ANAS by a Presidential Decree in 2016, ANAS-HTP supports the development of innovative and hi-tech industries, scientific research, and new technology development and development of an innovation complex. In 2017, BIC (Business Incubation Center) and TTP (Technology Transfer Office) were established in ANAS-HTP.

Oil residency has been located in the current ANAS-HTP since the Soviet times in 1974, and Azerbaijan's President established the testing center within ANAS-HTP along with the oil and chemical plants to maintain the scientific industrialization and functions of science.

Currently, there are 10 Residents* in ANAS-HTP, and the purpose of these companies is technological innovation. The companies mostly perform projects related to oil-chemistry, HTP, and national defense, and there are also several projects in the simulation stage.

* 1) ANAS Experimental – Industrial Plant LLC, 2) Millers Oils Azerbaijan LLC, 3) ETP, 4) EPC Group, 5) Alquoritmika LLC, 6) DN Technologies LLC, 7) Azeltech LLC, 8) Alximik-H1 LLC, 9) AZMONBAT, 10) BUTA ARMOR LLC

The main mission of ANAS-HTP is to create an environment for linkage between the basic science and applied science researchers working for private and state enterprises through technical transfer and commercialization. Another mission is to contribute to national economic development through scientific and technical innovation and to improve the national innovation system by successfully attracting projects and investments.

The vision is to become the hub of hi-tech product design and production in Southern Caucasus and Central Asia until 2030.

ANAS-HTP is still in the initial development stage of the organization and system, so there is great interest in the establishment of an institutional-level science and technology strategy, reinforcement of capabilities in the management and operation of the science & technology innovation complex, technical transfer and commercialization program, improvement of the relevant operation and management systems, etc.

9) Wrap-up Meeting: The project contact point, team organization, scope of activity, and qualification of candidates for the invitational training were discussed.

Both parties discussed the main agenda (contact point, team organization, scope of activity, and qualification of the candidates for the invitational training on the 2019 project).

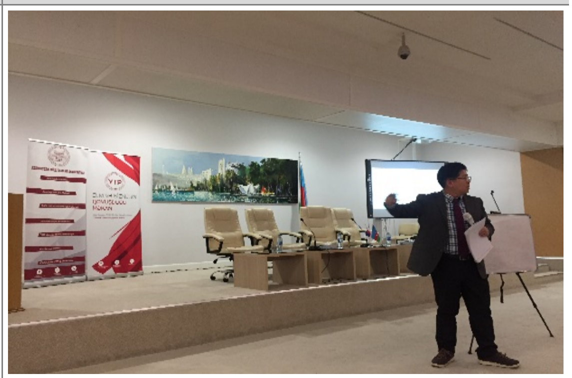
STEPI requested ANAS-HTP to make the final decision on the following discussions to be notified by mid-March ~ early April:

[Table 4-2] Main Discussions of the 2019 Azerbaijan Kick-off Workshop

Main Agenda	Discussions
1) Contact Point	Bunyamin Seyidov (Head of Division of Relations with Eastern Countries, ANAS Presidium): Bunyamin was selected since he is directly connected to the ANAS President and can be expected to give a quick response on promoting the project. - Decision Maker: Akif Alizadeh (ANAS President) - Prof. Esmirae Alirzayeva (Head of the International Cooperation Department) - Ms. Fatali Abdullayev (Head of the Administrative Department)
2) List of Research Teams (Counterpart) (End of March)	The participation of not only the HTP manager but also the technical transfer-specializing organization (Innoland, etc.) is promoted, and the target for reinforcing capability is selected in detail for the further development of HTP; participation must be enabled continuously for 2 years. **(ANAS)** - Bunyamin Seyidov - Rufat Afandiyev (Institute of Economics) **(ANAS-HTP)** - One person among the Deputy Directors - Vusal Suleymanli **(Innoland)** - Zafar Hasanov (Project Team Leader) - Mammad Karim (Managing Director) **(Innovation Center LLC)**

Main Agenda	Discussions
	- Aysel Abdullayeva (Specialist) - Samir Valiyev (Specialist): Currently working as consultant of the Central Bank of Azerbaijan
3) Fact finding on the Technical Transfer Ecosystem in Azerbaijan (mid-April)	Preparation of fact finding report on the technical transfer ecosystem in Azerbaijan - Preparation of about 20~25 pages until mid-April The contents (draft) must be discussed with STEPI before preparing the report (during March) (Drafter) - Aysel Abdullayeva (Specialist) - Samir Valiyev (Specialist)
4) Qualification and Selection of Candidates for Invitational Training (mid-March ~ early April)	**(Qualification of the Candidates)** - Person working at the Technology Innovation Center or Research Innovation Center - Person who can communicate in English (Listening, Writing, Speaking) - Person with outstanding capability in preparing the research plan (plan before/after the training) (The application method and performance disseminating method must be described; especially, preparation is required on the value point. For example, the value point can be on searching for the utilization method of the main technology of Azerbaijan on the naphthalene technology.) - Person who can actively participate in education and training, has the capability and attitude necessary for execution **(Advance Preparation)** - STEPI will prepare the letter of invitation and other administrative matters from the beginning of May.
5) Project Scope Activity (mid-March)	Clear awareness of the problem is required as to why collaboration is made with Korea and what is the goal for learning including the exclusive organization for technical transfer. Understanding of the concept of TP/TC differs, so the concept is currently not established when visiting Azerbaijan's TP. Therefore, the detailed demand as to the field of technical transfer and commercialization must be discussed between the relevant agencies to notify STEPI through the preliminary questionnaire (to be sent by STEPI before the business trip).

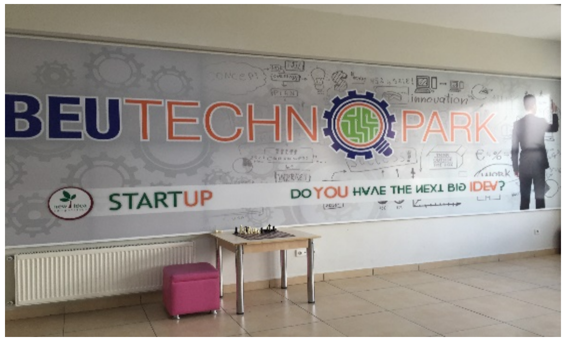
1.8 Photos

Group Photo of the Kick-off Workshop	Discussion Session
	
Lecture of Prof. Sang Hyuk Suh	**Lecture of Prof. Jong in Choi**
	
Presentation of ANAS-HTP	**Q&A**
	

<Research Team Seminar>	Meeting with the Korean Embassy and KOICA
	
Site Visit to STP	Meeting with STP
	
Meeting with ADA University	ADA Innovation Center of ADA University
	
Meeting with BEU	BEU Incubation Center (1)
	

BEU Incubation Center (2)

<BEU Incubation Center (3)



Innoland (1)

Innoland (2)

Innoland (3)

Group Photo in Innoland

ANAS-HTP (1)

ANAS-HTP (2)

ANAS-HTP (3)	ANAS-HTP (4)
	
Millers Oils Azerbaijan LLC (1)	**Millers Oils Azerbaijan LLC (2)**
	
Group Photo in ANAS-HTP	**Wrap-up Meeting**
	

1.9 Publicity Activities (Press Releases)

- STEPI-IICC Facebook (2019.2.21)
  (https://ko-kr.facebook.com/pages/category/Government-Organization/STEPI-IICC-625865667508308/)

【2019 K-Innovation ODA Program with Azerbaijan】 Kick-off Workshop for Promotion of Technology Transfer and Commercialization in Azerbaijan

STEPI held the Kick-off Workshop on Promotion of Technology Transfer and Commercialization of Azerbaijan in collaboration with ANAS-HTP* in Baku from 18th to 22th February 2019. This workshop was part of 2019 K-Innovation ODA Program with Azerbaijan**. This workshop is to diagnose the Azerbaijan technology transfer ecosystem and to define the scope of research for the year 2019. To achieve this purpose, STEPI's researchers and Korean experts shared Korea's relevant experiences with local stakeholders by conducting a workshop and site visits.
Moreover, the STEPI team visited ANAS –HTP, INNOLAND (Incubation and Acceleration Center), ADA university, Baku Engineering University and Sumgait Technology Park to interview practitioners, officials, entrepreneurs, professors and experts who are playing important roles in the field of technology and transfer and commercialization in Azerbaijan.
Through in-depth discussion with major innovation actors of Azerbaijan, the STEPI team could plan the process for the project management in an effective way.
* STEPI's 2019-2020 Policy Consultation Project by Korean government's ODA fund
**ANAS-HTP (Azerbaijan National Academy of Sciences-High Tech Park): ANAS-HTP was established in 2016 for the development of innovation and high-technology industries, scientific research and new technologies in order to increase state support for the creation of modern innovative complexes. BIC and TTO started operation in 2017.

- STEPI-IICC Homepage (2019.4.22)
(http://iicc.stepi.re.kr/0202/view/id/417)

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(Azerbaijan) Kick-off Workshop for Promotion of Technology Transfer and Commercialization in Azerbaijan

Name	홈페이지 관리자	Date	2019.04.22 10:37	Views	356

[2019 K-Innovation ODA Program with Azerbaijan]

STEPI held the Kick-off Workshop on Promotion of Technology Transfer and Commercialization of Azerbaijan in collaboration with ANAS-HTP* in Baku from 18th to 22th February 2019. This workshop was part of 2019 K-Innovation ODA Program with Azerbaijan**. This workshop is to diagnose the Azerbaijan technology transfer ecosystem and to define the scope of research for the year 2019. To achieve this purpose, STEPI's researchers and Korean experts shared Korea's relevant experiences with local stakeholders by conducting a workshop and site visits.

- ANAS-HTP (2019.2.18)
(http://science.az/gallery/open/9799)

ANA SƏHİFƏ » XƏBƏRLƏR » KONFRANSLAR, İCLASLAR

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FOTO
VIDEO (ELM TV)

18.02.2019 14:40

Azərbaycanda texnologiya transferi və elmin kommersiyalaşdırılmasına həsr olunan seminar keçirilib

AMEA-nın Yüksək Texnologiyalar Parkı (YT Park) və Cənubi Koreyanın Elm və Texnologiya Siyasəti İnstitutu (STEPİ) tərəfindən həyata keçirilən "Azərbaycanda texnologiya transferi və elmin kommersiyalaşdırılmasının təşviqi" layihəsi çərçivəsində seminar keçirilib.

Mərkəzi Elmi Kitabxanada baş tutan tədbirdə YT Parkın, STEPİ-nin, Cənubi Koreyanın Hoseo və Hanbat universitetlərinin əməkdaşları, o cümlədən AMEA-nın elmi müəssisə və təşkilatlarının nümayəndələri iştirak ediblər.

Seminarı YT Parkın direktoru, iqtisad üzrə fəlsəfə doktoru Nazim Şükürov açaraq AMEA-nın ölkəmizdə elm sahəsində dövlət siyasətini həyata keçirən ali təşkilat olduğunu deyib. Bildirib ki, bu gün Azərbaycan alimləri öz innovativ ideyaları və layihələri ilə respublikanın iqtisadiyyatına töhfə verməkdədirlər.

N.Şükürov 5 gün davam edəcək seminarın əsas məqsədinin Azərbaycanda texnologiyaların transferi və elmi nailiyyətlərin kommersiyalaşdırılması imkanlarının öyrənilməsi olduğunu bildirib.

Hazırda Cənubi Koreyanın Asiya qitəsinin ən inkişaf etmiş dövlətlərindən biri olduğunu deyən alim qeyd edib ki, sözügedən ölkədə adambaşına düşən ümumi daxili məhsul 30 min dollardan çoxdur və bu baxımdan Cənubi Koreyanın təcrübəsinin öyrənilməsi olduqca vacibdir.

Sonra YT Parkın Patent, sertifikatlaşdırma, marketinq və biznes inkubasiya şöbəsinin müdir müavini **Vüsal Süleymanlı** təmsil etdiyi qurumun fəaliyyət prosesi barədə təqdimatla çıxış edib. V.Süleymanlı qurumun rezidentləri tərəfindən hərbi və mülki təyinatlı dronların, qoruyucu jiletlərin, akkumulyatorların, sürtkü yağlarının və digər məhsulların istehsal olunduğunu bildirib. O, həmçinin YT Parkın rezidentlərinə qanunvericiliyə müvafiq qaydada vergi və gömrük güzəştlərinin tətbiq edildiyini vurğulayıb.

Bundan əlavə, V.Süleymanlı AMEA-nın elmi müəssisə və təşkilatları tərəfindən rəsmiləşdirilmiş ixtiraların patent bankının yaradıldığını diqqətə çatdırıb.

Daha sonra STEPİ-nin Beynəlxalq İnnovasiya Əməkdaşlıq Mərkəzinin nümayəndələri - təlimlər qrupunun rəhbəri **Eun Yo Kim**, layihənin koordinatoru **So Hyun Kvon**, Hoseo Universitetinin professoru **Sanq Hyuk Suh** və Hanbat Universitetinin professoru **Yonq İn Çoi** Azərbaycanda texnologiya transferi ekosisteminin hazırki statusunun qiymətləndirilməsi, bu sahədə fəaliyyət göstərən təşkilatlarla əməkdaşlığın müzakirəsi, eləcə də texnologiya transferi və elmin kommersiyalaşdırılması sahəsində Cənubi Koreya təcrübəsinə dair təqdimatlarla çıxış edib, iştirakçıların suallarını cavablandırıblar.

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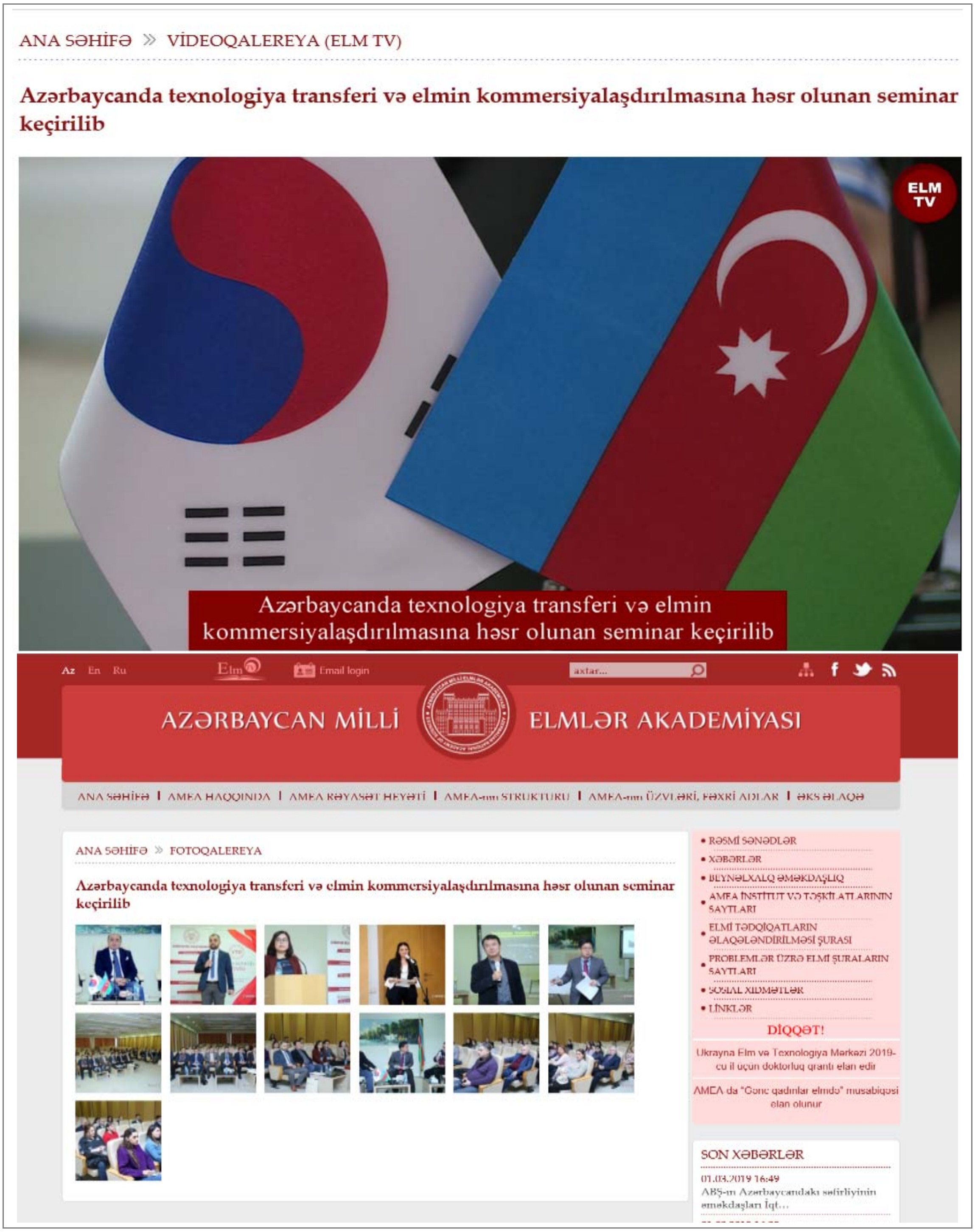
ANA SƏHİFƏ » VİDEOQALEREYA (ELM TV)
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ANA SƏHİFƏ » FOTOQALEREYA
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ELMİ TƏDQİQATLARIN ƏLAQƏLƏNDİRİLMƏSİ ŞURASI
PROBLEMLƏR ÜZRƏ ELMİ ŞURALARIN SAYTLARI
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Ukrayna Elm və Texnologiya Mərkəzi 2019-cu il üçün doktorluq qrantı elan edir
AMEA da "Gənc qadınlar elmdə" müsabiqəsi elan olunur
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2. STI Capacity Development Workshop: Invitational Training Program

2.1 Background

As the first stage of the project, the Korean researcher team specified the demand for the 2019-2020 project through a kick-off meeting with the relevant stakeholders and conducted fact finding on the status of the technology transfer ecosystem in Azerbaijan from February 16 to 23, 2019.

In order to promote the organizational and institutional development of ANAS-HTP, ANAS-HTP and STEPI analyzed the need for capacity development on HRD in the field of Technology Transfer and Commercialization. In particular, Azerbaijan's research team requested the training for sharing Korean experiences and practical skills on TT and TC.

Azerbaijan's 7 practitioners were invited to Korea for the Capacity Development Workshop. The Azerbaijani experts' team consisted of the appropriate trainees in the field of TT and TC in Azerbaijan including ANAS-HTP. To select the right target and to ensure effective training, STEPI planned and considered the application process, and ANAS recommended the applicants.

The 2019 training program was designed to improve the relevant knowledge and work skills in developing the TT and TC system at the institutional level. The training consisted of 6-day intensive curriculum with lectures, Q&A, group discussions, TT action planning, study visits and cultural visits, etc. from May 27 to June 1, 2019.

2.2 Overview of the Capacity Development Workshop

- Title: STI Capacity Development Workshop: Promotion of Technology Transfer and Commercialization in Azerbaijan
- Dates: May 27 (Mon.) – June 1 (Sat.), 2019
- Venue: Shilla Stay Gwanghwamun Hotel, Seoul, Republic of Korea
- Purpose

In this invitational training, the capabilities of the exclusive organization for technical transfer are to be improved by providing the practical methodology for technical transfer and commercialization through a series of lectures as well as Q&A related to technical transfer, commercialization, marketing, and startup and by promoting group practice (including preparation of action plan), final matching lecture, visit to institutions, etc.

- Subject of Invitational Training

STEPI placed emphasis on reinforcing sustainability and professionalism through the intensified training course instead of being a one-time, single training, with the qualification of the candidates for the training proposed under discussion with ANAS on selecting the trainees to compose the invitational training as those meeting the qualification.

[Table 4-3] Trainee Qualification

・Person recommended by ANAS-HTP
・Person working at the Technology Innovation Center or Research Innovation Center
・Person who can communicate in English (Listening, Writing, Speaking)
・Person with outstanding capability in preparing the research plan (plan before/after the training) (The application method and performance dissemination method must be described; especially, preparation is required on the value point. For example, the value point can be on searching for the utilization method of the main technology of Azerbaijan on the naphthalene technology.)
・Person who can participate actively in the education and training, has the capability and attitude necessary for execution

The STEPI research team shall review the Applications Forms I and II submitted by the applicants (11 people) with the qualifications above for the final selection of 7 trainees. The matters to be prepared in advance are notified to the seven trainees who are ultimately selected, and an official letter on the invitation to the training is sent.

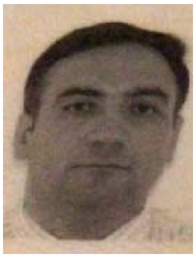
- Application Form I: Personal information, professional field and relevant work experience, language skills, etc.
- Application Form II: Purpose of applying for training, career, and experiences related to the training topic, challenge of the work, applicability to the actual work, plan for applying and disseminating the knowledge acquired after the training, expected results and suggestions, etc.

For internal stability in operating the training course and to improve the effectiveness of the training, the STEPI research team emphasizes trainee selection, program reflecting the needs of the trainees, training for trainers' program, qualified instructors and visit to the class, discussion and site, etc.

- STEPI Research Team

Organization	Name	Position	Role
Science and Technology Policy Institute (STEPI)	Eun Joo Kim	Head of External Cooperation and Public Relations	Project Manager
	So Hyun Kwon	Researcher of the Division of Global Innovation Strategy	Project Coordinator

- Participants (total of 7)

No.	Photo	Name	Organization	Position	Area of Expertise
1		Bunyamin Seyidov	Presidium of ANAS	Head of Relations with Eastern Countries	Focal point (Group leader) International relations in the field of TT and TC
2		Mammad Baghir-zada	ANAS-HTP	Deputy Director for Strategic Development	Strategic Planning on HTP
3		Zaur Ahmadzada	ANAS-HTP	Head of Organizational Issues	Law in the field of TT

No.	Photo	Name	Organization	Position	Area of Expertise
4		Vusal Suleymanli	ANAS-HTP	Deputy Head of Patent, Certification, Marketing, and Business Incubation	Founding Director of TT and Incubation
5		Isa Gasimov	Entrepreneurship Development Foundation of the Republic of Azerbaijan	Senior Specialist	Innovation and Entrepreneursh ip/TT and Incubation in Univ.
6		Samir Veliev	International Bank of Azerbaijan (OJSC)	Economist at the Economic Research Department under the Board of Directors	Economic Analysis on Innovation and Industry
7		Aysel Abdullayeva	Azerbaijan State University of Economics (UNEC)	Lecturer in Innovation Risk Management at UNEC Business School	Research on Innovation, Entrepreneursh ip, and Marketing

- Lecturers (total of 13)

No.	Name	Organization	Position	Lecture Name	Syllabus
1	Sang-Hyuk Suh	Hoseo Graduate School of Global Entrepreneurship	Professor	Technology Transfer: Overview, Support Programs & Implications for Azerbaijan	• Definition and overall process of TT/TC: Concept, Process, Success Factors • Related status and programs in Korea • Implications for Azerbaijan
				Technology Transfer Action Planning	• Grouping & Task Assignment of Trainees • Action Planning
				Technology Marketing as a New Paradigm of Technology Transfer	• New paradigm of TT • Principle, procedure, and points of technology marketing • New paradigm of TT • Definition of Techno Marketing
				Comparative Analysis of TT System between the 2 Countries & Group Presentation	• Future Plans of TT System in Azerbaijan • Group Discussion • Action Planning
2	Oong-Hyun Sung	Hanshin University	Professor	Evaluation Model for the Selection of Transferable Patent Technology	• Linking intellectual capital with corporate value • Factors affecting patent technology value • Patent selection evaluation for technology transfer
				Financial Support for Technology Transfer & Commercialization	• Financial demand for technology - commercialization • Change in the concept of financial support • Evaluation guarantee

No.	Name	Organization	Position	Lecture Name	Syllabus
					for TT and TC • Loan financing through technology credit bureau
3	Seong-Kun Kang	Chungnam Technopark	General Manager	Practice of Technology Transfer in Technopark	• Overview of methods and procedures for technology transfer in technopark • Methods and procedures for obtaining and analyzing information on technical needs • Technology transfer cases in Korea • Creation of technical needs by idea • Application of technology in other fields • Mediation & Negotiation
4	Gyeongsoo (John) Shim	Korea University Research and Business Foundation (KURBF)	Professor/ Chief Technical Officer	Current Status of Korean Universities' TTO and Challenge for the IP Globalization Business	• How a university's IP assets can be commercialized further • How to evaluable the value of each of the intellectual assets • Comparison between the US and South Korea's IP market policies and university technology transfer processes
5	Yong-Soo Hwang	Science and Technology Policy Institute (STEPI)	Senior Research Fellow Emeritus	Policies for Technology Transfer: Korean Experiences	• Channels of technology transfer • Changes in legislative framework and policy directions for technology transfer • Policies for international

No.	Name	Organization	Position	Lecture Name	Syllabus
					technology transfer • Policies for domestic technology transfer • Implications for developing countries
6	Kyung-Min Chung	Dowool IP Group	Representative Patent Attorney	IPR Management for Patent Transfer	• What is IPR (briefly) • IPR valuation What is IPR valuation? 4 factors (scope, technical, market, business) Evaluation methods and process • IPR management What is IPR management? Why do IPR management? • Marketing process
7	Hyun-Gon Shin	Sejong University	Professor	Evaluation and Management of Public Technology	• Technology assessment and evaluation • Factors for evaluating technologies • Introducing the evaluation tools in various organizations (including NTC) • Introducing the technology management systems in Korean public sectors
8	Seung-Ho Lee	Delta Tech-Korea Ltd.	CEO	Global Technology Transfer between Firms	• Firm's motive for trading technology globally • Type of global technology transfer • Global technology trade network: EEN • Possible technology cooperation b/t

No.	Name	Organization	Position	Lecture Name	Syllabus
					Azerbaijan and S. Korea
9	Byung-Jo Choi	Incheon National University	Professor of Startup Center	Startup Support Programs in Korea	• Startup environment and the government's policy direction for supporting startups in Korea • Support programs for each stage of the startup's growth • Introduction of Incheon National University Startup Center
10	Wang-Kyu Lim	Walton Technology	Adviser	Technology Startups: Strategy & Management to Facilitate	• Concept and characteristics of techno-startups • Technology trends • Success factors of technology entrepreneurship • Supporting Policy Suggestion
11	Sun-Young Lee	Hoseo Graduate School of Global Entrepreneurship	Associate Professor	Case analysis of successful technology spin-offs in Korea	• KolmaBNH Korea • Success factors: technology perspective & market perspective • Success case of Samsung Electronics C Lab (Mango Slave, Welt, Aimt, etc.) • Hyundai Motor "Venture Plaza": 9 success cases • Institute spin-off cases: newratek, i-SENS, Inc.
12	Young-Taeck Lim	Dowools International Patent Law Firm	Patent Lawyer	Patent Management: Overview & Evaluation	• Introduction: Why Apply for Patent (including several cases of IP systems)? • What is patent? • How to acquire

No.	Name	Organization	Position	Lecture Name	Syllabus
					patent? (Patentability requirement) • Patent management and basic IP portfolio
13	Sungchul Shin	Korea Institute of Science and Technology (KIST)	Principal Specialist	Commercialization of Innovative Technologies in KIST	• Introduction of the Technology Commercialization Team of KIST • Commercialization of Innovative Technologies in KIST

2.3 Objectives and Expected Outputs

- ANAS-HTP of Azerbaijan has the goals of accelerating technology transfer and commercialization.
  - Improving the operation and management system of Technology Transfer and Commercialization in Azerbaijan (2019)
  - Supporting technology transfer and commercialization program development in Azerbaijan (2020)
  - Setting the technology transfer and commercialization capacity building
  - STEPI and ANAS jointly conduct the diagnosis of technology transfer ecosystem and share Korean experiences relevant to Azerbaijan in order to support the promotion of technology transfer and commercialization in Azerbaijan.

2.4 Organizers

This workshop is organized by the Science and Technology Policy Institute (STEPI) in partnership with ANAS.

2.5 Training Methodology

The tailored training consisted of four methods: (1) lectures and Q&A, (2) case studies of best practices, (3) group discussions and trainees' reporting, and (4) action planning. During the workshop, it is strongly recommended that the participants keep on discussing with the Korean experts intensively as well as deriving implications and lessons from lectures and group discussions. For the development of a practical action plan, the trainers will give feedback consistently to respond to the trainees' curiosity and questions.

Methods	Module Details
Lectures and Q&A	• Lectures on Technology Transfer and Commercialization Capacity Development in 5 aspects; 1) Definition of TT and TC 2) Management of TT and TC at the institutional level 3) Evaluation of TT and TC 4) Tech Entrepreneurship and Startup Program 5) Global TT and TC
Case Studies of Best Practices	• Case study of Korea's TT and TC
Group Discussions and Trainees' Reporting	• To share trainees' views and opinions through presentation of individual study • Group discussions on the status analysis of Azerbaijan and feedback provided & facilitated by an expert • To develop the group's action plan
Action Planning	• To specify TT project formulations • To promote voluntary knowledge dissemination

2.6 Program Schedule

Day	Time	Program
May 25 (Saturday)	22:15 – 23:59	Departure from Baku (Qatar Airways, QR0354)
May 26 (Sunday)	02:00 – 16:55	Arrival at Incheon Airport (Qatar Airways, QR0858)
	20:00 – 21:00	Dinner
May 27 (Monday)	09:00 – 11:00	**[Lecture 1]** Policies for Technology Transfer: Korean Experiences Dr. Yong-Soo Hwang, Senior Research Fellow Emeritus, STEPI
	11:00 – 12:00	**[Orientation]** Opening Remarks and Introduction on Training Ms. Eun Joo Kim, Head, External Cooperation and Public Relations, STEPI Ms. So Hyun Kwon, Researcher, Division of Global Innovation Strategy, STEPI Introduction of the Workshop and Participants
	12:00 – 13:30	Welcome Luncheon
	13:30 – 16:30	**[Lecture 2]** Technology Transfer: Overview, Support Programs & Implications for Azerbaijan Dr. Sang-Hyuk Suh, Professor, Hoseo Graduate School of Global Entrepreneurship
	16:30 –17:00	Coffee Break
	17:00 – 19:00	**[Lecture 3]** Action Planning on Technology Transfer Dr. Sang-Hyuk Suh, Professor, Hoseo Graduate School of Global Entrepreneurship
	19:00 – 20:00	Dinner
May 28	09:00 –11:00	**[Lecture 4]**

Day	Time	Program
(Tuesday)		Global Technology Transfer between the Firms Mr. Seung-Ho Lee, CEO, Delta Tech-Korea Ltd.
	11:00 – 13:00	**[Lecture 5]** Evaluation Model for the Selection of Transferable Patent Technology Dr. Oong-Hyun Sung, Professor, Hanshin University
	13:00 –14:00	Lunch
	14:00 – 16:00	**[Lecture 6]** Current Status of Korean Universities' TTO and Challenge for the IP Globalization Business Dr. Gyeongsoo (John) Shim, Professor/Chief Technical Officer, Korea University Research and Business Foundation (KURBF)
	16:00 – 18:00	**[Lecture 7]** Startup Support Programs in Korea Dr. Byung-Jo Choi, Professor, Startup Center, Incheon National University
	18:00 – 19:00	Dinner
May 29 (Wednesday)	09:00 – 11:00	**[Lecture 8]** Patent Management: Overview & Evaluation Mr. Young-Taeck Lim, Patent Lawyer, Dowools International Patent Law Firm
	11:00 –13:00	**[Lecture 9]** IPR Management for Patent Transfer Mr. Kyung-Min Chung, Representative Patent Attorney, Dowool IP Group
	13:00 – 14:00	Lunch
	14:00 – 16:00	**[Lecture 10]**

Day	Time	Program
		Technology Marketing as a New Paradigm of Technology Transfer Dr. Sang-Hyuk Suh, Hoseo Graduate School of Global Entrepreneurship
	16:00 – 18:00	**[Lecture 11]** Evaluation and Management of Public Technology Dr. Hyun-Gon Shin, Professor, Sejong University
	18:00 – 20:00	Dinner
May 30 (Thursday)	09:00 – 11:00	**[Lecture 12]** Technology Startups: Strategy & Management to Facilitate Dr. Wang-Kyu Lim, Adviser, Walton Technology
	11:00 – 13:00	**[Lecture 13]** Practice of Technology Transfer in Technopark Mr. Seong-Kun Kang, General Manager, Chungnam Technopark
	13:00 – 14:00	Lunch
	14:00 – 16:00	**[Lecture 14]** Financial Support for Technology Transfer & Commercialization Dr. Oong-Hyun Sung, Professor, Hanshin University
	16:00 – 18:00	**[Lecture 15]** Comparative Analysis of TT System between the 2 Countries & Group Presentation Dr. Sang-Hyuk Suh, Professor, Hoseo Graduate School of Global Entrepreneurship
	18:00 – 19:00	Dinner
May 31	09:00 – 11:00	**[Lecture 16]** Case Analysis of Successful Technology Spin-offs in Korea

Day	Time	Program
(Friday)		Dr. Sun-Young Lee, Associate Professor, Hoseo Graduate School of Global Entrepreneurship
	11:00 – 12:00	**[Study Visit 1]** National Museum of Korean Contemporary History
	12:00 – 14:00	Lunch and Move to KIST
	14:00 – 15:00	**[Study Visit 2]** Korea Institute of Science and Technology (KIST): KIST History Museum (KOSTORIUM)
	15:00 – 16:00	**[Lecture 17]** Commercialization of Innovative Technologies in KIST Mr. Sungchul Shin, Principal Specialist, Korea Institute of Science and Technology (KIST)
	19:00 – 21:00	**[Embassy of the Republic of Azerbaijan to the Republic of Korea]** **Farewell Dinner**
June 1 (Saturday)	09:00 – 11:00	Check out of the Hotel
	11:00 – 12:30	Lunch
	12:30 – 17:30	**[Cultural Experience]**
	17:30 – 19:00	Dinner
	19:00 – 21:00	Leaving Korea
June 2 (Sunday)	01:15 – 05:15	Departure from Incheon (Qatar Airways, QR0859)
	07:30 – 11:20	Arrival at Baku Airport (Qatar Airways, QR0351)

2.7 Selection of Trainees

2.7.1 Invitation Letter

STEPI

5th, 6th, 7th Fl., Building B, Sejong National Research Complex 370,
Sicheong-daero Sejong, 339-007, Republic of Korea
Tel : +82-44-287-2000, Fax : +82-44-287-2068, www.stepi.re.kr, iicc.stepi.re.kr

April 23rd , 2019

Mammad Baghir-zada
Deputy Director on the Strategic Development
High Technologies Park (HTP)
Azerbaijan National Academy of Sciences (ANAS)
Street 622/9, Ahmadli settlement, Khatai dist., Baku, Azerbaijan

Dear Mammad Baghir-zada,

On behalf of STEPI, I am pleased to invite you to the Capacity Development Training Workshop. This workshop will take place between May 25th and June 2nd 2019 (including entry and departure) in the Republic of Korea. As part of *2019 K-Innovation ODA Program* titled *"Policy Consultation for Promotion of Technology Transfer and Commercialization in Azerbaijan"*, this workshop is to share Korean experience and practices which can be applied in technology transfer and commercialization.

STEPI will provide participants with round trip flight tickets and hotel accommodations and meals during their stay in Seoul.

This invitation may serve as proper documentation to apply for visa at the Embassy of the Republic of Korea in Azerbaijan. I and my colleagues at STEPI hope that your visit will provide you with valuable knowledge and experience that you can apply at work when you go back to Azerbaijan as well as unforgettable memories in Korea.

We strongly believe in power of collaborative partnership, and our project will further develop a sincere relationship between our institutes, and furthermore, between two countries.

Yours Sincerely,

Cho. Hwang hee

Hwang Hee Cho
President
Science and Technology Policy Institute (STEPI)

2.7.2 Application Form I

STEPI
5-7th Floor, Building B, Sejong National Research Complex,
370 Sicheong-daero, Sejong, Republic of Korea
T:+82-44-287-2000 F:+82-44-287-2099 www.stepi.re.kr

2019 K-Innovation ODA Program with Azerbaijan
Training on Technology Transfer and Commercialization

Application Form I

Personal Information	Name (As written on passport)	Last Name	First Name	Middle Name	
	Sex	Female □ / Male □	Nationality		
	Country of Birth		Date of Birth (MM/DD/YY)		
	Passport Number		Current Employer/Institute		
	Division		Position		
	Contact Information	Permanent Address :			
		Tel: Mobile:			
		FAX :			
		E-mail :			
		Emergency Contact:			
Invitational Training Program	Academic Degree	(e.g. PhD, MA, MSc, etc.)			
	Area of Expertise				
	Relevant Work Experience	(e.g. TT, TC, Incubation, etc.)			
Language Proficiency	English Proficiency (low-intermediate-high) ①-②-③-④-⑤	Speaking	①-②-③-④-⑤	Listening	①-②-③-④-⑤
		Reading	①-②-③-④-⑤	Writing	①-②-③-④-⑤

I certify that all information provided above is accurate and truthful. I understand that STEPI reserves the right to refuse acceptance or to terminate the program if it is discovered that any false statement has been made.

Signature ________________________________ Date ____________________

2.7.3 Application Form II

STEPI

5-7th Floor, Building B, Sejong National Research Complex
370 Sicheong-daero, Sejong, Republic of Korea
T:+82-44-287-2000 F:+82-44-287-2099 www.stepi.re.kr

2019 K-Innovation ODA Program with Azerbaijan
Training on Technology Transfer and Commercialization

Application Form II

You were recommended by ANAS-HTP as a candidate for the training on **"technology transfer & commercialization"** from **27-31 May** (arrival and departure dates not included) **in Seoul**. As the training is mostly composed of lectures and group discussion to come up with ways to improve TT/TC capacity at your institution, you are asked to fill out and submit the application forms to the STEPI organizers (Ms. Eun Joo Kim, ejkim@stepi.re.kr and Ms. Sohyun Kwon, shkwon@stepi.re.kr) **by no later than Friday, 5th April, 2019**. After reviewing the candidates' application forms, STEPI will send the final list of participants and guidelines for their participation to ANAS-HTP.

Name	
Position	
Organization	
1. Why do you need to participate in the training?	
2.How does this training relate to your work?	

3.What are your work challenges?	
4.What is your plan to apply and disseminate the learned knowledge during the training?	
5.Expected result & suggestions	

2.8 Workshop Book

2.9 Main Activities

- Orientation
  - Overall introduction on the purpose of the training and curriculum
  - Prior notice is given to the participants to have ideas on policies related to technical transfer and commercialization during the workshop period to prepare the action plan on the organizational role and implement the TT/TC promotion policy

- Lecture 1 (Policies for Technology Transfer)
  - This lecture aimed to introduce the major aspects of policies for technology transfer (TT) and Korean experiences on the theme, focusing on international TT and absorption of the transferred technologies.
  - Korea has been recognized as a notable country that had absorbed advanced technologies and had developed its own technologies successfully by exploiting various channels of international TT.

- The lecture provides basic understanding of TT and related policies. Based on these concepts. The experiences of Korea in policy implementation and subsequent technology transfer and technological learning processes will be analyzed and introduced.
- The lecture is composed of four parts: (a) Understanding of TT, (b) Policy Measures for TT (based on various channels of TT), (c) Korean Experiences (in light of channels, roles, policy measures in terms of technology transfer, and (d) Implications.
- This lecture is intended to help the workshop participants get lessons and ideas in designing and implementing related policies and managing related issues and paths.

▪ Lecture 2 (Technology Transfer: Overview, Support Programs & Implications for Azerbaijan)

- This presentation introduces the foundation of TT including the concept, process, and success factors of technology transfer ("TT"). After a brief presentation of the meaning, the importance, types of TT/TC, and factors influencing TT are introduced. In this regard, different programs to foster TT/TC in Korea for the past 2 decades are provided.
- For application in the context of Azerbaijan, some implications and questions are given, followed by several recommendations for Azerbaijan.

▪ Lecture 3 (Action Planning on Technology Transfer)

- Action Planning on Technology Transfer

▪ Lecture 4 (Global Technology Transfer Between Firms)

- This lecture aims to give Azerbaijani participants basic understanding of technology transfer (TT) mode and insight of firms' technology cooperation. The key contents are as follows:
  - Why a firm makes an entry into the international market based on TT
  - How to identify attractive technology and commercialize it

- 8 types of TT business model internationally
- Discussion of business opportunity in Korea from the perspective of Azerbaijan

▪ Lecture 5 (Evaluation Model for the Selection of Transferable Patent Technology)

• This evaluation model is developed for selecting transferable patent technology by the Korean Patent Office and has been used by various technology transfer organizations. This lecture consists of understanding the key factors and evaluation criteria for selecting the transferable patented technology. The following are the contents of this lecture:

- Linking Intellectual Capital with Corporate Value
- Factors Affecting Patent Technology Value
- Patent Selection Evaluation for Technology Transfer
- Evaluation Worksheets of the Patent Selection Evaluation Model
- Technology Evaluation Model for Business Feasibility

▪ Lecture 6 (Current Status of Korean Universities' TTO and Challenge for the IP Globalization Business)

• The mission of a university does not mirror the mission of a corporation or its metrics for measuring success or its value system. The mission of a business is to maximize economic return to shareholders. The university's mission is to create knowledge through research and development and to disseminate the IP assets mostly through education for the public good.

• Based on those topics, this presentation seeks to provide the key message of how a university's IP assets can be commercialized further and how to evaluate the value of each of the intellectual assets through the comparison between the US and South Korea IP market policies and university technology transfer processes.

▪ Lecture 7 (Startup Support Programs in Korea)

• Startup environment and the government's policy direction to support Startups in Korea

- Support programs for each stage of the Startup's growth in Korea
- Introduction of the Incheon National University Startup Center and support programs

▪ Lecture 8 (Patent Management: Overview & Evaluation)

- Patent is key to protecting innovation and financing of research and can be used as a token in technology transfer.
- Patent is an exclusive right having limited terms and should be managed.
- To get a patent right, the invention defined in claims must meet the patentability requirements including industrial applicability, novelty, and inventive step.
- Patent management and establishment of IP portfolio are very important for innovating and survival.
- To manage IP, plan a strategy for BM, segment the elements of BM, branch it, try to prevent competitors, and think of future value.

▪ Lecture 9 (IPR Management for Patent Transfer)

- Briefly discuss what IPR and IPR valuation are.
- First, compare IPR and Technology and find out what is so special about IPR. IPR can be traded independently, but its value is highly dependent on technology and business. IPR can be treated as an independent property in warrant, licensing, and calculation of compensation.
- There are various purposes for IPR valuation: trading, finance, contribution, strategy, litigation, and tax issue. The valuation process and method can be adjusted based on the purpose of the valuation. For the valuation of IPR, there are 3 known methods: market approach, revenue approach, and cost approach. The method can be selected based on many factors: purpose, technology step, when to use, kind of IPR.

- 4 factors concerning IPR valuation (Technical, Right, Marketability, Business). Among the 4 factors, I, as a patent attorney, concentrate on Right -- stability, scope of right, and matching.
- Discuss what IPR management is and why we need it. (1) Offensive interests – protection, royalties (2) Defensive interests – Cross-license (3) Marketing means – advertisement (4) Reducing the R&D cost
- How to do IPR management – patent portfolio, IP trading, holding IP analysis, IP marketing strategy, IP risk analysis, infringement monitoring

▪ Lecture 10 (Technology Marketing as a New Paradigm of Technology Transfer)

- The concept and some practices of technology marketing are to be presented as the new effective paradigm of TT.
- Major contents will be the New paradigm of TT, definition of techno-marketing, and prospecting and contact for successful TT.

▪ Lecture 11 (Evaluation and Management of Public Technology)

- The purpose of this lecture is to introduce a concept of technology evaluation and practical evaluation models that select technologies with great economic effects among the proposed technologies. This lecture will introduce the government management system of technologies. This lecture will help you understand what the key evaluation criteria and contents are in the public evaluation model.
- These evaluation systems have been developed for the selection of the most economically effective technologies by the Korean government and have been used by various technology transfer organizations. This lecture consists of understanding of the key factors and evaluation criteria to select the technologies that will have economical results. The following are the contents of this lecture:
  - Technology assessment and evaluation
  - Factors for evaluating Technologies
  - Introducing the evaluating tools in various organizations (including NTTC)

  - Introducing the technology management systems in Korean public sectors

- Lecture 12 (Technology Startups: Strategy & Management to Facilitate)
  - Concept and characteristics of techno-startups
    - Definition & Scope, Technology life cycle, Death valley & chasm, Overcoming crisis, Model for establishing techno-startups
  - Technology trends
    - Gartner presents the top ten strategic technology trends and has three themes: intelligent, digital, mesh
    - Understanding of the Fourth Industrial Revolution
    - Technology Startup Trend
  - Success Factors of Technology Entrepreneurship
    - Importance of preparing for technology startup, Success factors, Entrepreneurship, Technology Valuation
  - Supporting Policy Suggestion

- Lecture 13 (Practice of Technology Transfer in Technopark)
  - Overview of methods and procedures for technology transfer in technopark: Technopark's role in technology transfer, business process, how to find Technology, Key factors in the technology transfer process, etc.
  - Methods and procedures for obtaining and analyzing information on technical needs: Explanation on obtaining and analyzing technical demand information as key information of technology transfer
  - Technology transfer cases in Korea: As for the technology transfer cases promoted by Technopark, the topics of each case are as follows:
    - Creation of technical needs by idea, Application of technology in other fields, Mediation & Negotiation

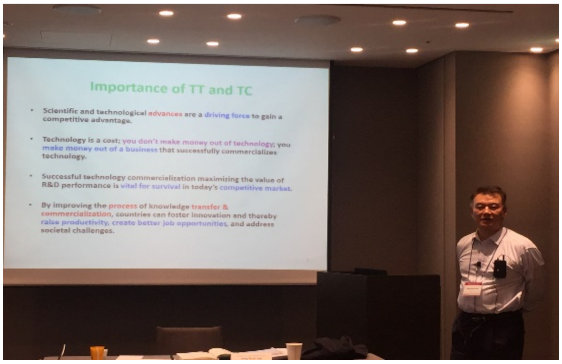
  - Global technology transfer cases: As cases promoted by technopark, Technology transfer of Mongolia and Korean companies, technology transfer of Ghana and Korean companies

- Lecture 14 (Financial Support for Technology Transfer & Commercialization)
  - This lecture is about the policy resources for technology transfer commercialization in Korea. Nowadays, the results of technology evaluation are being used as the basis of financial support. This lecture will provide the financial support policy and contents including investment, guarantee, and loan financing. The content of the lecture is as follows:
    - Financial Demand for Technology Commercialization
    - Investment (Equity) Financing for Startups
    - Evaluation Guarantee for TT & TC
    - Loan (Debt) Financing Through the Technology Credit Bureau
    - Conclusion

- Lecture 15 (Comparative Analysis of TT System Between the 2 Countries & Group Presentation)
  - Future plans of TT System in Azerbaijan
  - Action Planning

- Lecture 16 (Case Analysis of Successful Technology Spin-offs in Korea)
  - Introduction of Case Study
  - Spin-off from Government Research Lab: KolmaBNH Korea, APAK
  - Spin-offs from Corporate Venturing Activities: Samsung, SK, POSCO, Hyundai
  - Success Factors of Spin-off
  - Discussion

2.10 Photos

Capacity Development Workshop (1)	Capacity Development Workshop (2)
	
Capacity Development Workshop (3)	**Capacity Development Workshop (4)**
	
Capacity Development Workshop (5)	**Capacity Development Workshop (6)**
	

Capacity Development Workshop (7)	Capacity Development Workshop (8)
	
Group Photo (1)	**Group Photo (2)**
	
Group Photo (3)	**Capacity Development Workshop (9)**
	
Site Visit to KIST	**Site Visit to Seoul Global Center**
	

Wrap-up Meeting	Capacity Development Workshop (10)
	
Awarding of Certificates (1)	**Awarding of Certificates (2)**
	
Korean Cultural Experience (1)	**Korean Cultural Experience (2)**
	

2.11 Evaluation Survey

2.11.1 Daily Evaluation Survey

A satisfaction survey is conducted among the trainees after they finish each class and complete the overall training.

Daily Evaluation Survey
for STI Capacity Development Workshop
– Promotion of Technology Transfer and Commercialization in Azerbaijan–

Date: May 27, 2019

Name	
Organization	
Position	
E-mail	

Instructions: This questionnaire is designed to evaluate the lecture. The evaluation of lecture is meant to improve the quality of our training program.
Please indicate your level of agreement with the items listed below by checking the appropriate box and provide any comments.
Many thanks for your support!

Lecture 1		Lecturer			
	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. The topic covered in the lecture was profoundly linked in my work.	①	②	③	④	⑤
2. The lecturer's delivery is clear and easy to understand.	①	②	③	④	⑤
3. The knowledge learned can be applied directly to my work.	①	②	③	④	⑤
4. I would like to recommend this lecture to others and to invite to the upcoming dissemination workshop in Azerbaijan.	①	②	③	④	⑤
5. Please provide any comments or suggestions that might help improve this lecture in the future.					

2.11.2 Evaluation Survey

Evaluation Survey

on STI Capacity Development Workshop Program

-Promotion of Technology Transfer and Commercialization in Azerbaijan-

Date: 1 June, 2019

【Respondent Information】

Name	
Organization	
Position	
Years of working	
Work Experience in TT/TC (Duration and Area)	
Gender	Male ( ) Female ( )
Age	
E-mail	

Instructions: This questionnaire is designed to evaluate the process of the training program on how much you agree with the question on a scale rated one to five. Your valuable opinions would be incorporated in organizing our next programs for the upcoming dissemination workshop in Azerbaijan. This is intended for capacity development of key stake holders for STI policy and TT/TC policy development in Azerbaijan.
We appreciate your support and cooperation!

Questions	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. The objectives of the training were clearly defined.	①	②	③	④	⑤
2. The program was helpful in enhancing my capacity to perform the duties.	①	②	③	④	⑤
3. The program met the objective of institutional capacity building.	①	②	③	④	⑤
4. The contents of the program were organized well and easy to follow.	①	②	③	④	⑤
5. The materials distributed were helpful.	①	②	③	④	⑤
6. The topics covered were relevant to my country's needs for TT/TC development.	①	②	③	④	⑤
7. Enough participation opportunities and interactions were encouraged.	①	②	③	④	⑤
8. The program has influenced in changing my mindset.	①	②	③	④	⑤
9. The duration of the program (6 days) was sufficient.	①	②	③	④	⑤
10. The meeting room, accommodation and the other facilities were adequate and comfortable.	①	②	③	④	⑤

11. What did you like most about this training? What other topics/contents would you suggest to be dealt in the future?

12. How do you hope to change your practice as a result of this training?

13. Are there any suggestions for improvement or requests on this training? If so, please explain in detail.

2.12 Result of the Evaluation Survey

Item	Contents
Survey Target	Workshop participants (total of 7 participants)/ No. of respondents: 7 trainees
Managing Department	STEPI ODA Azerbaijan Project Research Team
Survey Period	May 31, 2019
Survey Method	Online (E-mail) Survey

No.	Evaluation Item	Overall Satisfaction Level* <*Satisfaction level is calculated by scoring from 1 point (very dissatisfied) to 5 points (very satisfied)>
1	Clarity of the training purpose	4.83
2	Expectation on work applicability and capability reinforcement	5
3	Compliance with the purpose of reinforcing the capability of the organization	4.83
4	Composition and understanding of the program details	4.67
5	Enhancement of understanding of the program contents through the source book	4.83
6	Conformity to the needs of the country for TT/TC development	5
7	Promotion of sufficient opportunities for participation and interaction	5
8	Change in the way of thinking through the program	5
9	Suitability of the program schedule (6 days)	4.17

No.	Evaluation Item	Overall Satisfaction Level* <*Satisfaction level is calculated by scoring from 1 point (very dissatisfied) to 5 points (very satisfied)>
10	Suitability and convenience of the conference room, accommodations, and other facilities	5
Average		4.83
11	The most favorite part in the training, topic/contents for future proposal	• An appropriate topic was drawn up, but more actual cases and experiences of Korea are expected. • Especially, the topic on TP management was very helpful. It will be better if there is time for visiting TP in Korea.
12	Improvement method on the application of training result to the work	• Overall, the training program is excellent and beneficial. • It is deemed necessary for a detailed action to be developed to apply the training results to the actual work. To enable this, the method of inviting a Korean expert is considered. • The knowledge obtained from the training is to be shared with the STI community.
13	Improvements and suggestions	• The program composition and operation are excellent, and the warm welcome is appreciated. • Nonetheless, considerations must be given to the time and period of the long-distance business trip. • There is a request for additional field trip, group work, case study method, etc. in the program.

In the composition and operation of the workshop program, the average score was 4.83 points, which is considered to be very satisfactory.

Especially, the expectation on work applicability and capability reinforcement, conformity to the needs of the country for TT/TC development, promotion of sufficient opportunities for participation and interaction, change in the ways of thinking through the program, and suitability and convenience of the conference room, accommodations, and other facilities were the 5 areas that garnered the highest score.

Second, in terms of the clarity of the training purpose and purpose of capability reinforcement of the organization, the composition of the training program scored 4.83 points — which is very suitable — and the evaluation item on enhancing understanding of the program contents through the source book received 4.83 points; thus showing high satisfaction level among trainees.

Note, however, that the suitability of the program schedule (6 days) received a relatively low rating, and the need to consider the time and period of the long-distance business trip between Azerbaijan and Korea was raised.

For improvements, there were suggestions on adding more field trips, group work, case study methods, etc. when organizing the training program.

2.13 Official Thank-You Letter by ANAS-HTP

AZƏRBAYCAN
MİLLİ ELMLƏR AKADEMİYASI
YÜKSƏK TEXNOLOGİYALAR PARKI

AZERBAIJAN
NATIONAL ACADEMY OF SCIENCES
HIGH TECHNOLOGIES PARK

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E-mail: office@ameaytp.az

№ 76/311

"02" July 2019

Dear Mr. Hwang Hee Cho!

There are good relations between the Azerbaijan National Academy of Sciences (ANAS) and the Science and Technology Policy Institute (STEPI), and there is close cooperation, which will have a good continuation, I am sure. Within the framework of this cooperation, a number of events were held both in Azerbaijan and in the Republic of Korea, such as seminars, trainings.

The Republic of Korea is one of the most advanced countries in the field of creation and development of high technologies, transfer and commercialization of technologies. One of the main research centers of South Korea in the field of science and technology is STEPI. ANAS is the main scientific center of Azerbaijan with over than 70 years of history. That is why we find our cooperation useful, mutually beneficial, and we hope that it will be fruitful.

The connecting link between ANAS and STEPI is the Azerbaijan National Academy of Sciences High Technologies Park (ANAS HTP). ANAS HTP - is a relatively new organization created by decree of the President of the Republic of Azerbaijan. ANAS HTP has been established to the provision of state support for the sustainable development and competitiveness of the economy, the expansion of branches of innovation and high technologies on the basis of scientific researches and technological achievements, creation of modern complexes for research and development of new technologies.

The direct participation of ANAS HTP in joint activities of ANAS and STEPI is useful for ANAS HTP from the point of view of obtaining new knowledge and the opportunity to communicate with well-known specialists from South Korea in the field of technology transfer and commercialization, to use the advices and consultations of Korean experts.

From 27th May to 1st June 2019, the Science Technology Innovation (STI) Capacity Development Workshop - "Promotion of technology transfer and technology commercialization in Azerbaijan" has been initiated and organized by STEPI and was held in the capital of the Republic of Korea, Seoul city. The employees of the ANAS HTP and the Presidium of ANAS took part in this workshop. This event was held at the high level, and was very useful.

Dear Mr. Hwang Hee Cho, from my own part and on behalf of the ANAS HTP's staff, I would like to thank you and all the STEPI's staff for the warm welcome and for the high level workshop, which were organized for our group in Seoul.

The cooperation between STEPI and ANAS HTP continues, and new joint projects, workshops and trainings await us, both in Azerbaijan and the Republic of Korea.

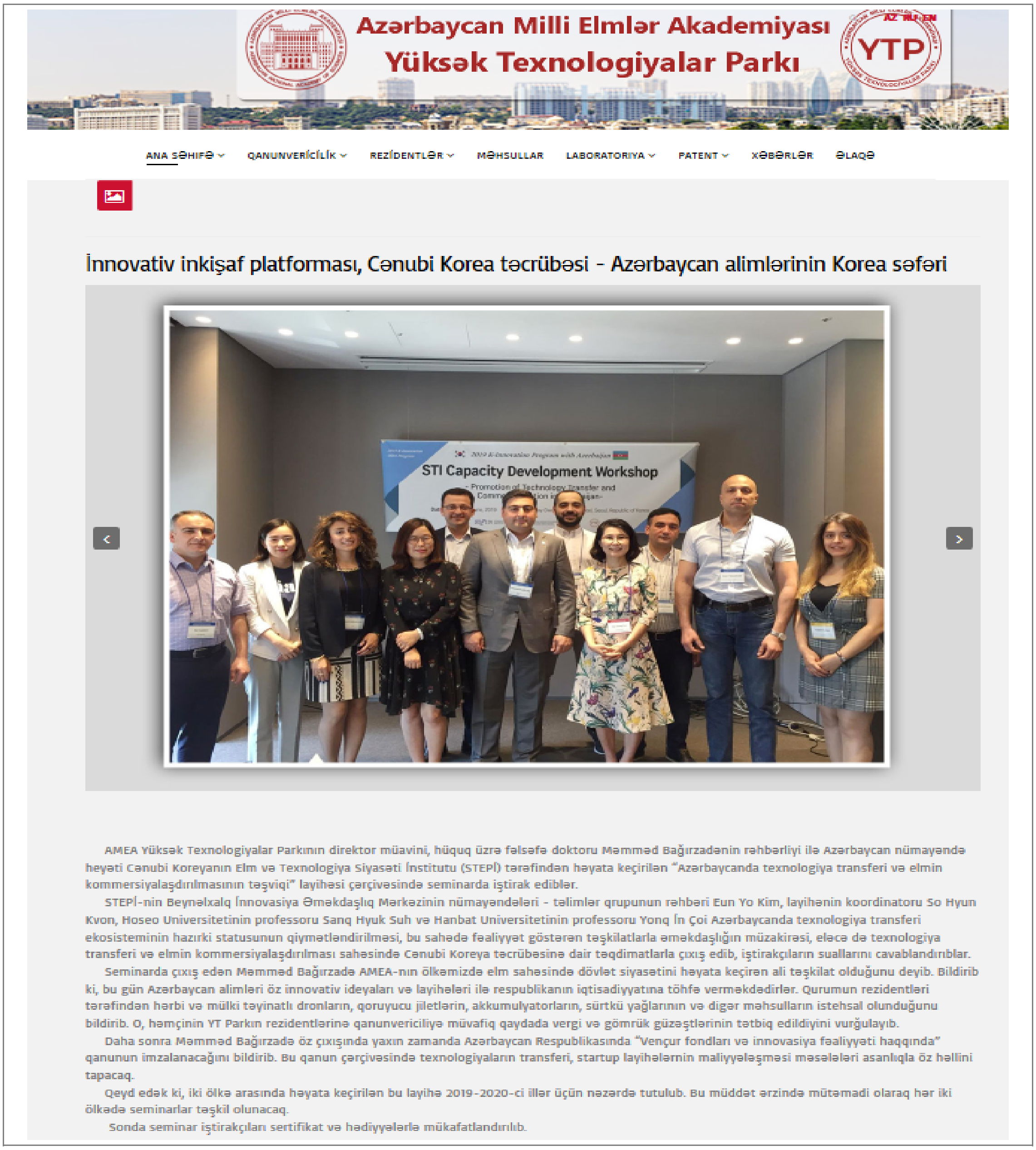
I wish you and all the STEPI's staff health, good luck, and many successful projects.

With the best regards,

Director Ph.D. in economics Nazim Shukurov

2.14 Publicity Activities (Press Releases)

- ANAS-HTP Homepage (2019.6.3)
  (http://ameaytp.az/index.php/component/content/article/88-son-x-b-r/546-amea-yt-park-n-nuemay-nd-hey-ti-c-nubi-koreyaya-isguezar-s-f-r-edibl-r?Itemid=539)



Azərbaycan Milli Elmlər Akademiyası
Yüksək Texnologiyalar Parkı

ANA SƏHİFƏ | QANUNVERİCİLİK | REZİDENTLƏR | MƏHSULLAR | LABORATORIYA | PATENT | XƏBƏRLƏR | ƏLAQƏ

İnnovativ inkişaf platforması, Cənubi Korea təcrübəsi - Azərbaycan alimlərinin Korea səfəri

AMEA Yüksək Texnologiyalar Parkının direktor müavini, hüquq üzrə fəlsəfə doktoru Məmməd Bağırzadənin rəhbərliyi ilə Azərbaycan nümayəndə heyəti Cənubi Koreyanın Elm və Texnologiya Siyasəti İnstitutu (STEPİ) tərəfindən həyata keçirilən "Azərbaycanda texnologiya transferi və elmin kommersiyalaşdırılmasının təşviqi" layihəsi çərçivəsində seminarda iştirak ediblər.

STEPİ-nin Beynəlxalq İnnovasiya Əməkdaşlıq Mərkəzinin nümayəndələri - təlimlər qrupunun rəhbəri Eun Yo Kim, layihənin koordinatoru So Hyun Kvon, Hoseo Universitetinin professoru Sanq Hyuk Suh və Hanbat Universitetinin professoru Yonq İn Çoi Azərbaycanda texnologiya transferi ekosisteminin hazırki statusunun qiymətləndirilməsi, bu sahədə fəaliyyət göstərən təşkilatlarla əməkdaşlığın müzakirəsi, eləcə də texnologiya transferi və elmin kommersiyalaşdırılması sahəsində Cənubi Koreya təcrübəsinə dair təqdimatlarla çıxış edib, iştirakçıların suallarını cavablandırıblar.

Seminarda çıxış edən Məmməd Bağırzadə AMEA-nın ölkəmizdə elm sahəsində dövlət siyasətini həyata keçirən ali təşkilat olduğunu deyib. Bildirib ki, bu gün Azərbaycan alimləri öz innovativ ideyaları və layihələri ilə respublikanın iqtisadiyyatına töhfə verməkdədirlər. Qurumun rezidentləri tərəfindən hərbi və mülki təyinatlı dronların, qoruyucu jiletlərin, akkumulyatorların, sürtkü yağlarının və digər məhsulların istehsal olunduğunu bildirib. O, həmçinin YT Parkın rezidentlərinə qanunvericiliyə müvafiq qaydada vergi və gömrük güzəştlərinin tətbiq edildiyini vurğulayıb.

Daha sonra Məmməd Bağırzadə öz çıxışında yaxın zamanda Azərbaycan Respublikasında "Vençur fondları və innovasiya fəaliyyəti haqqında" qanunun imzalanacağını bildirib. Bu qanun çərçivəsində texnologiyaların transferi, startup layihələrnin maliyyələşməsi məsələləri asanlıqla öz həllini tapacaq.

Qeyd edək ki, iki ölkə arasında həyata keçirilən bu layihə 2019-2020-ci illər üçün nəzərdə tutulub. Bu müddət ərzində mütəmadi olaraq hər iki ölkədə seminarlar təşkil olunacaq.

Sonda seminar iştirakçıları sertifikat və hədiyyələrlə mükafatlandırılıb.

3. The Dissemination Workshop and Field Study

3.1 Background

In the fourth quarter of this year, STEPI and ANAS-HTP created an opportunity for disseminating the outcomes of the 2019 policy consultation in Azerbaijan.

STEPI tried to derive an implementation action plan through a dissemination workshop aimed at developing TT/TC programs and strengthening linkage with Industry-University – Institute Collaboration.

The dissemination of the outcomes of 2019 was held on 16-17 October by STEPI and ANAS-HTP.

The dissemination workshop consisted of lectures and Q&A, case studies of best practices and discussions of Korean experts (Technology Entrepreneurship and Ideation-based Startup, Role of Technopark in Technology Commercialization and Chungnam TP Case Study, Korean Technopark (K-TP) Model as an Integrated Business Platform: GBTP Case), and Azerbaijan experts' reporting and presentation (cases of application and improvement of business capacity after the invitational training for STI capacity development in Korea).

In-depth discussion will be held on the lecture and practice by Korean experts (technological commercialization and technical idea-based startup, role of the technopark and startup/enterprise support program), including presentation by the local experts (case on the degree of application to the actual work after the invitational training and improvement of work capability) and on the relevant issues with related parties of the industry-university-institute.

Accordingly, STEPI plans to develop the technical transfer and commercialization promotion development program and to derive the detailed implementation method for linkage and collaboration between the entities (industry-university-institute) through workshop and field survey, etc. for related parties of the industry-university-institute of technical transfer/commercialization, startup, and technopark in Azerbaijan.

3.2 Overview of the Dissemination Workshop and Field Study

3.2.1 Dissemination Workshop

- Title: Dissemination Workshop for the Promotion of Technology Transfer and Commercialization in Azerbaijan
- Date: October 16 – 17 October (Wed. - Thur.), 2019
- Venue: ANAS-HTP Business Incubation Center, Baku, Azerbaijan
- Purpose:
  - Hold the 2019 workshop on disseminating the project outcome
    - Share experiences in the startup based on technological entrepreneurship and technical idea
    - Share cases on the technopark operating strategy and technical transfer/startup support in Korea
    - Presentation on the applicable cases and lessons on the local science & technology innovation and startup support in Azerbaijan (share experiences and lessons regarding the Korean innovative ecosystem and strategy through invitational training in Korea)
  - Roundtable meeting for the materialization of the implementation plan for the following year (Year 2020)
    - Discussions between the Korean expert in technical transfer/commercialization and technopark, director of ANAS-HTP, local research team, major technical transfer and commercialization agency and personnel of the responsible agency, and related parties of the industry-university-institute and relevant agency
  - Fact finding related to the local technology innovative ecosystem and inspection of the main industry-university-institute organizations

3.2.2 Field Study

- Name of Field Study: Field study for the implementation and application of technology transfer and commercialization ecosystem in Azerbaijan
- Period of Field Study: October 14 – 15 October (Mon. - Tue.), 2019
- Place of Field Study: Baku, Azerbaijan (ANAS-HTP, Institute of Information Technology, Institute of Control Systems, Entrepreneurship Development Fund, Baku Engineering University Technopark, Ministry of Transport, Communication & High Technologies, Azerbaijan Innovation Agency, etc.)
- Purpose of Field Study
  - Investigate the current status of TT/TC activities and programs in Azerbaijan
  - Identify the specific demand for the 2019-2020 consultation project
  - Interviews and meetings with TT/TC stakeholders

3.3 Research Team

- STEPI Research Team (total of 5)

Name	Position/Organization	Role/Area of Expertise
Eun Joo Kim	Head of External Cooperation and Public Relations, STEPI	Project Manager (PM)
So Hyun Kwon	Researcher, Division of Global Innovation Strategy, STEPI	Project Coordinator (PC)
Jong In Choi	Professor, Hanbat National University	Technology Entrepreneurship and Ideation-based Startup
Seong-Kun Kang	General Manager, Chungnam Technopark	Role of Technopark in Technology Commercialization and Chungnam TP Case Study
Seong-Keun Park	Team Manager, Gyeong-Buk Technopark	Korean Technopark (K-TP) Model as an Integrated Business Platform: GBTP Case

- Azerbaijan's Research Team (total of 7)

Name	Position/Organization	Role/Area of Expertise
Mammad Baghir-zada	Deputy Director for Strategic Development, ANAS-HTP	Strategic Planning on HTP
Bunyamin Seyidov	Chief, Department of Relations with Eastern Countries, ANAS Presidium	Global STI Partnership Program Development
Zaur Akhmedzadeh	Head, Technology Transfer and Business Incubator Department, ANAS-HTP	Law in the field of TT
Isa Gasimov	Senior Specialist, Entrepreneurship Development Foundation of the Republic of Azerbaijan	Innovation and Entrepreneurship/ TT and Incubation in Univ.
Samir Veliev	Economist of Economic Research Department under the Board of Directors, International Bank of Azerbaijan (OJSC)	Economic Analysis on Innovation and Industry
Aysel Abdullayeva	Lecturer in Innovation Risk Management of UNEC Business School, Azerbaijan State University of Economics (UNEC)	Research on Innovation, Entrepreneurship, and Marketing
Vusal Suleymanli	Chief Innovation Officer, Innova Ventures Group	Tech Innovation & Entrepreneurship

3.4 Objectives and Expected Outputs

- Disseminate the outcomes of the 2019 project
    - Presentations on Action Plans for the Promotion of TT and Tech Incubation Program by Azerbaijan's counterpart team
    - Presentations on Korean experiences and policy lessons for Azerbaijan

- Make an implementation action plan through the dissemination workshop aimed at developing TT/TC programs and strengthening linkage with Industry-University – Institute Collaboration.

- Discussion with policymakers, practitioners, researchers of TT- and TC-related ministries, institutions, research centers
  - An interview will be conducted on how to apply and implement Azerbaijan's technology transfer ecosystem, current policies, and programs.

- To specify the scope of activities in the 2020 project

3.5 Organizers

This workshop is organized by the Science and Technology Policy Institute (STEPI) in partnership with ANAS.

3.6 Methodology and Structure

The workshop consisted of four methods: (1) lectures and Q&A, (2) case studies of best practices, (3) discussions and feedback, and (4) action planning. During the workshop, it is strongly recommended that the participants keep on discussing with the Korean experts intensively as well as deriving implications and lessons from the lectures and discussions. For the development of a practical action plan, the related actors will give immediate feedback to the participants.

Methods	Module Details
Lectures and Q&A	• Lectures on the Ecosystem of Technology Entrepreneurship, Evolution, and Role of Korean Technopark and K-TP Model 1) Technology Entrepreneurship and Ideation-based Startup 2) Role of Technopark for Technology Commercialization and Case Study 3) Korean Technopark (K-TP) Model as an Integrated Business Platform

Methods	Module Details
Case Studies of Best Practices	• Case study of Korea's Technology Entrepreneurship and Ideation-based Startup • Case study of the Korean Technopark Model
Discussions and Feedback	• To share stakeholders' views and opinions • Discussions on the diagnosis and evaluation of the progress and achievement of the 2019 project and feedback provided & facilitated by Korean experts • To develop the action plan
Action Planning	• To specify TT/TC project formulations • To promote voluntary knowledge dissemination

3.7 Program Schedule

DATE	TIME	DESCRIPTION	VENUE
Saturday, October 12 - Sunday, October 13	13:40~22:30 - 02:15	Departure Arrival	Korea Azerbaijan
Monday, October 14	10:~12:00	Meeting between ANAS-HTP and STEPI • Introduction of the Korean research team • Introduction of the 2019 project progress and achievements • Discussion together with preparation for the dissemination workshop • Discussion on the scope of activities in 2020 with the ANAS-HTP research team	
	Lunch	Lunch	
	14:00 ~ 15:20	Site Visit and Interview • ANAS-Institute of Information Technology	
	15:30 ~ 16:30	Site Visit and Interview • ANAS-Institute of Control Systems	

DATE	TIME	DESCRIPTION	VENUE
Tuesday, October 15	10:00 ~ 11:30	Field study, Data collection, and Interview: • Entrepreneurship Development Fund	
	12:00 ~ 14:00	Field study, Data collection, and Interview • Baku Engineering University	
Wednesday, October 16	08:40 ~ 09:00	**[Registration]**	
	09:00 ~ 09:30	**[Opening Ceremony]** Opening Remarks • Samir Mammadov, Director, ANAS-HTP Introduction of STEPI's activities and outcomes • Eun Joo Kim, Head, Global Training Program Team, Office of STI ODA Project (IICC), STEPI • So Hyun Kwon, Researcher, Division of Global Innovation Strategy, STEPI	
	09:30 ~ 11:30	**[Session 1]** Technology Entrepreneurship and Ideation-based Startup Jong In Choi, Professor, Hanbat National University	
	11:30 ~ 12:00	Introduction of Korean Cultural Experience (1) (Sevinj Iskandarova, Lecturer, Khazar University)	
	12:00 ~ 13:30	Lunch	
	13:30 ~ 16:00	**[Session 2]** Presentation on the Progress and Achievement of the 2019 Project • Zaur Akhmedzadeh, Head, Technology Transfer and Business Incubator Department, ANAS-HTP Presentation on Azerbaijan expert's reporting (cases of application and improvement of business capacity after the invitational training for STI capacity development in Korea) • Vusal Suleymanli, Chief Innovation Officer,	

DATE	TIME	DESCRIPTION	VENUE
		Innova Ventures Group • Aysel Abdullayeva, Lecturer, Azerbaijan State University of Economics (UNEC) • Isa Gasimov, Senior Specialist, Entrepreneurship Fund	
Thursday, October 17	09:00 ~ 11:00	**[Session 3]** Role of Technopark for Technology Commercialization and Chungnam TP Case Study • Seong-Kun Kang, General Manager, Chungnam Technopark.	
	11:00 ~ 11:20	Introduction of Korean Cultural Experience (2) (Sevinj Iskandarova, Lecturer, Khazar University)	
	11:20 ~ 12:30	Lunch	
	12:30 ~ 14:30	**[Session 4]** Korean Technopark (K-TP) Model as an Integrated Business Platform: GBTP Case • Seong-Keun Park, Team Manager, Gyeong-Buk Technopark	
	14:30 ~ 14:50	**[Closing Ceremony]** Awarding of Certificates	
	15:00 ~ 15:20	Meeting with Azerbaijan's Research Team • Specify the scope of the 2020 project	
	16:00 ~ 16:30	Field study, Data collection, and Interview • Ministry of Transport, Communication & High Technologies	
	17:00 ~ 18:00	Field study, Data collection, and Interview • Azerbaijan Innovation Agency	
Friday, October 18 - Saturday, October 19		Wrap-up Meeting Departure Arrival	Azerbaijan Korea

3.8 Main Activities

1) Discuss the pre-inspection of the workshop on disseminating the outcome with the ANAS-HTP research team

- Introduction on the status and performance of promoting the 2019 project
  - (Kick-off meeting and field survey, Feb. 16~23, 2019): Confirmation of the project scope, field survey to identify the status of the innovation ecosystem in Azerbaijan
  - (Invitational workshop in Korea to reinforce capabilities, May 25 ~ Jun. 2, 2019): Derivation of methodology for technical transfer/commercialization and technology startup, transfer of experiences by the Korea Technopark and TLO (Technology Licensing Organization), and exchange of study and knowledge through the participation of local experts
  - (Local workshop to disseminate the results and field survey, Oct. 12~19, 2019): Sharing and dissemination of knowledge for the related parties in Azerbaijan, derivation of improvement methods on the level of application to the actual work, and concretization of the implementation plan for the following year (Year 2020)
- Workshop PR: Outcome to be disseminated through the Azerbaijan media (ELM TV, AZERTAC Video, ANAS Facebook, ANAS website, ANAS-HTP website, etc.)
  - Verification of other matters (workshop schedule, estimated participants, venue, source book, certificate and visit schedule to institutions, etc.)
- Materialization of project scope and activities in 2020
  - Hold invitational workshop in Korea
  - Provide policy channel for high-level dialogue
  - Hold workshop on disseminating the outcome

2) Interview with related parties and inspection of the ANAS-affiliated Institute of Information Technology

- The ANAS-affiliated Institute of Information Technology was established in 2002.

- The institute deals with national analysis, web technology, information security, social network, biometric system, information resource, ICT, geographic information, Big Data-related research and IT training, etc.
- The doctorate course was opened in 2015, and courses are mostly related to information security.
- Currently, there are 15 Internet Service Providers (ISP).
- There are a total of 400 employees (3 academicians, 2 correspondent members, 8 Doctor of Science (ScD) holders, 28 PhDs, 7 employees in the doctorate course, and 19 employees who completed the doctorate). At present, there are 18 departments including the education innovation center, e-library, multimedia, AzScienceNet, etc.

3) Interview with related parties and inspection of the ANAS-affiliated Institute of Control Systems

• The name of the ANAS-affiliated Institute of Control Systems was changed to its current name in 2014.

- The former name of this institute was the “Laboratory of Electro-modeling” affiliated with the “Petroleum Expedition of Azerbaijan Academy of Sciences (AS)” established in 1952. The institute was renamed “Computer Center (CC)” in 1958, “Institute of Cybernetics” in 1965, and finally “Institute of Control Systems” in 2014.
- At the end of the 1950s, the emergence of the general-purpose digital computer drew interest in researches regarding numerical analysis and programming in Azerbaijan. Among the Soviet states in the 1960s, Azerbaijan was second to Ukraine in terms of playing an important role in the development of cybernetics, computer engineering and other engineering development, etc.
- There are about 90 employees (4 academicians, 3 correspondent members, 25 doctors of science, 56 PhDs, etc.); at present, the institute has 6 departments and 26 labs.

- The main research fields include identification method and control system, technical process, mathematical modeling of the technical and ecological system, signal recognition method and technical diagnosis system method, theoretical control issue, telegraphic theory, intelligent and powerful hybrid system for control, diagnosis, prediction, and management, modeling and management in agriculture, modeling in the socio-economic system, etc.
  - The main projects include oil drilling improvement project, establishment of online early earthquake warning system and online early gas warning system (this system was introduced in the conference held recently in Turkmenistan to draw interest from 5 countries* around the Caspian Sea), and establishment of earthquake monitoring system and irrigation project, etc.

    * 5 countries around the Caspian Sea: Turkmenistan, Iran, Russia, Kazakhstan, and Azerbaijan

4) Interview with related parties of the Entrepreneurship Development Fund

- The Ministry of Economy-affiliated Entrepreneurship Development Fund was established in 2002 based on the “National Fund for Entrepreneurship Support” operated from 1997.
  - The approved fund amount is 1.9 billion Manat. 5,000 ~ 50,000 Manat for 3 years, 50,000 ~ 1 million Manat for 5 years, and 1 ~ 10 million Manat for 10 years can be loaned. The annual loan ratio is 5%.
  - This year, 160 million Manat is planned to be issued as Soft Loan, and 68% of the loan is allocated to the agricultural sector.
  - Nationally, the fund is provided mostly for entrepreneurs in the sectors of agriculture, tourism, and startup.
  - In the startup support field, 8 startup offices are opened, and intellectual services (training) are provided to entrepreneurs and students, etc.

5) Interview with related parties of BEU (Baku Engineering University)

- BEU (formerly Qafqaz University) is a national university established by a Presidential Decree two years ago in 2016.
  - BEU has 3 departments (Economics and Policymaking, Engineering, and Pedagogy); as the university is currently aiming to specialize in Engineering, the Department of Pedagogy will be closed after 2 years. There are a total of 16 majors in the Department of Engineering.
- 70% of the class is in English, 25%, in Azerbaijani language, and 5%, in Russian.
- There is great interest in the internationalization of the technology sector in the national level as well, and emphasis is placed on the necessity of technical transfer and HTP. As such, BEU is aiming to become a technology-specializing university centrally in English. More focus will be given on mechanical engineering, computer science, and IT.
  - Currently, BEU has a total of 4,700 students and almost 300 faculty members.
  - The current Rector of BEU was the former Rector of Technical University for 17 years, having participated in the 3-year project (Construction of the Electronic Management System in Technical University) with KOICA. During this project, the CEO of Korea Polytechnics Colleges also visited the Technical University.
- Currently, BEU has a Double Diploma (Master’s + Doctor’s) system established according to Azerbaijan’s Presidential Decree.
- BEU has great interest in the collaboration with HTP on a national level. HTP experts are invited as lecturers in BEU.
- BEU is operating a project referred to as NYC (New Idea Competition).
  - New Idea Competition (NYC) is a project hosted by BEU Technopark (Baku Engineering University Technopark), and support is provided to cultivate innovative entrepreneurship. In addition, NYC is contributing to the improvement of entrepreneurship of the youth and qualitative improvement of scientific research and manpower training.

- NYC plays the role of a bridge between the youth and entrepreneurs. Held once a year, this contest can be participated in by not only the main group of BEU but also by the youth all around Azerbaijan.
- Main Mission of NYC
  - ➢ Support innovative entrepreneurship
  - ➢ Contribute to providing the infrastructure for ecosystem to enable the development of cutting-edge technology by BEU Technopark
  - ➢ Support socio-economic improvement
  - ➢ Support ICT development
  - ➢ Support youth startup
  - ➢ Promote the entrepreneurial approach
- The purpose of NYC is to contribute to discovering innovative ideas in the technology area. These areas include startup idea, mobile APP, Internet, program provision, game, service program, corporate software, new business model, service robotics, information technology, consumer electronics, medical information technology, etc.

6) Workshop on Disseminating the Outcome

- Opening Ceremony (Samir Mammadov, Director of ANAS-HTP)
  - Opening address (Eun Joo Kim, PM of STEPI): Purpose of the workshop on disseminating the outcome, overall introduction on the program
- (Session 1) 「Technology Entrepreneurship and Ideation-based Startup」(Jong-in Choi, Professor, Hanbat National University)
  - Technology Entrepreneurship
    - Concept: Entrepreneurship
    - Ecosystem of Entrepreneurship

  - Entrepreneurship University
  - Entrepreneurship Case: Professor, CEO, Singer, et al
  - Key concept: INC model

- ·Ideation-based Startup
  - Key failure factors of startup
  - Ideation source: Needs and capability
  - Change trend: Conceptual age
  - Valley of death (TVOD)
  - Methodology: TRIZ
  - Altshuler of TRIZ
  - TPM linkage
  - Idea to Opportunity
  - Worksheet
  - Idea potential: Practice and presentation
  - Value proposition

- Korean National Entrepreneurship Plan
  - Industry-university- research center collaboration plan
  - University Entrepreneurship education plan
  - Hanbat National University case: Startup and graduate school of Entrepreneurship
  - INC model reality and alternative
  - 5 key factors of Entrepreneurship University

▪ (Session 2)

1) Presentation of the 「2019 Project Performance Results」

▪ (Zaur Akhmedzadeh, Head, Technology Transfer and Business Incubator Department, ANAS-HTP)

- ANAS has 50. The A0 patents, and ANAS-HTP is performing the role of transferring these patent rights NAS-HTP Business Incubator Center is supporting various startups.
- One of the issues that ANAS-HTP must gradually solve regarding technical transfer and commercialization in Azerbaijan is to find financial support for technical transfer -- in other words, to find financing as well as a sponsor for development.
- A good business plan must be prepared to secure the funding, and research on the potential market is required for future products.
- In addition, legislation (act, law, regulation, etc.) is required for the Technopark activity. This legislation should be able to reflect the issues related to technical transfer, commercialization, and startup support. Currently, the relevant laws on Technopark are as follows:
  - Regulations on Technoparks (2014.5.15)
  - Resolution of the Cabinet of Ministers on the Approval of Regulations for the Registration of Residents of the Technology Park (2015.7.8)
  - Order of the Cabinet of Ministers on the High Technology Park of the Academy of Sciences (2016.11.16)
  - Order of the President of Azerbaijan on the Establishment of the High Technology Park of the Academy of Sciences (2016.11.8)
- ANAS-HTP has the “right of the legislative initiative,” and ANAS-HTP participated in the document preparation of the recently adopted “Resolution of the Cabinet of Ministers on the approval of the Rules for placing government orders for products manufactured in the ANAS High Technologies Park.” When ANAS-HTP proposed the legislative initiative, the technical experience of advanced nations including the experience of Korea was cited.

2) Presentation & Discussion on the 「Local Application Case in Azerbaijan and Lessons (Experiences & Lessons of the Invitational Training)」

- (Vusal Suleymanli, Chief Innovation Officer, Innova Ventures Group)

- The case of applying the knowledge from the STEPI technical transfer & commercialization capability reinforcement training is as follows:
  - (ANAS-HTP Technology Commerce Platform Update) One of the technical transfer and commercialization promotion activities in Azerbaijan of the “Technology Commerce Platform” is performed by ANAS-HTP, which is integrating the existing patent information through the collaboration with ANAS. Currently, information on over 100 venture capital companies is being disclosed on the online platform. <Preliminary Survey Implementation & Adjustment: Mammad Baghir-zada and Zaur Akhmedzadeh>
  - (KOICA Smart Bridge Program Organization) As the innovative ecosystem promotion project of EDF (Entrepreneurship Development Fund), the Smart Bridge Program is organized to support the KOICA technical transfer and commercialization and entrepreneurship development. <Planning & Organization: Isa Gasimov>
  - (New Idea Program) Youth Entrepreneurship Development Program by BEU <Planning & Organization: Isa Gasimov>
  - (Innovative Entrepreneurship Program) Special initiative for innovative business development by SMEDA (Small and Medium Entrepreneurship Development Agency) <Program Organizer: Vusal Suleymanli>
  - (I2B Program) The From Idea to Business Entrepreneurship Program is operated in the form of entrepreneurship boot camp and workshop hosted by the Ministry of Transport, Communication & High Technologies. <Joint Participant: Vusal Suleymanli>
  - (Crowd Funding for Promoting the Innovative Entrepreneurship Ecosystem) Innovative ecosystem project support by the International Bank of Azerbaijan (IBAR) <Project Participants: Samir Valiev and Aysel Abdullayeva>
  - (Republic Innovation Competition) It is one of the largest programs for the development of entrepreneurship mindset in Azerbaijan, and the competition is overseen by ANAS, Ministry of Economy, Social Innovation Agency, and Ministry of Transport, Communication & High Technologies.

- (Aysel Abdullayeva, Lecturer, Azerbaijan State University of Economics (UNEC))
  - (Innovative economy diagnosis in Azerbaijan)
    - The goal of the Azerbaijan government is to improve the National Innovation System (NIS) through the indirect and direct reform and development program.
    - There is no single central organization to develop and implement the STI policy.
    - Performances in special fields (mathematics, chemistry, geology, and other priority fields) were good during the Soviet Period.
    - Innovative vision and policy strategy are insufficient.
  - (Suggestion)
    - ANAS-HTP must participate actively in the innovative strategy development process to contribute to the technical commercialization factors while preparing consistently with regard to the innovative strategy.
    - ANAS-HTP must establish effective, appropriate organizational structure and clear roles on the sub-unit.
    - ANAS must support ANAS-HTP in providing high-quality research while giving considerations to commercialization.

- (Isa Gasimov, Senior Specialist, Entrepreneurship Fund)
  - It was mentioned that there was help in actually applying and utilizing the experiences and knowledge obtained through the STEPI training on the work.
  - KOICA's Smart Bridge Program (technical transfer and commercialization and entrepreneurship development were supported) and BEU's New Idea Program (youth entrepreneurship development program) were programs that I directly planned and organized.

- (Session 3) 「Role of Technopark for Technology Commercialization and Chungnam TP Case Study」 (Seong-Kun Kang, General Manager, Chungnam Technopark)
  - Establishment and overview of Chungnam TP: CTP's History, Organization, and Function, Strategy for Development, Achievement, etc.

- Chungnam TP Business promotion program: Contact center program, Integrated Support program for technology commercialization, Techno CEO Training Program, Business incubation system, etc.
- Chungnam TP technology transfer support system: Know-how and process of technology transfer, Integration of technology demands, Technology commercialization support, Technology transfer case study in the Korea market
- Chungnam TP case of international cooperation: Overview of global commercialization, Korea-Mongolia (horse oil cosmetic), Korea-Ghana (meat processing facility)
- Conclusion and opinion for developing country TP operation: Need to link the Human resources, Business resources, and Acknowledge resources, Need to cooperate with a public organization to build up the governance system, Need to set up the strategy considering the nation's industry environment and situation

▪ (Session 4) 「Korean Technopark (K-TP) Model as an Integrated Business Platform: GBTP Case」(Seong-Keun Park, Team Manager, Gyeong-Buk Technopark)

- Introduction of K-TP Model: Background of the Korean Technopark project, concept and brief history, and
- Introduction of Gyeong-Buk Technopark (GBTP): Major functions, Facilities, and Achievements
- Evolution of GBTP – Globalization: Globalization of K-TP model for mutually beneficial global cooperation via ODA programs (Ethiopia, Uzbekistan…)
- Toward TP (“True Platform”): Lessons from the GBTP case, future challenges and strategic goal as a “True Platform” for regional industry promotion

7) Meeting with the Azerbaijan Research Team

▪ Discussion on the 2020 Project Plan

- Invitational workshop held in Korea
  - Period: March 2~6, 2020

- Training Target*: 6 participants from ANAS-HTP (including the entrepreneur recommended by ANAS-HTP), 1 participant from the Innovation Agency, 1 participant from the Ministry, 1 participant from the ANAS Presidium, and 1 participant from EDF (Entrepreneurship Development Fund), etc.

  * The trainees will be selected, and the participants in the training will be personnel and middle-rank decision makers in technical transfer and commercialization.

• MOU agreement on the STEPI-ANAS science & technology collaboration

• Hold high-level dialogue on the science & technology policy

- ANAS Presidium
- Cabinet of Ministers
- Ministerial level
- Azerbaijani Ambassador to Korea, Korean Ambassador to Azerbaijan, KOICA Director

8) Interview with related parties of Azerbaijan’s Ministry of Transport, Communication & High Technologies

• On November 6, 2018, the Presidential Decree on the establishment of the Innovation Agency affiliated with Azerbaijan’s Ministry of Transport, Communication & High Technologies was announced.

- Based on this, the purpose of the corporation is to support scientific research to promote innovative projects (including startup), support investment, grant, and concessional loan on the authorized capital (including venture financing), and promote innovative initiatives.

• The main goal of the Innovation Agency is to reinforce the sustainable development and competitiveness of the ICT field, expand the hi-tech industry, and promote the R&D of new technology.

- The purpose of establishment of the Innovation Agency is to enable the innovative development of the non-petroleum sector in Azerbaijan.
  - Product manufacture and provision services that are innovative and hi-tech are promoted, with the existing local brand aimed at advancing and expanding to the international market.
  - Digital innovation products and services identification, robot and cloud technology, large-scale data processing, and support for intellectual solution on AI are included.

- The Innovation Agency supports youth ideas and infrastructure for innovative ideas, and the Business Incubation (BI) Center is operated 24 hours a day, 7 days a week to develop and improve innovative products and cutting-edge technology.
- Azerbaijan's Ministry of Transport, Communication & High Technologies is planning to activate the process of improving the role and function of the Innovation Agency.
- Hosted by Azerbaijan's Ministry of Transport, Communication & High Technologies (also participated in by the Ministry of Youth & Sports, Ministry of Education, Heydar Aliyev Center, Azerbaijan Youth Foundation, UNDP, US Embassy in Azerbaijan, and Azerbaijan Telecommunication Company Azercell) to hold the innovative technology forum of the 1st InnoFest Innovation Festival in Baku during the period May 29 ~ June 4, 2019 with the goal of developing innovative technologies, implementing new projects, and attracting young entrepreneurs in the innovative technology field.
- During the Innovation Week (Oct. 18~23, 2019), "Eurasia Innovation Day" will be held by Azerbaijan's Ministry of Transport, Communication & High Technologies and Huawei on October 23.
  - This event is a dialogue platform participated in by politicians, scientists, representatives of the private sector, media, etc.

9) Interview with related parties of the Innovation Agency in Azerbaijan

- Recently, the Innovation Agency submitted a PCP (Project Concept Paper) to KOICA on the "Development of a Strategic Innovation Roadmap."

- The performance period of this project is 2 years (2021-2022).
- The goal includes: 1) Development of strategic innovation roadmap for the efficiency of the innovation policy in Azerbaijan; 2) Reinforcement of the capability of the Innovation Agency as a key player in the innovation ecosystem on developing and implementing the innovation policy; 3) Benchmarking of the Korean experience to establish a venture capital fund, and 4) Search for methods of constructing the innovation ecosystem through the collaboration between the Innovation Agency and Industry-University-Institute.
- This project was selected as a project* subject to preliminary survey by KOICA.

  * Once the PCP is registered, KOICA will perform preliminary review, and projects that passed the pre-review are subjected to preliminary survey to prepare the PD (Project Document). This PD includes the details of the project, budget, schedule, PDM, etc. KOICA conducts the final project review on the projects prepared with PD; in the case of this project, KOICA conducted the preliminary survey, and domestic professionals who will prepare the PD will be dispatched to Azerbaijan.

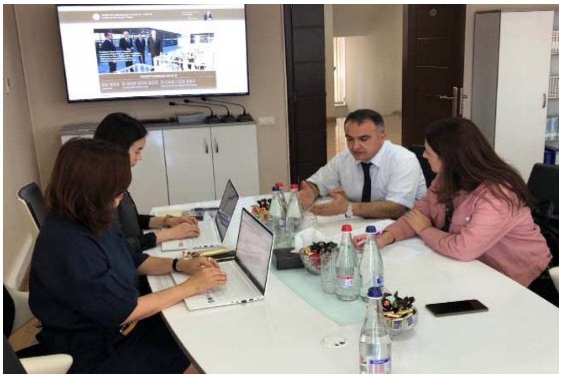
- On October 21 ~ 29, KOICA will send the preliminary survey team to Azerbaijan.
- The purpose of the preliminary survey is to calculate the project period, budget, TOR, and project cost and discuss the project performance indicator setting.
- The preliminary survey team will visit the Innovation Agency, Ministry of Transport, Communication & High Technologies, Azerbaijan Technical University, Innoland, ANAS-HTP, AzIntelecom Data Center, etc. for investigation.

• Mr. Seong-Keun Park mentioned that it is important to select a specialized industry with high added value in developing the innovation roadmap to analyze the innovation capability of the local strategic industry and establish the RIS (Regional Innovation Scheme).

  - Azerbaijan is developed with basic science, so the most important thing is to find the strategic field for collaboration with Korea.
  - Especially, technology that can cope with the technical paradigm changes must be secured.

3.9 Photos

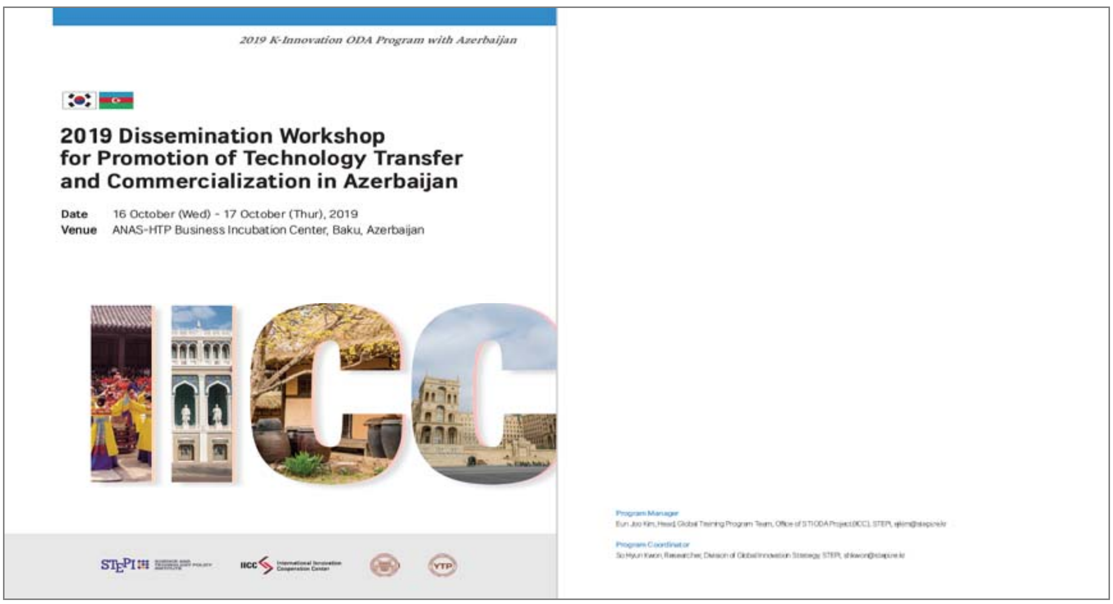
Meeting with ANAS-HTP	Meeting with the Institute of Information Technology
	
Group Photo at the Institute of IT	**Site Visit to the Institute of IT**
	
Meeting with the Institute of Control System	**Site Visit to the Institute of Control System**
	

Meeting with the Entrepreneurship Development Fund	Meeting with BEU
	
Dissemination Workshop (1)	**Dissemination Workshop (2)**
	
Dissemination Workshop (3)	**Dissemination Workshop (4)**
	

Dissemination Workshop (5)	Dissemination Workshop (6)
	
Dissemination Workshop (7)	**Dissemination Workshop (8)**
	
Dissemination Workshop (9)	**Dissemination Workshop (10)**
	

Meeting with the Ministry of Transport, Communication & High Technologies	Meeting with the Azerbaijan Innovation Agency
	

3.10 Workshop Book



3.11 Public Activities (Press Releases)

- ANAS Homepage (2019.10.16)
  (http://science.gov.az/news/open/11297

AZƏRBAYCAN MİLLİ ELMLƏR AKADEMİYASI

ANA SƏHİFƏ | AMEA HAQQINDA | AMEA RƏYASƏT HEYƏTİ | AMEA-nın STRUKTURU | AMEA-nın ÜZVLƏRİ, FƏXRİ ADLAR | ƏKS ƏLAQƏ

ANA SƏHİFƏ » XƏBƏRLƏR » KONFRANSLAR, İCLASLAR

A+ a-

FOTO
VİDEO (ELM TV)

16.10.2019 14:10

Azərbaycanda texnologiyaların transferi imkanlarının öyrənilməsinə həsr olunan seminar keçirilib

Oktyabrın 16-da AMEA-nın Yüksək Texnologiyalar Parkı (YT Park) və Cənubi Koreyanın Elm və Texnologiya Siyasəti İnstitutunun (STEPİ) birgə təşkilatçılığı ilə "Azərbaycanda texnologiya transferi və elmin kommersiyalaşdırılmasının təşviqi" layihəsi çərçivəsində növbəti seminar keçirilib.

Mərkəzi Elmi Kitabxanada baş tutan tədbirdə YT Parkın direktoru, yer elmləri üzrə fəlsəfə doktoru, dosent Samir Məmmədov, AMEA Rəyasət Heyəti aparatının Beynəlxalq əlaqələr idarəsinin Şərq ölkələri ilə əlaqələr şöbəsinin müdiri Bünyamin Seyidov, STEPİ-nin nümayəndələri - təlimlər qrupunun rəhbəri Eun Yo Kim, layihənin koordinatoru So Hyun Kvon, Hanbat Universitetinin professoru Yonq İn Çoi, o cümlədən AMEA-nın elmi müəssisə və təşkilatlarının əməkdaşları iştirak ediblər.

Tədbiri YT Parkın direktoru S.Məmmədov açaraq seminarın artıq üçüncü dəfə baş tutduğunu və burada əsas məqsədin koreyalı alimlərlə birgə Azərbaycanda texnologiyaların transferi və elmi nailiyyətlərin kommersiyalaşdırılması imkanlarının öyrənilməsi olduğunu bildirib.

- RƏSMİ SƏNƏDLƏR
- XƏBƏRLƏR
- BEYNƏLXALQ ƏMƏKDAŞLIQ
- AMEA İNSTİTUT VƏ TƏŞKİLATLARININ SAYTLARI
- ELMİ TƏDQİQATLARIN ƏLAQƏLƏNDİRİLMƏSİ ŞURASI
- PROBLEMLƏR ÜZRƏ ELMİ ŞURALARIN SAYTLARI
- SOSİAL XİDMƏTLƏR
- LİNKLƏR

SON XƏBƏRLƏR

18.10.2019 16:58
"Folklorun təsnifi problemi: janr, yoxsa performan…

18.10.2019 16:11
Akademik Akif Əlizadə paleontologiyanın inkişafına…

18.10.2019 15:53
Dövlət Müstəqilliyi Günü Riyaziyyat və Mexanika İn…

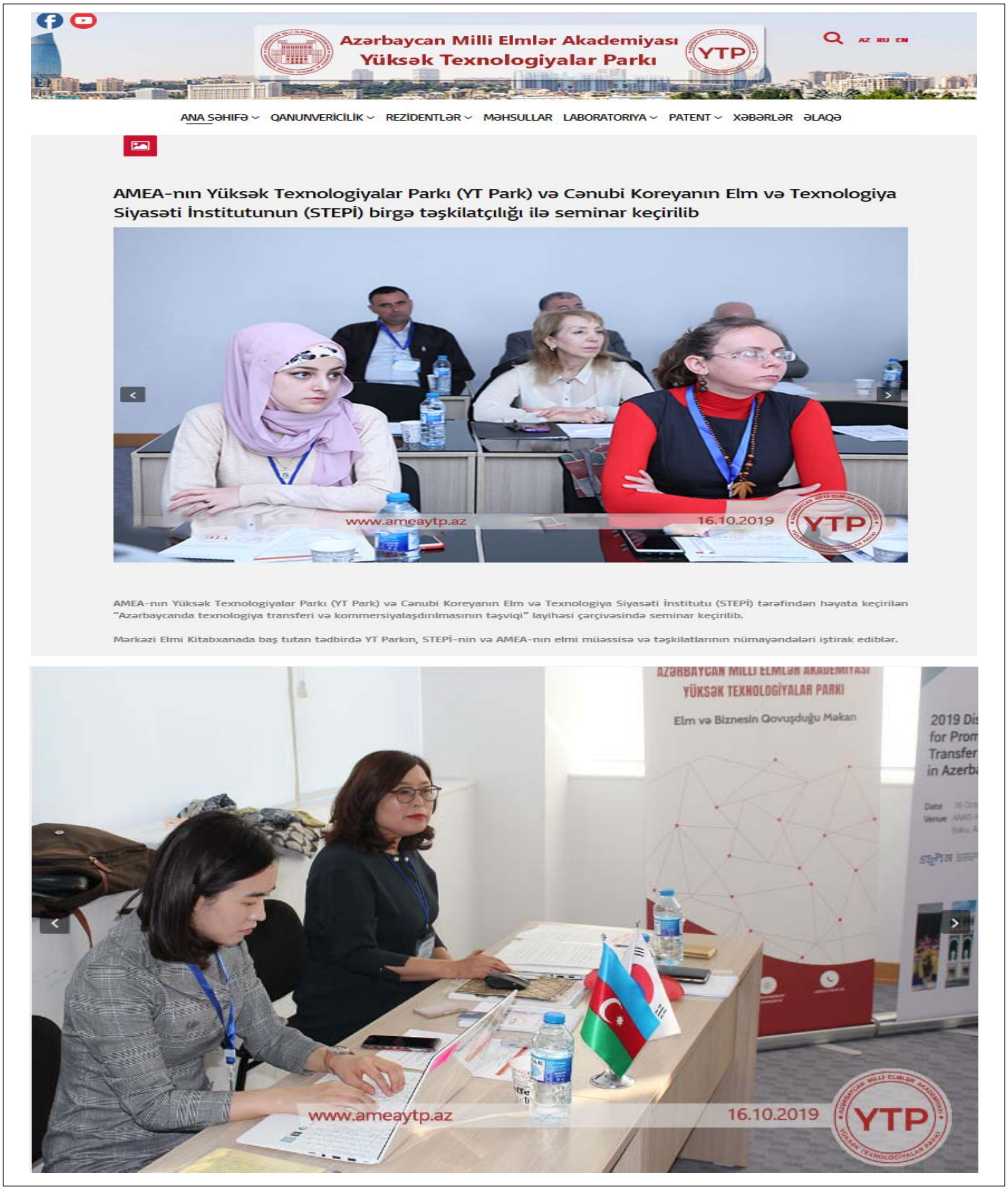
18.10.2019 15:53
Bakıda VI Beynəlxalq Uşaq və Gənclik

- Azerbaijan ELM TV (2019.10.16)
  (http://science.gov.az/video/open/news_11297)

ANA SƏHİFƏ » VİDEOQALEREYA (ELM TV)

Azərbaycanda texnologiyaların transferi imkanlarının öyrənilməsinə həsr olunan seminar keçirilib

- ANAS-HTP Homepage (2019.10.16)
  (http://ameaytp.az/index.php/component/content/article/88-son-x-b-r/634-amea-n-n-yueks-k-texnologiyalar-park-yt-park-v-c-nubi-koreyan-n-elm-v-texnologiya-siyas-ti-institutunun-stepi-birg-t-skilatc-l-g-il-seminar-kecirilib?Itemid=539)



- ANAS-HTP Facebook (2019.10.16)
(https://www.facebook.com/770709229636847/posts/3095402070500873/)

AMEA Yüksək Texnologiyalar Parkı
10월 16일 오후 2:29 ·

AMEA-nın Yüksək Texnologiyalar Parkı (YT Park) və Cənubi Koreyanın Elm və Texnologiya Siyasəti İnstitutunun (STEPİ) birgə təşkilatçılığı ilə seminar keçirilib

AMEA-nın Yüksək Texnologiyalar Parkı (YT Park) və Cənubi Koreyanın Elm və Texnologiya Siyasəti İnstitutu (STEPİ) tərəfindən həyata keçirilən “Azərbaycanda texnologiya transferi və kommersiyalaşdırılmasının təşviqi” layihəsi çərçivəsində seminar keçirilib.

Mərkəzi Elmi Kitabxanada baş tutan tədbirdə YT Parkın, STEPİ-nin və AMEA-nın elmi müəssisə və təşkilatlarının nümayəndələri iştirak ediblər.

Tədbiri giriş nitqi ilə açan AMEA Yüksək Texnologiyalar Parkının direktoru Samir Məmmədov seminar iştirakçılarını salamladıqdan sonra STEPİ və AMEA YT Parkın birgə təşkilatçılığı ilə Azərbaycanda keçirilən seminarın ilk olmadığını və davamlı keçirildiyini qeyd etdi.
O, ölkə iqtisadiyyatının əsas prioritet istiqamətinin elm, texnologiya və innovasiya potensialının artırılması olduğunu bildirdi. Bu sahədə beynəlxalq təcrübənin öyrənilməsinin vacibliyini vurğulayaraq qeyd edib ki, seminarın keçirilməsində əsas məqsəd Koreyanın innovasiyaların sənayedə tətbiqi sahəsində təcrübəsinin öyrənilməsi, həmçinin Koreya ilə Azərbaycan arasında elm və texnologiya sahəsində davamlı inkişaf edən əməkdaşlığa nail olmaqdır.

- Institute of Information Technology Homepage (2019.10.14)
  (https://ict.az/az/news/4704)

- BEU, MektebGushesi, Modern.az, Qafqazinfo.az, Muallim.edu.az (2019.10.15)

(http://www.beu.edu.az/az/news/xeberler/bmu-ile-koreyanin-aparici-tedqiqat-merkezi-arasinda-emekdasligin-esas-istiqametleri-mueyyen-olunub-10228/)

BMU BAKI MÜHƏNDİSLİK UNİVERSİTETİ TMS PMS E-mail

Xəbərlər Təhsil Elmi-tədqiqat Tələbə Həyatı BMU Haqqında

Ana səhifə > Xəbərlər

BMU ilə Koreyanın aparıcı tədqiqat mərkəzi arasında əməkdaşlığın əsas istiqamətləri müəyyən olunub

Bakı Mühəndislik Universitetinin (BMU) rektoru, professor Havar Məmmədov Cənubi Koreyanın Elm və Texnologiya Siyasəti İnstitutunun (STEPİ) Beynəlxalq İnnovasiya Əməkdaşlıq Mərkəzinin nümayəndələri - təlimlər qrupunun rəhbəri Eun Joo Kim, layihənin koordinatoru So Hyun Kvon və Hanbat Universitetinin professoru Yonq İn Çoi ilə görüşüb.

Rektor H.Məmmədov Azərbaycanla Koreya arasında əksər sahələrdə, o cümlədən təhsil sahəsində uğurlu əməkdaşlıq əlaqələrinin olduğunu vurğulayıb. Son illər ölkədə Prezident cənab İlham Əliyevin rəhbərliyi ilə insan kapitalının inkişaf etdirilməsi, keyfiyyətli əmək bazarının yaradılması istiqamətində görülən işlərə toxunan H.Məmmədov Koreya ilə peşə təhsili sahəsində olan səmərəli

MektebGushesi(https://mektebgushesi.az/2019/10/15/bmu-ilə-koreyanin-aparici-tədqiqat-mərkəzi-arasinda-əməkdasligin-əsas-istiqamətləri-muəyyən-olunub/?fbclid=IwAR0bU4M7juifswxcLKo6WMGPcVg_QNCHlv51iMqBc_QsoanK4yMwe4mlzwM)

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Qafqazinfo.az(https://qafqazinfo.az/news/detail/bmu-ile-koreyanin-tedqiqat-merkezi-arasinda-muzakireler-olub-266749?fbclid=IwAR2FCUpU974a-nPPkjbHAy6Vw7-ZKqVlbQj_kMPUR8HAGj5aZnsjtHV7HJc)

Muallim.edu.az(https://www.muallim.edu.az/news.php?id=7874&fbclid=IwAR0p7NeoinO7ul66rKxPVlad6wktYbQGA0vODYBb-S9Qj7rUU43sjSlccDo)

CHAPTER 5
Conclusion

Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan

Chapter 5. Conclusion

1. Implications and Suggestions

1.1 Suggestions of a New Paradigm of Technology Transfer and Commercialization

Too much importance is placed on academic contribution, but too little attention is given to commercialization knowledge in the public organization.

- The skills necessary for successful technology transfer and commercialization are different from the skills necessary to do good science.
- Technology transfer and commercialization can be compatible with -- and in fact enhance -- the traditional missions and roles of a university or a research institute.
- More research is needed on how to strengthen the linkages between knowledge creation, diffusion, and application.
- Put more emphasis on cooperation among various bodies to enhance the nation's performance in commercializing the public sector.

More comprehensive effort seems to be needed, instead of promotion of inward transfer of technologies,

- Such as infrastructure of TT/TC, evaluation & analysis of technology demand, development of domestic absorptive capacity to digest, assimilate, and improve upon the transferred technologies to adopt.

- In order for technologies made from the public sector R&D to enter the market, pre-evaluation as well as follow-up of R&D focusing on marketability is necessary.
- Developing domestic absorptive capacity to digest, assimilate, and improve upon the transferred technologies
- New forms and models of technology transfer offices (TTOs) are recommended. Explore more effective business models in several countries, such as Yeda, Yissum, Oxford OTL, etc.

The shift of mindset and new paradigm to Techno-Marketing is highly recommended as well.

- Note that technology transfer and commercialization require a dedicated effort to be successful.
- A major shift of mindset is needed to change the perception and improve the performance dramatically.
- Best effort is needed to eliminate the bureaucracy, as it is the biggest obstacle reducing the performance.
- Technology marketing is very important to get a super performance of Technology transfer. Education on marketing and technology marketing is very desirable to shift from supply-oriented to demand-oriented, and from R&D-oriented to commercialization-oriented.
- Boost the cooperation atmosphere among related bodies to enhance the nation's performance in commercialization.

1.2 Implications and Recommendations for Technology Entrepreneurship, Startup, and Incubation in Azerbaijan

The main implication for Azerbaijan is that the strength of entrepreneurial education in a regional innovation system reinforces the open innovation capability and performance of companies. It has been shown that entrepreneurship in particular promotes open innovation in companies involved in new value creation (Choi and Markham, 2019; Chesbrough, 2006; Nambisan, et al, 2018; Yun, 2015).

Corporate entrepreneurship education recommendations are a collaboration between the regional ecosystem and the university's entrepreneurial programs. The successful program "Campus Wide Entrepreneurship Education" (CWEE), which is defined at the university level, includes several schools and colleges that are interconnected and diverse student teams composed of and directed by professors and mentors, and it is important for the diverse corporate needs of entrepreneurial thinking. We found that CWEE must include the five factors: entrepreneurial leadership; faculty champions; student-focused policies; engagement with the community; and decentralized, autonomous structure of entrepreneurship programs. CWEE has a close relationship with different kinds of industry, large companies, and human and capital resources.

In conclusion, we can sum up the five success components by applying the Triple Helix model to categorize the interview results by university, industry, and government for Azerbaijan (Choi and Markham, 2019).

1.2.1 Entrepreneurial Leadership by Entrepreneurship Education Plan

First is Entrepreneurial Leadership by Entrepreneurship Education Plan. University entrepreneurial leadership from the administration and faculty is very important for the ecosystem. The leadership of administration and faculty provides a top-down, bottom-up approach to campus-wide entrepreneurship education. Entrepreneurial leadership creates a culture for industry involvement and administration support for cooperative programs. The faculty engages in programs and course creation and takes charge of thought leadership and skills training. A faculty champion paves the way to entrepreneurship education and supports corporate entrepreneurship.

Industry leadership can serve as a cornerstone for corporate entrepreneurship through the deep involvement and role of champions. They can lead the university and government, especially through champion proposals for university engagement with a company. Industry entrepreneurial leadership initiates the creation of corporate entrepreneurship programs at universities. It connects leading corporate entrepreneurs with university programs, faculty, and students.

A government entrepreneurial leader has an important role as a policy champion. He commissions research and economic development studies and policy champions for tax and regulatory issues. Additionally, these leaders initiate and fund technical and business programs in emerging areas of strength and opportunity.

1.2.2 Student Focus Policy

Second is Student Focus. The student is a core asset of corporate entrepreneurship, and the university has to develop and provide a corporate entrepreneurship program to students at an adequate level for their undergraduate and graduate degrees. Universities should offer corporate entrepreneurship education opportunities (courses, curriculum, majors, minors, certificates) and extracurricular activities (games, visits, competitions). The student has a career vision, and he/she is a key member of development programs. Students can establish clubs and activities to increase corporate entrepreneurship confidence.

The industry should collaborate to create corporate entrepreneurship programs and host visits by university students. It should also establish hiring and promotion criteria for corporate entrepreneurship and develop job descriptions for corporate entrepreneurs so that students can see a new window of opportunities. This includes internship opportunities so that students can experience the real issue and to narrow the gap between the theory-based classrooms and practical large companies' activities.

Governments expand internship programs with government agencies (TP, Innopolis) to include corporate entrepreneurship opportunities. They create education-funding programs and craft policy to include women and minorities.

1.2.3 Faculty Champion Development

Third is Faculty Champion. University faculties are often hesitant to get involved in entrepreneurship education because of their interests as well as time constraints, traditional evaluation criteria, and sometimes ignorance of the industry. The faculty identify and support corporate entrepreneurship for students and corporate businesses. Universities provide time and leave policies to participate with the industry and make entrepreneur-friendly evaluation criteria. They need course development support as well as to provide faculty corporate entrepreneurship funding. For instance, the Korean government provides a “faculty leave system” for startups. The faculty should have time to develop new courses on corporate entrepreneurship.

The industry participates in course development and assigns mentors (or Executives in Residence) to courses and projects. They provide access to company projects for courses and also offer “faculty internships” at companies. The industry can create professorships in corporate entrepreneurship.

Governments can sponsor economic development studies and provide funding for positions to hire and retain high-profile faculty. They support centers and institutes for university/industry cooperation and corporate entrepreneurship development. They integrate economic development infrastructure between universities and industry.

1.2.4 Community Engagement Strategy

Fourth is Community Engagement. Community engagement is the process by which community interests and individuals form a lasting, permanent relationship to apply the collective vision for community benefit. Communities create joint programs, such as speakers, games, and competitions, with university and university extension services for corporate entrepreneurship. They contribute to regional economic development and establish entrepreneurship clinics and project spaces.

The industry can act as a sponsor and enable joint programs, such as speakers, competitions, and visits to companies. It convenes technical and business user groups and has the job creation programs integrated within the university training in corporate entrepreneurship. It also supports community entrepreneurship programs.

Governments create and sustain business relationship agencies and include universities. They support places for faculty, students, and industry to meet -- for instance, Daedeok Innopolis. They pass a tax policy to encourage industry and university involvement and create co-investment funds, etc. State governments recruit companies to the region.

1.2.5 De/Centralized Structures

Fifth is De/Centralized Structures. The structure makes corporate entrepreneurship education sustainable. Universities establish central support for decentralized programs for sustainability. They encourage corporate entrepreneurship programs across the entire university and provide corporate entrepreneurship portals for communication between the university and industry collaborators.

The industry creates a university involvement office and sponsors corporate entrepreneurship centers and programs in different departments around the campus. It emphasizes involvement from a diversified major population base. Governments support structures for university involvement in government economic development projects. They create grants for entrepreneurship structures at universities and provide corporate entrepreneurship training programs for university faculty, students, and administrators for industry cooperation.

Large, mature, capital, and technology-intensive organizations struggle to keep up with the global competition, changing demographics, and rapid technology advancements. Universities are offering a wide array of entrepreneurial programs that range from art entrepreneurship to non-profit companies as well as family-owned businesses. These are all worthy approaches to entrepreneurial education, but they do not prepare the people for entrepreneurial activity in large companies. Startups and corporate entrepreneurial thinking have many similarities, but some critical differences must be understood.

2. Future Directions

This project intends to support ANAS-HTP in its institutional capacity development in technology transfer and commercialization. Since its inception in 2016, however, ANAS-HTP has gone through frequent change of leadership and lacked personnel with adequate qualification in the management and operation of the technology park. To achieve the mission of ANAS-HTP as a national center for technology transfer and commercialization, ANAS-HTP should also create an environment for networking among diverse TT/TC stakeholders and promote international cooperation.

To this end, this project for next year will provide more opportunities through training and workshops for Azerbaijan and Korean stakeholders in technology transfer and commercialization to share their experiences and pursue strategic partnership in research and technology commercialization. This project will also try to instill an innovation mindset among researchers, academicians, and TT managers of technology parks in Azerbaijan through workshops on technological innovation and entrepreneurship.

- Reinforcement of collaboration for establishing the infrastructure and empowerment of technical transfer and commercialization in Azerbaijan
  - Continuous support through the 2-year project to activate local technical commercialization and startup
  - Seek mentoring and initial investment possibility for technical startup in Azerbaijan through collaboration with global startup support agencies and professionals in Korea
- Modification and Supplementation of the Operating Strategy of ANAS-HTP
  - The operation strategy document drafted primarily by ANAS-HTP is reviewed by domestic experts for modification and supplementation prior to completion of the final version.
  - The Technopark model of Korea is used to support the development of business support program appropriate for Azerbaijan.

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Chapter 3. Sharing Korea’s Experiences in Technology Transfer System and Policy Support

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2. **Technology Transfer and Commercialization of Public R&D Organization in Korea: Status and Limitations**
3. **Suggestions of a New Paradigm of Technology Transfer and Commercialization: Technology Marketing**

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